

COPPERBELT UNIVERSITY

COMPUTER SCIENCE DEPARTMENT

INTRODUCTION TO COMPUTING

BASICS OF COMPUTER

Introduction

Nowadays, computers are an integral part of our lives. They are used for the reservation of tickets for airplanes and railways, payment of telephone and electricity bills, deposit and withdrawal of money from banks, processing of business data, forecasting of weather conditions, diagnosis of diseases, searching for information on the Internet, etc. Computers are also used extensively in schools, universities, organizations, music industry, movie industry, scientific research, law firms, fashion industry, etc. The term computer is derived from the word *compute*. The word *compute* means *to calculate*. A *computer* is an electronic machine that accepts data from the user, processes the data by performing calculations and operations on it, and generates the desired output results. Computer performs both simple and complex operations, with speed and accuracy. This chapter discusses the history and evolution of computer, the concept of input-process-output and the characteristics of computer. This chapter also discusses the classification of digital computers based on their size and type, and the application of computer in different domain areas.

Digital and Analog Computers

A *digital computer* uses distinct values to represent the data internally. All information are represented using the digits 0s and 1s. The computers that we use at our homes and offices are digital computers. *Analog computer* is another kind of a computer that represents data as variable across a continuous range of values. The earliest computers were analog computers. Analog computers are used for measuring of parameters that vary continuously in real time, such as temperature, pressure and voltage. Analog computers may be more flexible but generally less precise than digital computers. Slide rule is an example of an analog computer. This book deals only with the *digital computer* and uses the term *computer* for them.

Characteristics of Computer

Speed, accuracy, diligence, storage capability and versatility are some of the key characteristics of a computer.

A brief overview of these characteristics are-

- *Speed* The computer can process data very fast, at the rate of millions of instructions per second. Some calculations that would have taken hours and days

to complete otherwise, can be completed in a few seconds using the computer. For example, calculation and generation of salary slips of thousands of employees of an organization, weather forecasting that requires analysis of a large amount of data related to temperature, pressure and humidity of various places, etc.

- *Accuracy* Computer provides a high degree of accuracy. For example, the computer can accurately give the result of division of any two numbers up to 10 decimal places.
- *Diligence* When used for a longer period of time, the computer does not get tired or fatigued. It can perform long and complex calculations with the same speed and accuracy from the start till the end.
- *Storage Capability* Large volumes of data and information can be stored in the computer and also retrieved whenever required. A limited amount of data can be stored, temporarily, in the primary memory. Secondary storage devices like floppy disk and compact disk can store a large amount of data permanently.
- *Versatility* Computer is versatile in nature. It can perform different types of tasks with the same ease. At one moment you can use the computer to prepare a letter document and in the next moment you may play music or print a document.

Computers have several limitations too. Computer can only perform tasks that it has been programmed to do. Computer cannot do any work without instructions from the user. It executes instructions as specified by the user and does not take its own decisions.

Data and Information

What is Data?

The term data refers to individual or raw facts and statistics collected together for reference or analysis. Examples include lists of the raw materials that are combined into a finished product called a bill of materials, records of who owes the organization money called accounts receivable. There are also records of what was produced, how much was sold and to whom, who works for an organization, where the organization's branch locations are, the addresses of the organization's customers, and so on.

What is Information?

When we combine data to gain additional insight, we are creating information. Therefore, information refers to processed facts; data with added meaning or interpretation. Information often consists of summarized data (such as total sales), or data compared across two or more entities (Lusaka versus Kitwe).

Alternatively, Information refers to data that have been shaped into a form that is meaningful and useful to human beings. Data, in contrast, are streams of raw facts representing events occurring in organizations or the physical environment before they have been organized and arranged into a form that people can understand and use.

Qualitative and Quantitative Data

- Qualitative information deals with opinions, policies and narrative descriptions of activities or problems.
- Quantitative information, on the other hand, deals with numbers, frequencies, or quantities.

Characteristics of Data and Information

There are several characteristics of data and information that affect their usefulness to the organization.

Source

The source is important because knowledge of the source affects the credibility of the information. Information from internal sources is often more highly regarded than information from external sources. External sources have different levels of credibility. When more than one source reports the same data, its credibility increases.

Scope

The scope of data is also important. The market trends of Lusaka province, Copperbelt province, are important if that is your entire market, but they are less important if your market is all of Zambia. As with source, it is important to report scope to those who are using the information.

Accuracy

Accuracy refers to the precision of the data – does it correspond to facts and is it free from error? Accuracy is relative, and perfect accuracy is often expensive to achieve, depending on the type of data. We expect almost perfect accuracy in our cash transactions, but we are willing to accept less than perfect accuracy in estimates of the time to complete a project. As data are processed to yield information, the accuracy of the information diminishes because the errors of the individual data are combined.

Frequency

The frequency of data collection and processing should match the needs of the manager. Daily reports of the organization's raw material inventory level might be important to production line managers, but certainly not to the executive in charge of long-term planning. However, that person might want to know typical inventory levels for the past three years.

Timeliness

The timeliness of reporting is another attribute of information. A daily report, received the morning after, is far less useful than one received at the end of the business day.

Value

Information has value. Sometimes the value is obvious. Sometimes the value is much less obvious.

Cost

Every one of the characteristics just discussed carries a cost. For instance, to obtain greater accuracy we must survey a wider sample, or use more precise instruments, or check the data more carefully. More frequent reports cost more to produce. External data usually costs more than internal data.

Generations of Computer

The computer has evolved from a large-sized simple calculating machine to a smaller but much more powerful machine. The evolution of computer to the current state is defined in terms of the generations of computer. Each generation of computer is designed based on a new technological development, resulting in better, cheaper and smaller computers that are more powerful, faster and efficient than their predecessors. Currently, there are five generations of computer. In the following subsections, we will discuss the generations of computer in terms of—

1. The technology used by them (hardware and software),
2. Computing characteristics (speed, i.e., number of instructions executed per second),
3. Physical appearance and
4. Their applications.

First Generation (1940 to 1956): Using Vacuum Tubes

Hardware Technology The first generation of computers used vacuum tubes for circuitry and magnetic drums for memory. The input to the computer was through punched cards and paper tapes. The output was displayed as printouts.

Software Technology The instructions were written in machine language. Machine language uses 0s and 1s for coding of the instructions. The first generation computers could solve one problem at a time.

Computing Characteristics The computation time was in milliseconds.

Physical Appearance These computers were enormous in size and required a large room for installation.

Application They were used for scientific applications as they were the fastest computing device of their time.

Examples UNIVersal Automatic Computer (UNIVAC), Electronic Numerical Integrator And Calculator (ENIAC), and Electronic Discrete Variable Automatic Computer (EDVAC).

The first generation computers used a large number of vacuum tubes and thus generated a lot of heat. They consumed a great deal of electricity and were expensive to operate. The machines were prone to frequent malfunctioning and required constant maintenance. Since first generation computers used machine language, they were difficult to program.

Second Generation (1956 to 1963): Using Transistors

Hardware Technology Transistors replaced the vacuum tubes of the first generation of computers. Transistors allowed computers to become smaller, faster, cheaper, energy efficient and reliable. The second generation computers used *magnetic core technology* for primary memory. They used magnetic tapes and magnetic disks for secondary storage. The input was still through punched cards and the output using printouts. They

used the concept of a stored program, where instructions were stored in the memory of computer.

Software Technology The instructions were written using the *assembly language*. Assembly language uses mnemonics like ADD for addition and SUB for subtraction for coding of the instructions. It is easier to write instructions in assembly language, as compared to writing instructions in machine language. High-level programming languages, such as early versions of COBOL and FORTRAN were also developed during this period.

Computing Characteristics The computation time was in microseconds.

Physical Appearance Transistors are smaller in size compared to vacuum tubes, thus, the size of the computer was also reduced.

Application The cost of commercial production of these computers was very high, though less than the first generation computers. The transistors had to be assembled manually in second generation computers.

Examples PDP-8, IBM 1401 and CDC 1604

Second generation computers generated a lot of heat but much less than the first generation computers. They required less maintenance than the first generation computers.

Third Generation (1964 to 1971): Using Integrated Circuits

Hardware Technology The third generation computers used the *Integrated Circuit (IC)* chips. In an IC chip, multiple transistors are placed on a silicon chip. Silicon is a type of semiconductor. The use of IC chip increased the speed and the efficiency of computer, manifold. The keyboard and monitor were used to interact with the third generation computer, instead of the punched card and printouts.

Software Technology The keyboard and the monitor were interfaced through the *operating system*. Operating system allowed different applications to run at the same time. *High-level languages* were used extensively for programming, instead of machine language and assembly language.

Computing Characteristics The computation time was in nanoseconds.

Physical Appearance The size of these computers was quite small compared to the second generation computers.

Application Computers became accessible to mass audience. Computers were produced commercially, and were smaller and cheaper than their predecessors.

Examples IBM 370, PDP 11

The third generation computers used less power and generated less heat than the second generation computers. The cost of the computer reduced significantly, as individual components of the computer were not required to be assembled manually. The maintenance cost of the computers was also less compared to their predecessors.

Fourth Generation (1971 to present): Using Microprocessors

Hardware Technology They use the *Large Scale Integration (LSI)* and the *Very Large Scale Integration (VLSI)* technology. Thousands of transistors are integrated on a small silicon chip using LSI technology.

VLSI allows hundreds of thousands of components to be integrated in a small chip. This era is marked by the development of microprocessor.

Microprocessor is a chip containing millions of transistors and components, and, designed using LSI and VLSI technology. This generation of computers gave rise to Personal Computer (PC). Semiconductor memory replaced the earlier magnetic core memory, resulting in fast random access to memory. Secondary storage device like magnetic disks became smaller in physical size and larger in capacity. The *linking of computers* is another key development of this era. The computers were linked to form networks that led to the emergence of the Internet. This generation also saw the development of pointing devices like mouse, and handheld devices.

Software Technology Several new operating systems like the MS-DOS and MS-Windows developed during this time. This generation of computers supported *Graphical User Interface (GUI)*. GUI is a user friendly interface that allows user to interact with the computer via menus and icons. High-level programming languages are used for the writing of programs.

Computing Characteristics The computation time is in picoseconds.

Physical Appearance They are smaller than the computers of the previous generation. Some can even fit into the palm of the hand.

Application They became widely available for commercial purposes. Personal computers became available to the home user.

Examples The Intel 4004 chip was the first microprocessor. The components of the computer like Central Processing Unit (CPU) and memory were located on a single chip. In 1981, IBM introduced the first computer for home use. In 1984, Apple introduced the Macintosh.

The microprocessor has resulted in the fourth generation computers being smaller and cheaper than their predecessors. The fourth generation computers are also portable and more reliable. They generate much lesser heat and require less maintenance compared to their predecessors. GUI and pointing devices facilitate easy use and learning on the

computer. Networking has resulted in resource sharing and communication among different computers.

Fifth Generation (Present and Next): Using Artificial Intelligence

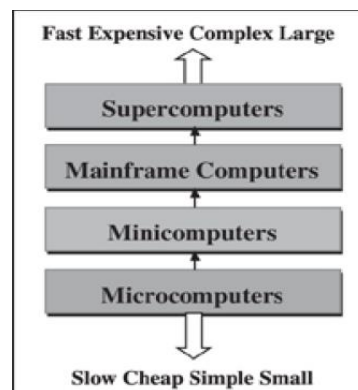
The goal of fifth generation computing is to develop computers that are capable of learning and self-organization. The fifth generation computers use *Super Large Scale Integrated (SLSI)* chips that are able to store millions of components on a single chip. These computers have large memory requirements.

This generation of computers uses *parallel processing* that allows several instructions to be executed in parallel, instead of serial execution. Parallel processing results in faster processing speed. The Intel dual-core microprocessor uses parallel processing.

The fifth generation computers are based on *Artificial Intelligence (AI)*. They try to simulate the human way of thinking and reasoning. Artificial Intelligence includes areas like Expert System (ES), Natural Language Processing (NLP), speech recognition, voice recognition, robotics, etc.

Classification of Computer

The digital computers that are available nowadays vary in their sizes and types. The computers are broadly classified into four categories (Figure 1.8) based on their size and type—(1) Microcomputers, (2) Minicomputers, (3) Mainframe computers, and (4) Supercomputer.



Microcomputers

Microcomputers are small, low-cost and single-user digital computer. They consist of CPU, input unit, output unit, storage unit and the software. Although microcomputers are stand-alone machines, they can be connected together to create a network of computers that can serve more than one user.

IBM PC based on Pentium microprocessor and Apple Macintosh are some examples of microcomputers. Microcomputers include desktop computers, notebook computers or laptop, tablet computer, handheld computer, smart phones and net book.

Desktop Computer or Personal Computer (PC) is the most common type of microcomputer. It is a stand-alone machine that can be placed on the desk. Externally, it consists of three units—keyboard, monitor, and a system unit containing the CPU,

memory, hard disk drive, etc. It is not very expensive and is suited to the needs of a single user at home, small business units, and organizations. Apple, Microsoft, HP, Dell and Lenovo are some of the PC manufacturers.

Notebook Computers or Laptops resemble a notebook. They are portable and have all the features of a desktop computer. The advantage of the laptop is that it is small in size (can be put inside a briefcase), can be carried anywhere, has a battery backup and has all the functionality of the desktop. Laptops can be placed on the lap while working (hence the name). Laptops are costlier than the desktop machines.

Netbook These are smaller notebooks optimized for low weight and low cost, and are designed for accessing web-based applications. Starting with the earliest netbook in late 2007, they have gained significant popularity now. Netbooks deliver the performance needed to enjoy popular activities like streaming videos or music, emailing, Web surfing or instant messaging. The word *netbook* was created as a blend of *Internet* and *notebook*.

Tablet Computer has features of the notebook computer but it can accept input from a stylus or a pen instead of the keyboard or mouse. It is a portable computer. Tablet computer are the new kind of PCs.

Handheld Computer or Personal Digital Assistant (PDA) is a small computer that can be held on the top of the palm. It is small in size. Instead of the keyboard, PDA uses a pen or a stylus for input. PDAs do not have a disk drive. They have a limited memory and are less powerful. PDAs can be connected to the Internet via a wireless connection. Casio and Apple are some of the manufacturers of PDA. Over the last few years, PDAs have merged into mobile phones to create smart phones.

Smart Phones are cellular phones that function both as a phone and as a small PC. They may use a stylus or a pen, or may have a small keyboard. They can be connected to the Internet wirelessly. They are used to access the electronic-mail, download music, play games, etc.

Blackberry, Apple, HTC, Nokia and LG are some of the manufacturers of smart phones.

Minicomputers

Minicomputers are digital computers, generally used in multiuser systems. They have high processing speed and high storage capacity than the microcomputers. Minicomputers can support 4–200 users simultaneously. The users can access the minicomputer through their PCs or terminal. They are used for real-time applications in industries, research centers, etc. PDP 11, IBM (8000 series) are some of the widely used minicomputers.

Mainframe Computers

Mainframe computers are multi-user, multi-programming and high performance computers. They operate at a very high speed, have very large storage capacity and can handle the workload of many users. Mainframe computers are large and powerful systems generally used in centralized databases. The user accesses the mainframe

computer via a terminal that may be a dumb terminal, an intelligent terminal or a PC. A *dumb terminal* cannot store data or do processing of its own. It has the input and output device only. An *intelligent terminal* has the input and output device, can do processing, but, cannot store data of its own. The dumb and the intelligent terminal use the processing power and the storage facility of the mainframe computer. Mainframe computers are used in organizations like banks or companies, where many people require frequent access to the same data. Some examples of mainframes are CDC 6600 and IBM ES000 series.

Supercomputers

Supercomputers are the fastest and the most expensive machines. They have high processing speed compared to other computers. The speed of a supercomputer is generally measured in FLOPS (FLoating point Operations Per Second). Some of the faster supercomputers can perform trillions of calculations per second. Supercomputers are built by interconnecting thousands of processors that can work in parallel.

Supercomputers are used for highly calculation-intensive tasks, such as, weather forecasting, climate research (global warming), molecular research, biological research, nuclear research and aircraft design. They are also used in major universities, military agencies and scientific research laboratories. Some examples of supercomputers are IBM Roadrunner, IBM Blue gene and Intel ASCI red. PARAM is a series of supercomputer assembled in India by CDAC (Center for Development of Advanced Computing), in Pune. PARAM Padma is the latest machine in this series. The peak computing power of PARAM Padma is 1 Tera FLOP (TFLOP).

The Computer System

Computer is an electronic device that accepts data as input, processes the input data by performing mathematical and logical operations on it, and gives the desired output. The computer system consists of four parts—(1) Hardware, (2) Software, (3) Data, and (4) Users.

Hardware consists of the mechanical parts that make up the computer as a machine. The hardware consists of physical devices of the computer. The devices are required for input, output, storage and processing of the data. Keyboard, monitor, hard disk drive, floppy disk drive, printer, processor and motherboard are some of the hardware devices.

Software is a set of instructions that tells the computer about the tasks to be performed and how these tasks are to be performed. *Program* is a set of instructions, written in a language understood by the computer, to perform a specific task. A set of programs and documents are collectively called software. The hardware of the computer system cannot perform any task on its own. The hardware needs to be instructed about the task to be performed. Software instructs the computer about the task to be performed. The hardware carries out these tasks. Different software can be loaded on the same hardware to perform different kinds of tasks.

Data are isolated values or raw facts, which by themselves have no much significance. For example, the data like 29, January, and 1994 just represent values. The data is provided as input to the computer, which is processed to generate some meaningful information. For example, 29, January and 1994 are processed by the computer to give the date of birth of a person.

Users are people who write computer programs or interact with the computer. Programmers, data entry operators, system analyst and computer hardware engineers fall into this category

The Input-Process-Output Concept

A computer is an electronic device that (1) accepts data, (2) processes data, (3) generates output, and (4) stores data. The concept of generating output information from the input data is also referred to as *input-process-output* concept.

The input-process-output concept of the computer is explained as follows—

Input The computer accepts input data from the user via an input device like keyboard. The input data can be characters, word, text, sound, images, document, etc.

Process The computer processes the input data. For this, it performs some actions on the data by using the instructions or program given by the user of the data. The action could be an arithmetic or logic calculation, editing, modifying a document, etc. During processing, the data, instructions and the output are stored temporarily in the computer's main memory.

Output The output is the result generated after the processing of data. The output may be in the form of text, sound, image, document, etc. The computer may display the output on a monitor, send output to the printer for printing, play the output, etc. **Storage** The input data, instructions and output are stored permanently in the secondary storage devices like disk or tape. The stored data can be retrieved later, whenever needed.

Components of Computer Hardware

The computer system hardware comprises of three main components —

1. Input/Output (I/O) Unit,
2. Central Processing Unit (CPU), and
3. Memory Unit.

The I/O unit consists of the input unit and the output unit. CPU performs calculations and processing on the input data, to generate the output. The memory unit is used to store the data, the instructions and the output information. The figure below illustrates the typical interaction among the different components of the computer.

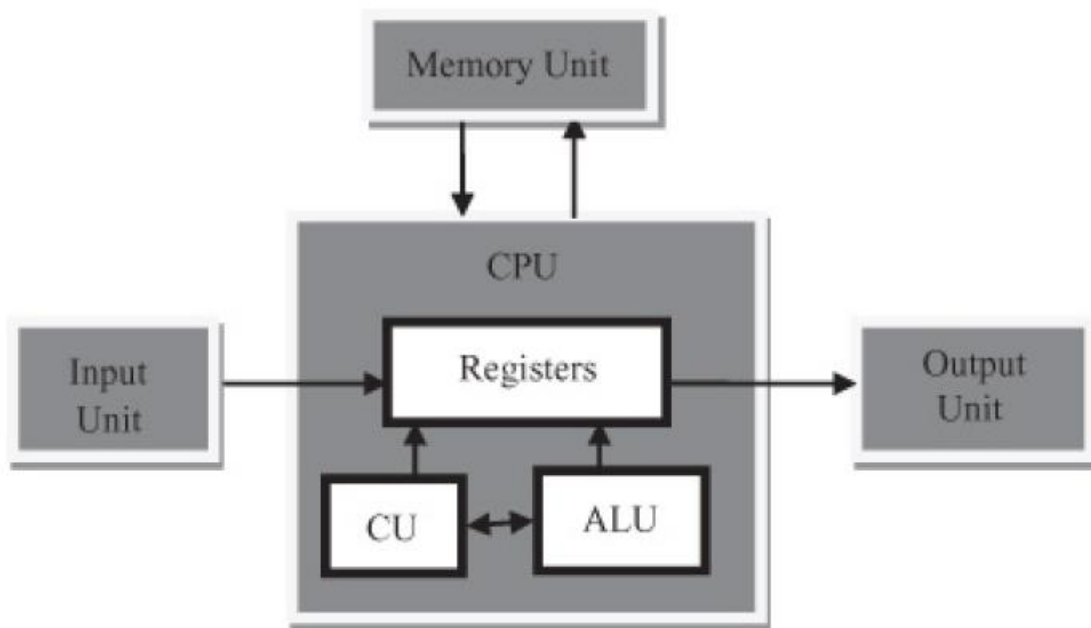


Figure: The computer system interaction

Input/Output Unit The user interacts with the computer via the I/O unit. The Input unit accepts data from the user and the Output unit provides the processed data i.e. the information to the user. The Input unit converts the data that it accepts from the user, into a form that is understandable by the computer. Similarly, the Output unit provides the output in a form that is understandable by the user. The input is provided to the computer using input devices like keyboard, trackball and mouse. Some of the commonly used output devices are monitor and printer.

Central Processing Unit CPU controls, coordinates and supervises the operations of the computer. It is responsible for processing of the input data. CPU consists of Arithmetic Logic Unit (ALU) and Control Unit (CU). ALU performs all the arithmetic and logic operations on the input data.

CU controls the overall operations of the computer i.e. it checks the sequence of execution of instructions, and, controls and coordinates the overall functioning of the units of computer. Additionally, CPU also has a set of *registers* for temporary storage of data, instructions, addresses and intermediate results of calculation.

Memory Unit Memory unit stores the data, instructions, intermediate results and output, *temporarily*, during the processing of data. This memory is also called the *main memory* or *primary memory* of the computer. The input data that is to be processed is brought into the main memory before processing. The instructions required for processing of data and any intermediate results are also stored in the main memory. The output is stored in memory before being transferred to the output device. CPU can work with the information stored in the main memory. Another kind of storage unit is also referred to as the *secondary memory* of the computer. The data, the programs and the output are stored

permanently in the storage unit of the computer. Magnetic disks, optical disks and magnetic tapes are examples of secondary memory.

Application of Computers

Computers have proliferated into various areas of our lives. For a user, computer is a tool that provides the desired information, whenever needed. You may use computer to get information about the reservation of tickets (railways, airplanes and cinema halls), books in a library, medical history of a person, a place in a map, or the dictionary meaning of a word. The information may be presented to you in the form of text, images, video clips, etc.

Some of the application areas of the computer are listed below—

Education Computers are extensively used, as a tool and as an aid, for imparting education. Educators use computers to prepare notes and presentations of their lectures. Computers are used to develop computer based training packages, to provide distance education using the e-learning software, and to conduct online examinations. Researchers use computers to get easy access to conference and journal details and to get global access to the research material.

Entertainment Computers have had a major impact on the entertainment industry. The user can download and view movies, play games, chat, book tickets for cinema halls, use multimedia for making movies, incorporate visual and sound effects using computers, etc. The users can also listen to music, download and share music, create music using computers, etc.

Sports A computer can be used to watch a game, view the scores, improve the game, play games (like chess, etc.) and create games. They are also used for the purposes of training players.

Advertising Computer is a powerful advertising media. Advertisement can be displayed on different websites, electronic-mails can be sent and reviews of a product by different customers can be posted. Computers are also used to create an advertisement using the visual and the sound effects. For the advertisers, computer is a medium via which the advertisements can be viewed globally. Web advertising has become a significant factor in the marketing plans of almost all companies. In fact, the business model of Google is mainly dependent on web advertising for generating revenues.

Medicine Medical researchers and practitioners use computers to access information about the advances in medical research or to take opinion of doctors globally. The medical history of patients is stored in the computers. Computers are also an integral part of various kinds of sophisticated medical equipments like ultrasound machine, CAT scan machine, MRI scan machine, etc. Computers also provide assistance to the medical surgeons during critical surgery operations like laparoscopic operations, etc.

Science and Engineering Scientists and engineers use computers for performing complex scientific calculations, for designing and making drawings (CAD/CAM applications) and also for simulating and testing the designs. Computers are used for storing the complex data, performing complex calculations and for visualizing 3-dimensional objects. Complex scientific applications like the launch of the rockets, space exploration, etc., are not possible without the computers.

Government The government uses computers to manage its own operations and also for e-governance. The websites of the different government departments provide information to the users. Computers are used for the filing of income tax return, paying taxes, online submission of water and electricity bills, for the access of land record details, etc. The police department uses computers to search for criminals using fingerprint matching, etc.

Home Computers have now become an integral part of home equipment. At home, people use computers to play games, to maintain the home accounts, for communicating with friends and relatives via Internet, for paying bills, for education and learning, etc. Microprocessors are embedded in house hold utilities like, washing machines, TVs, food processors, home theatres, security devices, etc.

The list of applications of computers is so long that it is not possible to discuss all of them here. In addition to the applications of the computers discussed above, computers have also proliferated into areas like banks, investments, stock trading, accounting, ticket reservation, military operations, meteorological predictions, social networking, business organizations, police department, video conferencing, book publishing, web newspapers, and information sharing.

COPPERBELT UNIVERSITY

COMPUTER SCIENCE DEPARTMENT

INTRODUCTION TO COMPUTING

UNIT 2: The Computer System Hardware

Introduction

When we talk of computer hardware, the three related terms that require introduction are—computer architecture, computer organization and computer design. Computer architecture refers to the structure and behavior of the computer. It includes the specifications of the components, for example, instruction format, instruction set and techniques for addressing memory, and how they connect to the other components. Given the components, computer organization focuses on the organizational structure. It deals with how the hardware components operate and the way they are connected to form the computer. Given the system specifications, computer design focuses on the hardware to be used and the interconnection of parts. Different kinds of computer, such as a PC or a mainframe computer may have different organization; however, basic organization of the computer remains the same.

A computer consists of three main components—(1) Input/output (I/O) Unit, (2) Central Processing Unit (CPU), and (3) Memory Unit. The computer user interacts with the computer via the I/O unit. The purpose of I/O unit is to provide data and instructions as input to the computer and to present relevant information as output from the computer. CPU controls the operations of the computer and processes the received input to generate the relevant output. The memory unit stores the instructions and the data during the input activity, to make instructions readily available to CPU during processing. It also stores the processed output. This chapter discusses the hardware components of the computer and the interaction between them.

Central Processing Unit

Central Processing Unit (CPU) or the processor is also often called the brain of computer. CPU ([Figure 1](#)) consists of Arithmetic Logic Unit (ALU) and Control Unit (CU). In addition, CPU also has a set of registers which are temporary storage areas for holding data, and instructions. ALU performs the arithmetic and logic operations on the data that is made available to it. CU is responsible for organizing the processing of data and instructions. CU controls and coordinates the activity of the other units of computer. CPU uses the registers to store the data, instructions during processing.

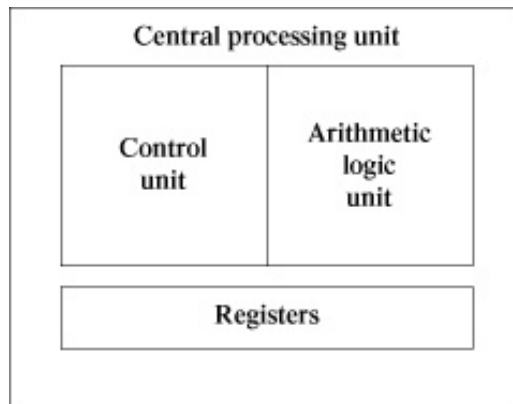


Figure 1 CPU

CPU executes the stored program instructions, i.e. instructions and data are stored in memory before execution. For processing, CPU gets data and instructions from the memory. It interprets the program instructions and performs the arithmetic and logic operations required for the processing of data. Then, it sends the processed data or result to the memory. CPU also acts as an administrator and is responsible for supervising operations of other parts of the computer.

The CPU is fabricated as a single Integrated Circuit (IC) chip, and is also known as the microprocessor. The microprocessor is plugged into the motherboard of the computer (Motherboard is a circuit board that has electronic circuit etched on it and connects the microprocessor with the other hardware components).

Arithmetic Logic Unit

- ALU consists of two units—arithmetic unit and logic unit.
- The arithmetic unit performs arithmetic operations on the data that is made available to it. Some of the arithmetic operations supported by the arithmetic unit are—addition, subtraction, multiplication and division.
- The logic unit of ALU is responsible for performing logic operations. Logic unit performs comparisons of numbers, letters and special characters. Logic operations include testing for greater than, less than or equal to condition.
- ALU performs arithmetic and logic operations, and uses registers to hold the data that is being processed.

Registers

- Registers are high-speed storage areas within the CPU, but have the least storage capacity. Registers are not referenced by their address, but are directly accessed and manipulated by the CPU during instruction execution.
- Registers store data, instructions, addresses and intermediate results of processing. Registers are often referred to as the CPU's working memory.
- The data and instructions that require processing must be brought in the registers of CPU before they can be processed. For example, if two numbers are to be added, both numbers are brought in the registers, added and the result is also placed in a register.

- Registers are used for different purposes, with each register serving a specific purpose. Some of the important registers in CPU (Figure 2) are as follows—
 - Accumulator (ACC) stores the result of arithmetic and logic operations.
 - Instruction Register (IR) contains the current instruction most recently fetched.
 - Program Counter (PC) contains the address of next instruction to be processed.
 - Memory Address Register (MAR) contains the address of next location in the memory to be accessed.
 - Memory Buffer Register (MBR) temporarily stores data from memory or the data to be sent to memory.
 - Data Register (DR) stores the operands and any other data.



Figure 2 CPU registers

- The number of registers and the size of each (number of bits) register in a CPU help to determine the power and the speed of a CPU.
- The overall number of registers can vary from about ten to many hundreds, depending on the type and complexity of the processor.
- The size of register, also called word size, indicates the amount of data with which the computer can work at any given time. The bigger the size, the more quickly it can process data. The size of a register may be 8, 16, 32 or 64 bits. For example, a 32-bit CPU is one in which each register is 32 bits wide and its CPU can manipulate 32 bits of data at a time. Nowadays, PCs have 32-bit or 64-bit registers.
- 32-bit processor and 64-bit processor are the terms used to refer to the size of the registers. Other factors remaining the same, a 64-bit processor can process the data twice as fast as one with 32-bit processor.

Control Unit

- The control unit of a computer does not do any actual processing of data. It organizes the processing of data and instructions. It acts as a supervisor and, controls and coordinates the activity of the other units of the computer.
- CU coordinates the input and output devices of a computer. It directs the computer to carry out stored program instructions by communicating with the ALU and the registers. CU uses the instructions in the Instruction Register (IR) to decide which circuit needs to be activated. It also instructs

the ALU to perform the arithmetic or logic operations.

When a program is run, the Program Counter (PC) register keeps track of the program instruction to be executed next.

- CU tells when to fetch the data and instructions, what to do, where to store the results, the sequencing of events during processing etc.
- CU also holds the CPU's Instruction Set, which is a list of all operations that the CPU can perform.

The function of a (CU) can be considered synonymous with that of a conductor of an orchestra. The conductor in an orchestra does not perform any work by itself but manages the orchestra and ensures that the members of orchestra work in proper coordination.

Memory Unit

The memory unit consists of cache memory and primary memory. Primary memory or main memory of the computer is used to store the data and instructions during execution of the instructions. Random Access Memory (RAM) and Read Only Memory (ROM) are the primary memory. In addition to the main memory, there is another kind of storage device known as the secondary memory. Secondary memory is non-volatile and is used for permanent storage of data and programs. A program or data that has to be executed is brought into the RAM from the secondary memory.

Cache Memory

- The data and instructions that are required during the processing of data are brought from the secondary storage devices and stored in the RAM. For processing, it is required that the data and instructions are accessed from the RAM and stored in the registers. The time taken to move the data between RAM and CPU registers is large. This affects the speed of processing of computer, and results in decreasing the performance of CPU.
- Cache memory is a very high speed memory placed in between RAM and CPU. Cache memory increases the speed of processing.
- Cache memory is a storage buffer that stores the data that is used more often, temporarily, and makes them available to CPU at a fast rate. During processing, CPU first checks cache for the required data. If data is not found in cache, then it looks in the RAM for data.
- To access the cache memory, CPU does not have to use the motherboard's system bus for data transfer. (The data transfer speed slows to the motherboard's capability, when data is passed through system bus. CPU can process data at a much faster rate by avoiding the system bus.)
- Cache memory is built into the processor, and may also be located next to it on a separate chip between the CPU and RAM. Cache built into the CPU is faster than separate cache, running at the speed of the microprocessor itself. However, separate cache is roughly twice as fast as RAM.

- The CPU has a built-in Level 1 (L1) cache and Level 2 (L2) cache, as shown in [Figure 3](#). In addition to the built-in L1 and L2 cache, some CPUs have a separate cache chip on the motherboard. This cache on the motherboard is called Level 3 (L3) cache. Nowadays, high-end processor comes with built-in L3 cache, like in Intel core i7. The L1, L2 and L3 cache store the most recently run instructions, the next ones and the possible ones, respectively. Typically, CPUs have cache size varying from 256KB (L1), 6 MB (L2), to 12MB (L3) cache.

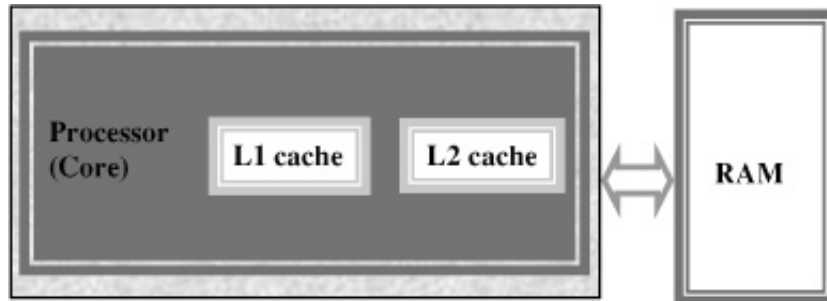


Figure 3 Illustration of cache memory

- Cache memory is very expensive, so it is smaller in size. Generally, computers have cache memory of sizes 256 KB to 2 MB.

Primary Memory

- Primary memory is the main memory of computer. It is used to store data and instructions during the processing of data. Primary memory is semiconductor memory.
- Primary memory is of two kinds—Random Access Memory (RAM) and Read Only Memory (ROM).
- RAM is volatile. It stores data when the computer is on. The information stored in RAM gets erased when the computer is turned off. RAM provides temporary storage for data and instructions.
- ROM is non-volatile memory, but is a read only memory. The storage in ROM is permanent in nature, and is used for storing standard processing programs that permanently reside in the computer. ROM comes programmed by the manufacturer.
- RAM stores data and instructions during the execution of instructions. The data and instructions that require processing are brought into the RAM from the storage devices like hard disk. CPU accesses the data and the instructions from RAM, as it can access it at a fast speed than the storage devices connected to the input and output unit ([Figure 4](#)).

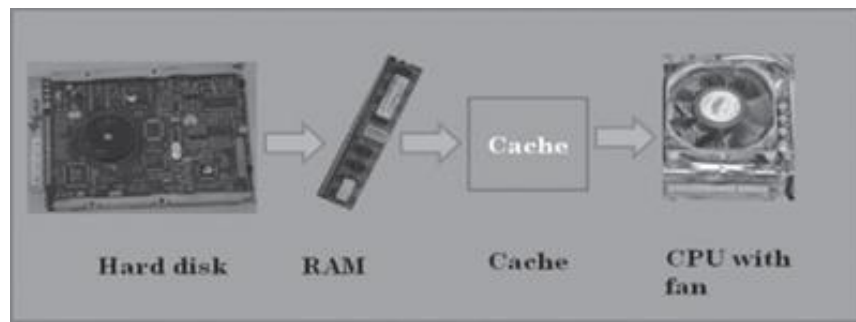


Figure 4 Interaction of CPU with memory

- The input data that is entered using the input unit is stored in RAM, to be made available during the processing of data. Similarly, the output data generated after processing is stored in RAM before being sent to the output device. Any intermediate results generated during the processing of program are stored in RAM.
- RAM provides a limited storage capacity, due to its high cost.

Secondary Memory

- The secondary memory stores data and instructions permanently. The information can be stored in secondary memory for a long time (years), and is generally permanent in nature unless erased by the user. It is a non-volatile memory.
- It provides back-up storage for data and instructions. Hard disk drive, floppy drive and optical disk drives are some examples of storage devices.
- The data and instructions that are currently not being used by CPU, but may be required later for processing, are stored in secondary memory.
- Secondary memory has a high storage capacity than the primary memory.
- Secondary memory is also cheaper than the primary memory.
- It takes longer time to access the data and instructions stored in secondary memory than in primary memory.

Magnetic tape drives, disk drives and optical disk drives are the different types of storage devices.

Instruction Cycle

The primary responsibility of a computer processor is to execute a sequential set of instructions that constitute a program. CPU executes each instruction in a series of steps, called instruction cycle (Figure 8).

- An instruction cycle involves four steps (Figure 9)—
 - Fetching

The processor fetches the instruction from the memory. The fetched instruction is placed in the Instruction Register. Program Counter holds the address of next instruction to be fetched and is incremented after each fetch.

- Decoding
The instruction that is fetched is broken down into parts or decoded. The instruction is translated into commands so that they correspond to those in the CPU's instruction set. The instruction set architecture of the CPU defines the way in which an instruction is decoded.
- Executing
The decoded instruction or the command is executed. CPU performs the operation implied by the program instruction. For example, if it is an ADD instruction, addition is performed.
- Storing
CPU writes back the results of execution, to the computer's memory.



Figure 8 Instruction cycle

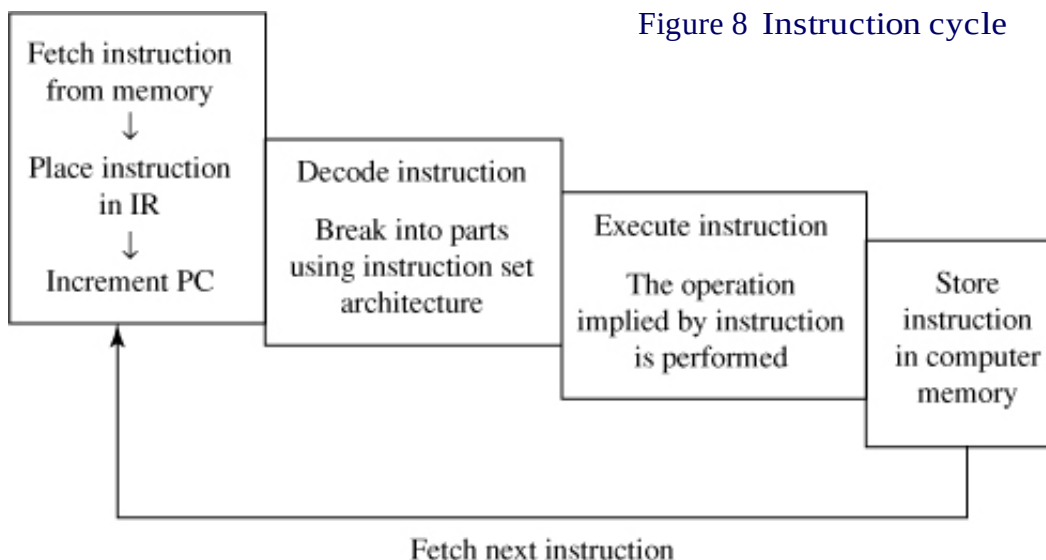


Figure 9 Steps in instruction cycle

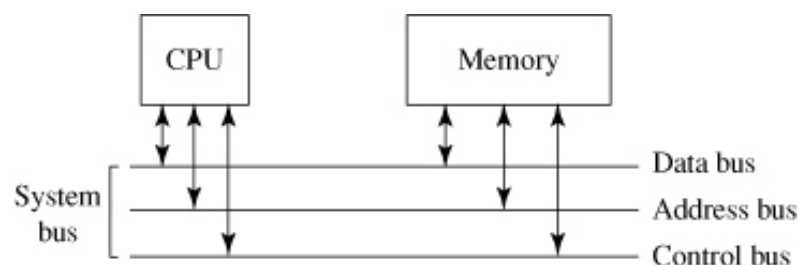
- Instructions are of different categories. Some categories of instructions are—
 - Memory access or transfer of data between registers.
 - Arithmetic operations like addition and subtraction.
 - Logic operations such as AND, OR and NOT.
 - Control the sequence, conditional connections, etc.

A CPU performance is measured by the number of instructions it executes in a second, i.e., MIPS (million instructions per second), or BIPS (billion instructions per second).

Interconnecting the Units of a Computer

CPU sends data, instructions and information to the components inside the computer as well as to the peripherals and devices attached to it. Bus is a set of electronic signal pathways that allows information and signals to travel between components inside or outside of a computer. The different components of computer, i.e., CPU, I/O unit, and memory unit are connected with each other by a bus. The data, instructions and the signals are carried between the different components via a bus. The features and functionality of a bus are as follows—

- A bus is a set of wires used for interconnection, where each wire can carry one bit of data.
- A bus width is defined by the number of wires in the bus.
- A computer bus can be divided into two types—Internal Bus and External Bus.
- The Internal Bus connects components inside the motherboard like, CPU and system memory. It is also called the System Bus. [Figure 10](#) shows interaction between processor and memory.



[Figure 10](#) Interaction between CPU and memory

- The External Bus connects the different external devices, peripherals, expansion slots, I/O ports and drive connections to the rest of computer. The external bus allows various devices to be attached to the computer. It allows for the expansion of computer's capabilities. It is generally slower than the system bus. It is also referred to as the Expansion Bus.

- A system bus or expansion bus comprise of three kinds of buses - data bus, address bus and control bus.
- The interaction of CPU with memory and I/O devices involves all the three buses.
 - The command to access the memory or the I/O device is carried by the control bus.
 - The address of I/O device or memory is carried by the address bus.
 - The data to be transferred is carried by the data bus.

Figure 11 shows interaction between processor, memory and the peripheral devices.

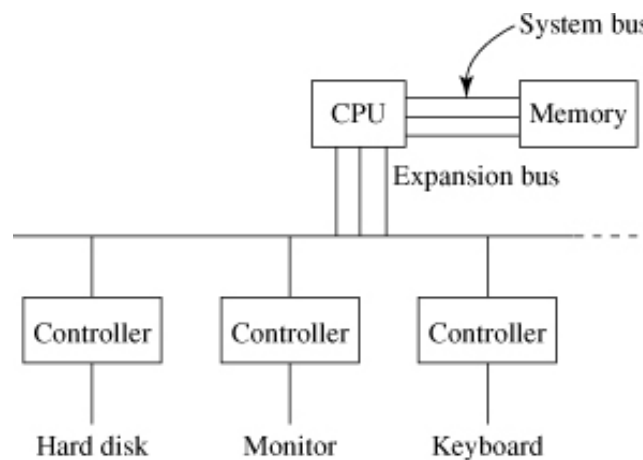


Figure 11 Interaction between CPU, memory and peripheral devices

System Bus

The functions of data bus, address bus and control bus, in the system bus, are as follows—

- **Data Bus** transfers data between the CPU and memory. The bus width of a data bus affects the speed of computer. The size of data bus defines the size of the processor. A processor can be 8, 16, 32 or 64-bit processor. An 8-bit processor has 8 wire data bus to carry 1 byte of data. In a 16-bit processor, 16-wire bus can carry 16 bits of data, i.e., transfer 2 bytes, etc.
- **Address Bus** connects CPU and RAM with set of wires similar to data bus. The width of address bus determines the maximum number of memory locations the computer can address. Currently, Pentium Pro, II, III, IV have 36-bit address bus that can address 2^{36} bytes or 64 GB of memory.
- **Control Bus** specifies whether data is to be read or written to the memory, etc.

Expansion Bus

The functions of data bus, address bus and control bus, in the expansion bus, are as follows—

- The expansion bus connects external devices to the rest of computer. The external devices like monitor, keyboard and printer connect to ports on the back of computer. These ports are actually a part of the small circuit board or expansion card that fits into an expansion slot on the motherboard. Expansion slots are easy to recognize on the motherboard.
- Expansion slots make up a row of long plastic connectors at the back of the computer with tiny copper ‘finger slots’ in a narrow channel that grab the connectors on the expansion cards. The slots are attached to tiny copper pathways on the motherboard (the expansion bus), which allows the device to communicate with the rest of computer.
- Data Bus is used to transfer data between I/O devices and CPU. The exchange of data between CPU and I/O devices is according to the industry standard data buses. The most commonly used standard is Extended Industry Standard Architecture (EISA) which is 32-bit bus architecture. Some of the common bus technologies are:
 - Peripheral Component Interconnect (PCI) bus for hard disks, sound cards, network cards and graphics cards,
 - Accelerated Graphics Port (AGP) bus for 3-D and full motion video,
 - Universal Serial Bus (USB) to connect and disconnect different devices.
- Address Bus carries the addresses of different I/O devices to be accessed like the hard disk, CD ROM, etc.
- Control Bus is used to carry read/write commands, status of I/O devices, etc.

External Ports

The peripheral devices interact with the CPU of the computer via the bus. The connections to the bus from the peripheral devices are made via the ports and sockets provided at the sides of the computer. The different ports and sockets facilitate the connection of different devices to the computer. Some of the standard port connections available on the outer sides of the computer are—port for mouse, keyboard, monitor, network, modem, and, audio port, serial port, parallel port and USB port. The different ports are physically identifiable by their different shapes, size of contact pins and number of pins. [Figure 12](#) shows the interaction of serial and parallel port interfaces with the devices.

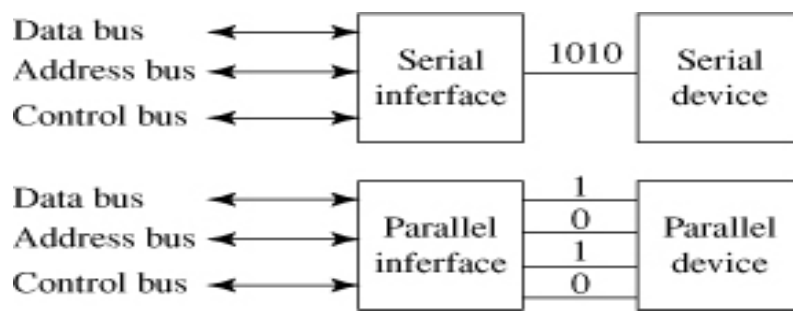


Figure 12 Interaction of serial and parallel port interfaces

Performance of a Computer

There are a number of factors involved that are related to the CPU and have an effect on the overall speed and performance of the computer. Some of the factors that affect the performance of the computer include —

- **Registers**

The size of the register (word size) indicates the amount of data with which the computer can work at any given time. The bigger the size, the more quickly it can process data. A 32-bit CPU is one in which each register is 32 bits wide.

- **RAM**

It is used to store data and instructions during execution of the instructions. Anything you do on your computer requires RAM. When the computer is switched on, the operating system, device drivers, the active files and running programs are loaded into RAM. If RAM is less, then the CPU waits each time the new information is swapped into memory from the slower devices. Larger the RAM size, the better it is. PCs nowadays usually have 1 GB to 4 GB of RAM.

- **System Clock**

The clock speed of a CPU is defined as the frequency with which a processor executes instructions or the data is processed. Higher clock frequencies mean more clock ticks per second. The computer's operating speed is linked to the speed of the system clock. The clock frequency is measured in millions of cycles per second or megahertz (MHz) or gigahertz (GHz) which is billions of cycles per second. A CPU's performance is measured by the number of instructions it executes in a second, i.e., MIPS or BIPS. PCs nowadays come with a clock speed of more than 1 GHz. In Windows OS, you can select the System Properties dialog box to see the processor name and clock frequency.

- **Bus Data**

Bus is used for transferring data between CPU and memory. The data bus width affects the speed of computer. In a 16-bit processor, 16-bit wire bus can carry 16 bits of data. The bus speed is measured in MHz. Higher the bus speed the better it is. Address bus connects CPU and RAM with a set of wires similar to data bus. The address bus width determines the maximum number of memory locations the computer can address. Pentium Pro, II, III, IV have 36-bit address bus that can address 2^{36} bytes or 64 GB of memory. PCs nowadays have a bus speed varying from 100 MHz to 400 MHz.

- **Cache**
Memory Two of the main factors that affect a cache's performance are its size (amount of cache memory) and level L1, L2 and L3. Larger the size of cache, the better it is. PCs nowadays have a L1 cache of 256KB and L2 cache of 1MB.

Inside a Computer Cabinet

The computer cabinet encloses the components that are required for the running of the computer. The components inside a computer cabinet include the power supply, motherboard, memory chips, expansion slots, ports and interface, processor, cables and storage devices.

Motherboard

The computer is built up around a motherboard. The motherboard is the most important component in the PC. It is a large Printed Circuit Board (PCB), having many chips, connectors and other electronics mounted on it. The motherboard is the hub, which is used to connect all the essential components of a computer. The RAM, hard drive, disk drives and optical drives are all plugged into interfaces on the motherboard. The motherboard contains the processor, memory chips, interfaces and sockets, etc.

The motherboard may be characterized by the form factor, chipset and type of processor socket used. Form factor refers to the motherboard's geometry, dimensions, arrangement and electrical requirements. Different standards have been developed to build motherboards, which can be used in different brands of cases. Advanced Technology Extended (ATX) is the most common design of motherboard for desktop computers. Chipset is a circuit, which controls the majority of resources (including the bus interface with the processor, cache memory and RAM, expansion cards, etc.) Chipset's job is to coordinate data transfers between the various components of the computer (including the processor and memory). As the chipset is integrated into the motherboard, it is important to choose a motherboard, which includes a recent chipset, in order to maximize the computer's upgradeability.

The processor socket may be a rectangular connector into which the processor is mounted vertically (slot), or a square-shaped connector with many small connectors into which the processor is directly inserted (socket). The Basic Input Output System (BIOS) and Complementary Metal-Oxide Semiconductor (CMOS) are present on the motherboard.

- **BIOS**
It is the basic program used as an interface between the operating system and the motherboard. The BIOS ([Figure 14](#)) is stored in the ROM and cannot be rewritten. When the computer is switched on, it needs instructions to start. BIOS contain the instructions for the starting up of the computer. The BIOS runs when the computer is switched on.

It performs a Power On Self Test (POST) that checks that the hardware is functioning properly and the hardware devices are present. It checks whether the operating system is present on the hard drive. BIOS invokes the bootstrap loader to load the operating system into memory. BIOS can be configured using an interface named BIOS setup, which can be accessed when the computer is booting up (by pressing the DEL key).

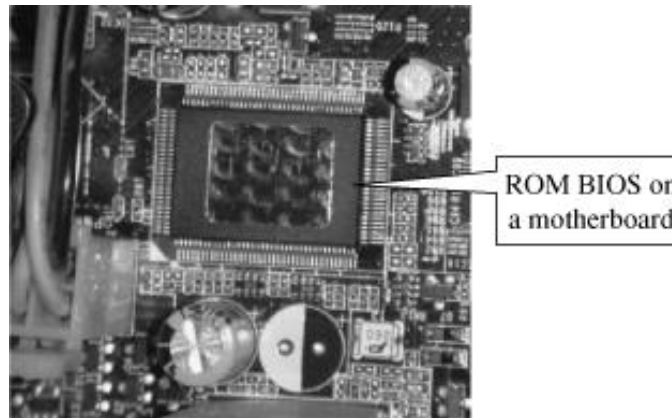


Figure 14 ROM BIOS

- **CMOS Chip**

BIOS ROMs are accompanied by a smaller CMOS (CMOS is a type of memory technology) memory chip. When the computer is turned off, the power supply stops providing electricity to the motherboard. When the computer is turned on again, the system still displays the correct clock time. This is because the CMOS chip saves some system information, such as time, system date and essential system settings. CMOS is kept powered by a button battery located on the motherboard (Figure 15). The CMOS chip is working even when the computer power is switched off. Information of the hardware installed in the computer (such as the number of tracks or sectors on each hard drive) is stored in the CMOS chip.

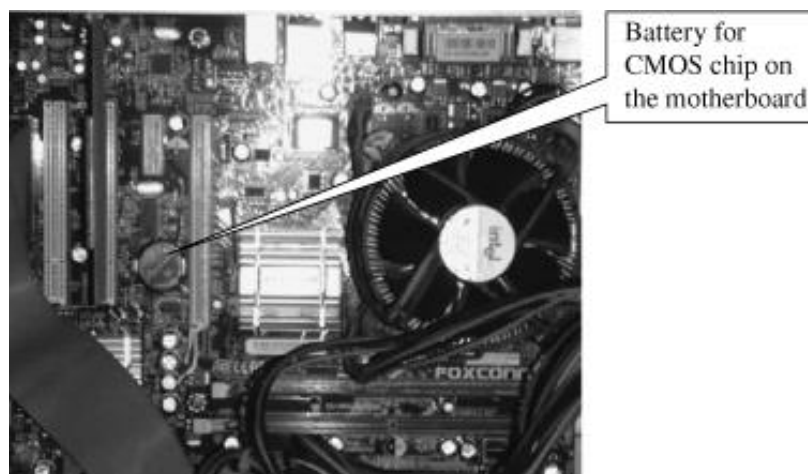


Figure 15 Battery for CMOS chip

Ports and Interfaces

Motherboard has a certain number of I/O sockets that are connected to the ports and interfaces found on the rear side of a computer (Figure 16). You can connect external devices to the ports and interfaces, which get connected to the computer's motherboard.

- Serial Port: to connect old peripherals.
- Parallel Port: to connect old printers.



Figure 16 Ports on the rear side of a PC

- USB Ports—to connect newer peripherals like cameras, scanners and printers to the computer. It uses a thin wire to connect to the devices, and many devices can share that wire simultaneously.
- Firewire is another bus, used today mostly for video cameras and external hard drives.
- RJ45 connector (called LAN or Ethernet port) is used to connect the computer to a network. It corresponds to a network card integrated into the motherboard.
- VGA connector - for connecting a monitor. This connector interfaces with the built-in graphics card.
- Audio plugs (line-in, line-out and microphone), for connecting sound speakers and the microphone. This connector interfaces with the built-in sound card.
- PS/2 port - to connect mouse and keyboard into PC.
- SCSI port for connecting the hard disk drives and network connectors.

Expansion Slots

The expansion slots (Figure 17) are located on the motherboard. The expansion cards are inserted in the expansion slots. These cards give the computer new features or increased performance. There are several types of slots:

- ISA (Industry Standard Architecture) slot: To connect modem and input devices.
- PCI (Peripheral Component Interconnection) slot: To connect audio, video and graphics. They are much faster than ISA cards.
- AGP (Accelerated Graphic Port) slot: A fast port for a graphics card.
- PCI (Peripheral Component Interconnection) Express slot: Faster bus architecture than AGP and PCI buses.
- PC Card: It is used in laptop computers. It includes Wi-Fi card, network card and external modem.

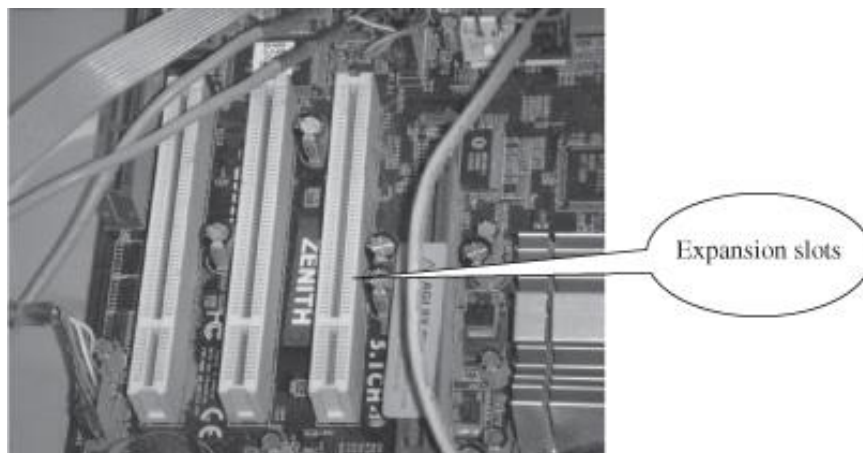


Figure 17 Expansion slots

Ribbon Cables

Ribbon cables (Figure 18) are flat, insulated and consist of several tiny wires moulded together that carry data to different components on the motherboard. There is a wire for each bit of the word or byte and additional wires to coordinate the activity of moving information. They also connect the floppy drives, disk drives and CD-ROM drives to the connectors in the motherboard. Nowadays, Serial Advanced Technology Attachment (SATA) cables have replaced the ribbon cables to connect the drives to the motherboard.

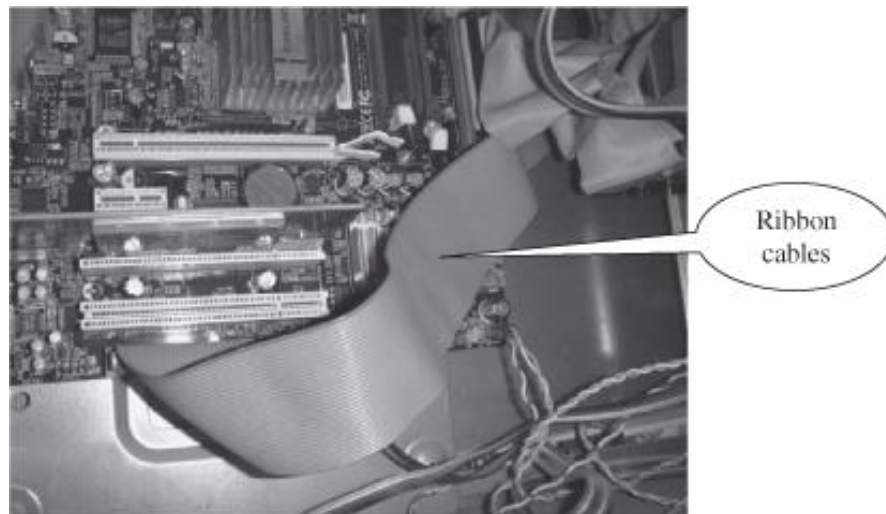


Figure 18 Ribbon cables inside a PC

Memory Chips

The RAM consists of chips on a small circuit board (Figure 19). Two types of memory chips—Single In-line Memory Module (SIMM) and Dual In-line Memory Module (DIMM) are used in desktop computers. The CPU can retrieve information from DIMM chip at 64 bits compared to 32 bits or 16 bits transfer with SIMM chips. DIMM chips are used in Pentium 4 onwards to increase the access speed.

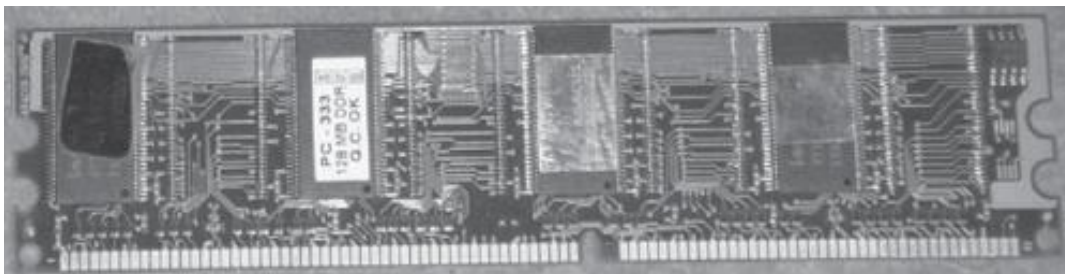


Figure 19 RAM memory chip

Storage Devices

The disk drives are present inside the machine. The common disk drives in a machine are hard disk drive, floppy drive (Figure 20 (i & ii)) and CD drive or DVD drive. High-storage devices like hard disk, floppy disk and CDs (Figure 20 (iii) & (iv)) are inserted into the hard disk drive, floppy drive and CD drive, respectively. These storage devices can store large amounts of data, permanently.



Figure 20 Storage devices (i) Hard disk drive, (ii) DVD drive, (iii) Floppy disk, (iv) CD

Processor

The processor or the CPU is the main component of the computer. Select a processor based on factors like its speed, performance, reliability and motherboard support. Pentium Pro, Pentium 2 and Pentium 4 are some of the processors.

COPPERBELT UNIVERSITY

COMPUTER SCIENCE DEPARTMENT

INTRODUCTION TO COMPUTING

UNIT 3: MEMORY

INTRODUCTION

The computer's memory stores data, instructions required during the processing of data, and output results. Storage may be required for a limited period of time, instantly, or, for an extended period of time. Different types of memories, each having its own unique features, are available for use in a computer. The cache memory, registers, and RAM are fast memories and store the data and instructions temporarily during the processing of data and instructions. The secondary memories like magnetic disks and optical disks have large storage capacities and store the data and instructions permanently, but are slow memory devices. The memories are organized in the computer in a manner to achieve high levels of performance at the minimum cost.

In this unit, we discuss different types of memories, their characteristics and their use in the computer.

MEMORY REPRESENTATION

The computer memory stores different kinds of data like input data, output data, intermediate results, etc., and the instructions. Binary digit or bit is the basic unit of memory. A bit is a single binary digit, i.e., 0 or 1. A bit is the smallest unit of representation of data in a computer. However, the data is handled by the computer as a combination of bits. A group of 8 bits form a byte. One byte is the smallest unit of data that is handled by the computer. One byte can store 28, i.e., 256 different combinations of bits, and thus can be used to represent 256 different symbols. In a byte, the different combinations of bits fall in the range 00000000 to 11111111. A group of bytes can be further combined to form a word. A word can be a group of 2, 4 or 8 bytes.

1 bit = 0 or 1

1 Byte (B) = 8 bits

1 Kilobyte (KB) = 2^{10} = 1024 bytes

1 Megabyte (MB) = 2^{20} = 1024KB

1 Gigabyte (GB) = 2^{30} = 1024 MB = 1024 * 1024 KB

1 Terabyte (TB) = 2^{40} = 1024 GB = 1024 * 1024 * 1024 KB

Memory is logically organized as a linear array of locations. For a processor, the range of the memory addresses is 0 to the maximum size of memory.

MEMORY HIERARCHY

The memory is characterized on the basis of two key factors—capacity and access time. Capacity is the amount of information (in bits) that a memory can store. Access time is the time interval between the read/write request and the availability of data. The lesser the access time, the faster is the speed of memory. Ideally, we want the memory with fastest speed and largest capacity. However, the cost of fast memory is very high. The computer uses a hierarchy of memory that is organized in a manner to enable the fastest speed and largest capacity of memory. The hierarchy of the different memory types is shown in Figure 1.

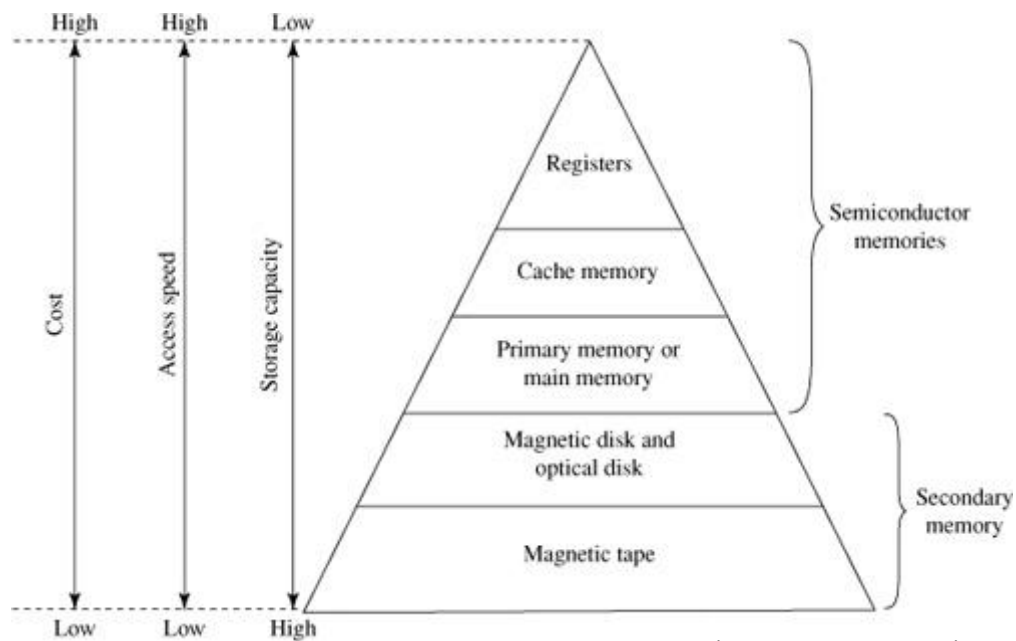


Figure 1 Memory Hierarchy

The internal memory and external memory are the two broad categories of memory used in the computer. The internal memory consists of the CPU registers, cache memory and primary memory. The internal memory is used by the CPU to perform the computing tasks. The external memory is also called the secondary memory. The secondary memory is used to store the large amount of data and the software.

Internal Memory: The key features of internal memory are—

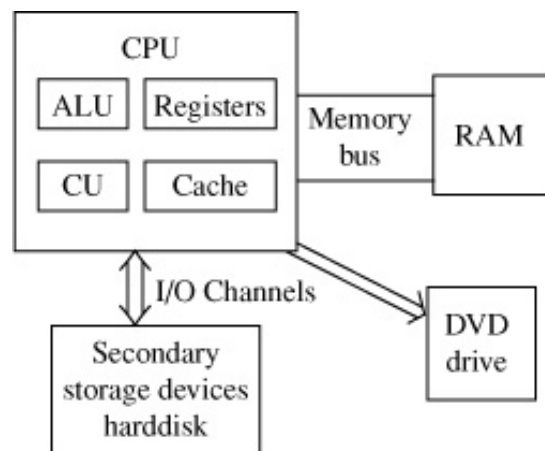
- i. Limited storage capacity,
- ii. Temporary storage,
- iii. Fast access, and
- iv. High cost.

Registers, cache memory, and primary memory constitute the internal memory. The primary memory is further of two kinds—RAM and ROM. Registers are the fastest and the most expensive among all the memory types. The registers are located inside the CPU, and are directly accessible by the CPU. The speed of registers is between 1–2 ns (nanosecond). The sum of the size of registers is about 200B. Cache memory is next in the hierarchy and is placed between the CPU and the main memory. The speed of cache is between 2–10 ns. The cache size varies between 32 KB to 8MB. Any program or data that has to be executed must be brought into RAM from the secondary memory. Primary memory is relatively slower than the cache memory. The speed of RAM is around 60ns. The RAM size varies from 512KB to 6GB.

Secondary Memory: The key features of secondary memory storage devices are—

- i. very high storage capacity,
- ii. permanent storage (non-volatile), unless erased by user,
- iii. relatively slower access,
- iv. stores data and instructions that are not currently being used by CPU but may be required later for processing, and
- v. Cheapest among all memory.

Figure 2 shows the interaction between CPU and memory.



To get the fastest speed of memory with largest capacity and least cost, the fast memory is located close to the processor. The secondary memory, which is not as fast, is used to store information permanently, and is placed farthest from the processor. With respect to CPU, the memory is organized as follows:

- Registers are placed inside the CPU (small capacity, high cost, very high speed)
- Cache memory is placed next in the hierarchy (inside and outside the CPU)
- Primary memory is placed next in the hierarchy

- Secondary memory is the farthest from CPU (large capacity, low cost, low speed)

***NOTE:** The speed of memories is dependent on the kind of technology used for the memory. The registers, cache memory and primary memory are semiconductor memories. They do not have any moving parts and are fast memories. The secondary memory is magnetic or optical memory, has moving parts and has slow speed.*

CPU REGISTERS

Registers are very high-speed storage areas located inside the CPU. After CPU gets the data and instructions from the cache or RAM, the data and instructions are moved to the registers for processing. Registers are manipulated directly by the control unit of CPU during instruction execution. That is why registers are often referred to as the CPU's working memory. Since CPU uses registers for the processing of data, the number of registers in a CPU and the size of each register affect the power and speed of a CPU. The more the number of registers (ten to hundreds) and bigger the size of each register (8 bits to 64 bits), the better it is.

CACHE MEMORY

Cache memory is placed in between the CPU and the RAM. Cache memory is a fast memory, faster than the RAM. When the CPU needs an instruction or data during processing, it first looks in the cache. If the information is present in the cache, it is called a cache hit, and the data or instruction is retrieved from the cache. If the information is not present in cache, then it is called a cache miss and the information is then retrieved from RAM. The content of cache is decided by the cache controller (a circuit on the motherboard). The most recently accessed information or instructions help the controller to guess the RAM locations that may be accessed next. To get good system performance, the number of hits must far outnumber the misses. The two main factors that affect the performance of cache are its size and level (L1, L2 and L3).

PRIMARY MEMORY

Primary memory is the main memory of computer. It is a chip mounted on the motherboard of computer. Primary memory is categorized into two main types-

- Random Access Memory (RAM), and
- Read Only Memory (ROM)

RAM is used for the temporary storage of input data, output data and intermediate results. The input data entered into the computer using the input device, is stored in RAM for processing. After processing, the output data is stored in RAM before being sent to the output device. Any intermediate results generated during the processing of program are also stored in RAM. Unlike

RAM, the data once stored in ROM either cannot be changed or can only be changed using some special operations.

Therefore, ROM is used to store the data that does not require a change. Flash memory is another form of rewritable read-only memory that is compact, portable, and requires little energy.

Random Access Memory

- RAM is used to store data and instructions during the operation of computer.
 - The data and instructions that need to be operated upon by CPU are first brought to RAM from the secondary storage devices like the hard disk.
 - CPU interacts with RAM to get the data and instructions for processing.
- RAM loses information when the computer is powered off. It is a volatile memory. When the power is turned on, again, all files that are required by the CPU are loaded from the hard disk to RAM. Since RAM is a volatile memory, any information that needs to be saved for a longer duration of time must not be stored in RAM.
- RAM provides random access to the stored bytes, words, or larger data units. This means that it requires same amount of time to access information from RAM, irrespective of where it is located in it.
- RAM can be read from and written to with the same speed.
- The size of RAM is limited due to its high cost. The size of RAM is measured in MB or GB.
- The performance of RAM is affected by—
 - Access speed (how quickly information can be retrieved). The speed of RAM is expressed in nanoseconds.
 - Data transfer unit size (how much information can be retrieved in one request).
- RAM affects the speed and power of a computer. More the RAM, the better it is. Nowadays, computers generally have 512 MB to 4 GB of RAM.
- RAM is a microchip implemented using semiconductors.
- There are two categories of RAM, depending on the technology used to construct a RAM—
 - Dynamic RAM (DRAM), and
 - Static RAM (SRAM).
- DRAM is the most common type of memory chip. DRAM is mostly used as main memory since it is small and cheap.
 - It uses transistors and capacitors. The transistors are arranged in a matrix of rows and columns. The capacitor holds the bit of information 0 and 1. The transistor and capacitor are paired to make a memory cell. The transistor acts as a switch that lets the control circuitry on the memory chip read the capacitor or change its state.

- DRAM must be refreshed continually to store information. For this, a memory controller is used. The memory controller recharges all the capacitors holding a 1 before they discharge. To do this, the memory controller reads the memory and then writes it right back.
- DRAM gets its name from the refresh operation that it requires to store the information; otherwise it will lose what it is holding. The refresh operation occurs automatically thousands of times per second. DRAM is slow because the refreshing takes time.
- Access speed of DRAM ranges from 50 to 150 ns.

SRAM chip is usually used in cache memory due to its high speed.

- SRAM uses multiple transistors (four to six), for each memory cell. It does not have a capacitor in each cell.
- A SRAM memory cell has more parts so it takes more space on a chip than DRAM cell.
- It does not need constant refreshing and therefore is faster than DRAM.
- SRAM is more expensive than DRAM, and it takes up more space.
- It stores information as long as it is supplied with power.
- SRAM are easier to use and very fast. The access speed of SRAM ranges from 2–10 nanosecond.
- The memory chips are available on a separate Printed Circuit Board (PCB) that is plugged into a special connector on the motherboard. Memory chips are generally available as part of a card called a memory module. There are generally two types of RAM modules—Single In-line Memory Module (SIMM) and Dual In-line Memory Module (DIMM).
 - SIMM modules have memory chip on one side of the PCB. SIMM modules can store 8 bits to 32 bits of data simultaneously.
 - DIMM modules have memory chips on both sides of the PCB. DIMM format are 64-bit memories. Smaller modules known as Small Outline DIMM (SO DIMM) are designed for portable computers. SO DIMM modules have 32-bit memory.

Read Only Memory

ROM is a non-volatile primary memory. It does not lose its content when the power is switched off. The features of ROM are described as follows—

- ROM, as the name implies, has only read capability and no write capability. After the information is stored in ROM, it is permanent and cannot be corrected.
- ROM comes programmed by the manufacturer. It stores standard processing programs that permanently reside in the computer. ROM stores the data needed

for the start up of the computer. The instructions that are required for initializing the devices attached to a computer are stored in ROM.

- The ROM memory chip stores the Basic Input Output System (BIOS). BIOS provide the processor with the information required to boot the system. It provides the system with the settings and resources that are available on the system. BIOS is a permanent part of the computer. It does not load from disk but instead is stored in a ROM memory chip. The program code in the BIOS differs from ordinary software since it acts as an integral part of the computer. When the computer is turned on, the BIOS does the following things—
- Power On Self Test (POST) is a program that runs automatically when the system is booted. BIOS performs the power-on self-test. It checks that the major hardware components are working properly.
- BIOS setup program, which is a built-in utility in BIOS, lets the user set the many functions that control how the computer works. BIOS displays the system settings and finds the bootable devices. It loads the interrupt handlers and device drivers. It also initializes the registers.
- Bootstrap Loader is a program whose purpose is to start the computer software for operation when the power is turned on. It loads the operating system into RAM and launches it. It generally seeks the operating system on the hard disk. The bootstrap loader resides in the ROM. The BIOS initiates the bootstrap sequence.
- ROMs are of different kinds. They have evolved from the fixed read only memory to the ones that can be programmed and re-programmed. They vary in the number of re-writes and the method used for the re-writing. Programmable ROM (PROM), Erasable Programmable ROM (EPROM) and Electrically Erasable Programmable ROM (EEPROM) are some of the ROMs. All the different kinds of ROM retain their content when the power is turned off.
- PROM can be programmed with a special tool, but after it has been programmed the contents cannot be changed. PROM memories have thousands of fuses (or diodes). High voltage (12 V) is applied to the fuses to be burnt. The burnt fuses correspond to 0 and the others to 1.
- EPROM can be programmed in a similar way as PROM, but it can be erased by exposing it to ultra violet light and re-programmed. EPROM chips have to be removed from the computer for re-writing.
- EEPROM memories can be erased by electric charge and re- programmed. EEPROM chips do not have to be removed from the computer for re-writing.

Flash Memory is a kind of semiconductor-based non-volatile, rewritable computer memory that can be electrically erased and reprogrammed. It is a specific type of EEPROM.

It combines the features of RAM and ROM. It is a random access memory and its content can be stored in it at any time. However, like ROM, the data is not lost when the machine is turned off or the electric power is cut. Flash memory stores bits of data in memory cells.

Flash memories are high-speed memories, durable, and have low- energy consumption. Since flash memory has no moving part, it is very shock-resistant. Due to these features, flash memory is used in devices such as digital camera, mobile phone, printer, laptop computer, and record and play back sound devices, such as MP3 players.

SECONDARY MEMORY

In the previous section, we saw that RAM is expensive and has a limited storage capacity. Since it is a volatile memory, it cannot retain information after the computer is powered off. Thus, in addition to primary memory, an auxiliary or secondary memory is required by a computer. The secondary memory is also called the storage device of computer. In this unit, the terms secondary memory and storage device are used interchangeably. In comparison to the primary memory, the secondary memory stores much larger amounts of data and information (for example, an entire software program) for extended periods of time. The data and instructions stored in secondary memory must be fetched into RAM before processing is done by CPU.

Magnetic tape drives, magnetic disk drives, optical disk drives and magneto- optical disk drives are the different types of storage devices.

ACCESS TYPES OF STORAGE DEVICES

The information stored in storage devices can be accessed in two ways—

- a. Sequential access
- b. Direct access

Sequential Access Devices

Sequential access means that computer must run through the data in sequence, starting from the beginning, in order to locate a particular piece of data. Magnetic tape is an example of sequential access device. Let us suppose that magnetic tape consists of 80 records. To access the 25th record, the computer starts from first record, then reaches second, third etc. until it reaches the 25th record. Sequential access devices are generally slow devices.

Direct Access Devices

Direct access devices are the ones in which any piece of data can be retrieved in a non-sequential manner by locating it using the data's address. It accesses the data directly, from a desired location. Magnetic disks and optical disks are examples of direct access devices. There is no

predefined order in which one can read and write data from a direct access device. In a magnetic disk consisting of 80 records, to access the 25th record, the computer can directly access the 25th record, without going past the first 24 records. Based on access, magnetic tapes are sequential access devices, and, magnetic disks, optical disk and magneto-optical disks are direct access devices.

MAGNETIC TAPE

Magnetic tape is a plastic tape with magnetic coating. It's a storage medium on a large open reel or in a smaller cartridge or cassette (like a music cassette). Magnetic tapes are cheaper storage media. They are durable, can be written, erased, and re-written. Magnetic tapes are sequential access devices, which mean that the tape needs to rewind or move forward to the location where the requested data is positioned in the magnetic tape. Due to their sequential nature, magnetic tapes are not suitable for data files that need to be revised or updated often. They are generally used to store back-up data that is not frequently used or to transfer data from one system to other.

The working of magnetic tape is explained as follows—

- Magnetic tape is divided horizontally into tracks (7 or 9) and vertically into frames. A frame stores one byte of data, and a track in a frame stores one bit. Data is stored in successive frames as a string with one data (byte) per frame.
- Data is recorded on tape in the form of blocks, where a block consists of a group of data also called as records. Each block is read continually. There is an Inter-Record Gap (IRG) between two blocks that provides time for the tape to be stopped and started between records.
- Magnetic tape is mounted on a magnetic tape drive for access. The basic magnetic tape drive mechanism consists of the supply reel, take-up reel, and the read/write head assembly. The magnetic tape moves on tape drive from the supply reel to take-up reel, with its magnetic coated side passing over the read/write head.
- Tapes are categorized based on their width - $\frac{1}{4}$ inch, $\frac{1}{2}$ inch, etc.
- The storage capacity of the tape varies greatly. A 10-inch diameter reel of tape which is 2400 feet long can store up to 180 million characters.

The features of magnetic tape are—

- Inexpensive storage device
- Can store a large amount of data
- Easy to carry or transport
- Not suitable for random access data
- Slow access device
- Needs dust prevention, as dust can harm the tape
- Suitable for back-up storage or archiving

Magnetic Disk

Magnetic disk is a direct access secondary storage device. It is a thin plastic or metallic circular plate coated with magnetic oxide and encased in a protective cover. Data is stored on magnetic disks as magnetized spots. The presence of a magnetic spot represents the bit 1 and its absence represents the bit 0.

The working of magnetic disk is explained as follows—

The surface of disk is divided into concentric circles known as tracks. The outermost track is numbered 0 and the innermost track is the last track. Tracks are further divided into sectors. A sector is a pie slice that cuts across all tracks. The data on disk is stored in sector. Sector is the smallest unit that can be read or written on a disk. A disk has eight or more sectors per track (Figure 4).

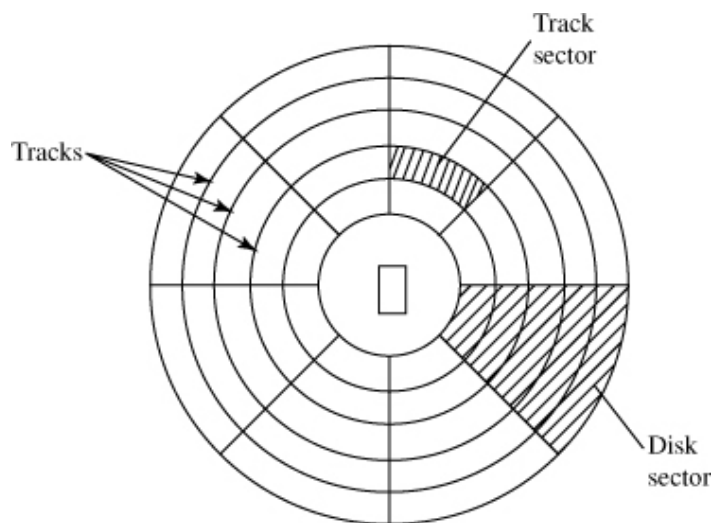


Figure 4 Tracks and sectors of a disk

- Magnetic disk is inserted into a magnetic disk drive for access. The drive consists of a read/write head that is attached to a disk arm, which moves the head. The disk arm can move inward and outward on the disk.
- During reading or writing to disk, the motor of disk drive moves the disk at high speed (60 – 150 times/sec.)
- Accessing data on the disk requires the following—
 - The read/write head is positioned to the desired track where the data is to be read from or written to. The time taken to move the read/write head to the desired track is called the seek time.
 - Once the read/write head is at the right track, then the head waits for right sector to come under it (disk is moving at high speed). The time taken for desired sector of the track to come under read/write head is called the latency time.

- Once the read/write head is positioned at the right track and sector, the data has to be written to disk or read from disk. The rate at which data is written to disk or read from disk is called data transfer rate.
- The sum of seek time, latency time and time for data transfer is the access time of the disk.
- The storage capacity of disk drive is measured in gigabytes (GB).
- Large disk storage is created by stacking together multiple disks. A set of same tracks on all disks forms a cylinder. Each disk has its own read/write head which work in coordination.
- A disk can also have tracks and sectors on both sides. Such a disk is called double-sided disk.

The features of magnetic disk are—

- Cheap storage device
- Can store a large amount of data
- Easy to carry or transport
- Suitable for frequently read/write data
- Fast access device
- More reliable storage device
- To be prevented from dust, as the read/write head flies over the disk. Any dust particle in between can corrupt the disk.

Finding data on a magnetic disk is as follows—

- In order to use a disk, it has to be formatted. Formatting includes assigning addresses to various locations on disk, assigning location of root directory and checking for defects on the surface of disk.
- During formatting, the tracks and sectors of a disk are labeled, which provides an address to each location of the disk.
- There are different methods to format a disk. File Allocation Table (FAT) is the commonly used logical format for disk formatting performed by Windows.
- Four areas are created when a disk is formatted using FAT—
 - i. Boot Sector: It contains the program that runs when the computer is started. The program checks if the disk has files required to run the operating system. It then transfers control to an operating system program which continues the startup process. Boot sector also contains information about the disk, like number of bytes per sector and number of sectors per track. This information is required by the operating system to access the data on the disk.

- ii. **File Allocation Table** It records the location of each file and status of each sector. While reading or writing to disk, operating system checks the FAT to find free area or locate where data is stored on disk, respectively.
- iii. **Root Directory** This is the main folder of disk. It contains other folders in it, creating a hierarchical system of folders. The root directory contains information about all folders on the disk.
- iv. **Data Area** The remaining area of the disk (after boot sector, FAT, root directory) is the data area. It stores the program files and data files that are stored on the disk.

Floppy disk, hard disk and zip disk are the different types of magnetic disks.

OPTICAL DISK

Optical disk (Figure 5) is a flat and circular disk which is coated with reflective plastic material that can be altered by laser light. Optical disk does not use magnetism. The bits 1 and 0 are stored as spots that are relatively bright and light, respectively.



Figure 5 Optical disk

An optical disk consists of a single spiral track that starts from the edge to the centre of disk. Due to its spiral shape, it can access large amount of data sequentially, for example music and video. The random access on optical disk is slower than that of magnetic disk, due to its spiral shape.

The tracks on optical disk are further divided into sectors which are of same length. Thus, the sectors near the centre of disk wrap around the disk longer than the sectors on the edges of disk. Reading the disk thus requires spinning the disk faster when reading near the centre and slower when reading near the edge of disk. Optical disks are generally slower than hard disks. Figure 6 shows the tracks and sectors in a magnetic disk and optical disk.

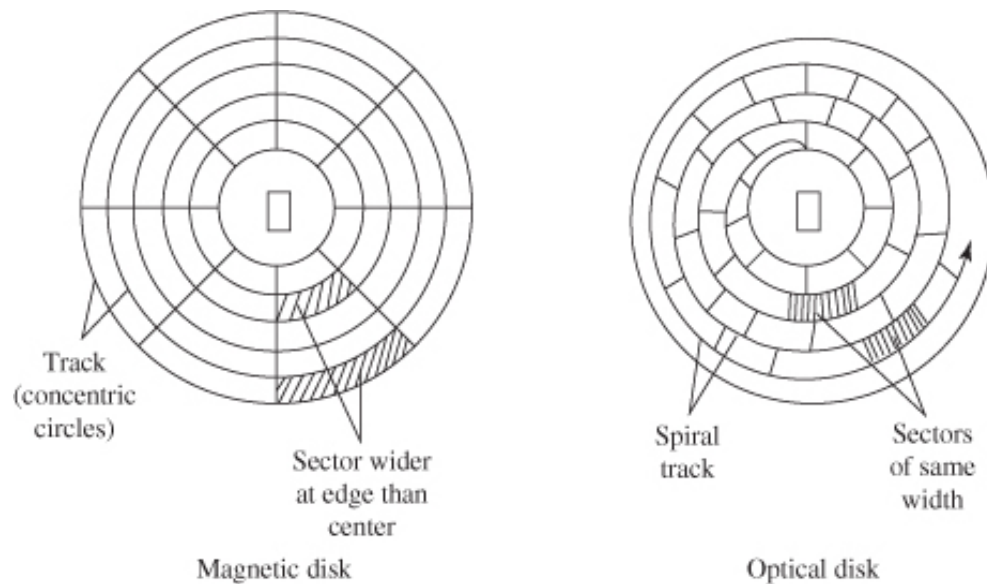


Figure 6 Sectors and track in magnetic disk and optical disk

Optical disks can store large amount of data, up to 6 GB, in a small space. Commonly used optical disks store 600–700 MB of data.

The access time for an optical disk ranges from 100 to 200 ms. There are two most common categories of optical disks—read-only optical disks and recordable optical disks.

CD-ROM

Originally, Compact Disk (CD) was a popular medium for storing music. Now, it is used in computers to store data and is called Compact Disk- Read Only Memory (CD-ROM).

As the name suggests, CD-ROM is an optical disk that can only be read and not written on. CD-ROM is written on by the manufacturer of the CD-ROM using the laser light.

A CD-ROM drive reads data from the compact disk. Data is stored as pits (depressions) and lands (flat area) on CD-ROM disk. When the laser light is focused on the disk, the pits scatter the light (interpreted as 0) and the lands reflect the light to a sensor (interpreted as 1).

As CD-ROM is read only, no changes can be made into the data contained in it.

Since there is no head touching the disk, but a laser light, CD-ROM does not get worn out easily.

The storage density of CD-ROM is very high and cost is low as compared to floppy disk and hard disk.

Access time of CD-ROM is less. CD-ROM drives can read data at 150Kbps. They come in multiples of this speed like—8x, 16x, 52x, 75x, etc.

It is a commonly used medium for distributing software and large data.

DVD-ROM

Digital Video Disk-Read Only Memory (DVD-ROM) is an optical storage device used to store digital video or computer data.

- DVDs look like CDs, in shape and physical size.
- It improves on CD technology.
- It is a high-density medium with increased track and bit density.
- DVD-ROM uses both sides of the disk and special data compression technologies. The tracks for storing data are extremely small.
- A full-length movie can be stored on a single disk.
- Each side of DVD-ROM can store 4.7 GB of data, so a single DVD can store 9.4 GB of data.
- New DVD-ROMs use layers of data track, to double its capacity. Such dual layer disks can store 17 GB of data.

Recordable Optical Disk

In addition to the read only CDs and DVDs, recordable optical disks are also available. Users can record music, video, audio and data on it. The recordable optical disks are—

- Compact Disk-Recordable (CD-R) is a Write Once-Read Many (WORM) disk. A CD-R disk allows the user to write data permanently on to the disk. Once the data is written, it cannot be erased. CD-R disk uses a laser that burns pits into the disk surface. It looks like a CD disk externally. To write to a CD-R disk, a device named CD-Writer or CD- burner is required. A CD-R disk can store 700 MB of data that can run for 80 minutes. CD-R is used to create music CDs in home computers, back up data from other storage devices, archives of large data, etc.
- Compact Disk-Rewritable (CD-RW) allows data to be written, erased and re-written on. The capacity of CD-RW is same as a CD. They generally do not play on all CD-ROM drives.
- Digital Video Disk-Recordable (DVD-R) allows recording of data on a DVD. A DVD writer device is required to write the data to DVD. The data once written on a DVD cannot be erased or changed.

Blu-ray Disc

Blu-ray Disc (BD) is an optical disc storage medium designed to supersede the DVD format. The plastic disc is 120 mm in diameter and 1.2 mm thick, the same size as DVDs and CDs. Conventional (pre-BD-XL) Blu-ray Discs contain 25 GB per layer, with dual layer discs (50 GB) being the industry standard for feature-length video discs. Triple layer discs (100 GB) and quadruple layers (128 GB) are available for BD-XL re-writer drives.

The name Blu-ray Disc refers to the blue laser used to read the disc, which allows information to be stored at a greater density than is possible with the longer-wavelength red laser used for DVDs. The major application of Blu-ray Discs is as a medium for video material such as feature films. Besides the hardware specifications, Blu-ray Disc is associated with a set of multimedia formats. Generally, these formats allow for the video and audio to be stored with greater definition than on DVD.

MAGNETO-OPTICAL DISK

- Magneto-optical disks use laser beam to read data and magnetic field to write data to disk.
- These are optical disks where data can be written, erased and re-written.
- They are expensive and outdated. They were used during the mid 1990s. They have now been replaced by CD-RW and DVD-R.

USING THE COMPUTER MEMORY

The computer starts using the memory from the moment the computer is switched on, till the time it is switched off. The list of steps that the computer performs from the time it is switched on are—

- Turn the computer on.
- The computer loads data from ROM. It makes sure that all the major components of the computer are functioning properly.
- The computer loads the BIOS from ROM. The BIOS provides the most basic information about storage devices, boot sequence, security, plug and play capability and other items.
- The computer loads the OS from the hard drive into the system's RAM. CPU has immediate access to the OS as the critical parts of the OS are maintained in RAM as long as the computer is on. This enhances the performance and functionality of the overall system.
- Now the system is ready for use.

When you load or open an application it is loaded in the RAM. Since the CPU looks for information in the RAM, any data and instructions that are required for processing (read, write or

update) is brought into RAM. To conserve RAM usage, many applications load only the essential parts of the program initially and then load other pieces as needed. Any files that are opened for use in that application are also loaded into RAM.

The CPU requests the data it needs from RAM, processes it and writes new data back to RAM in a continuous cycle. The shuffling of data between the CPU and RAM happens millions of times every second.

When you save a file and close the application, the file is written to the secondary memory as specified by you. The application and any accompanying files usually get deleted from RAM to make space for new data.

If the files are not saved to a storage device before being closed, they are lost.

Sometimes, when you write a program and the power goes off, your program is lost if you have not saved it. This is because your program was in the RAM and was not saved on the secondary memory; the content of the RAM gets erased when the power is switched off.

COPPERBELT UNIVERSITY
COMPUTER SCIENCE DEPARTMENT

INTRODUCTION TO COMPUTING

UNIT 4: INPUT AND OUTPUT DEVICES (Notes)

INTRODUCTION

A computer interacts with the external environment via the input-output (I/O) devices attached to it. Input device is used for providing data and instructions to the computer. After processing the input data, computer provides output to the user via the output device. The I/O devices that are attached, externally, to the computer machine are also called peripheral devices. Different kinds of input and output devices are used for different kinds of input and output requirements. In this chapter, we shall discuss different kinds of input devices and output devices.

INPUT-OUTPUT UNIT

An I/O unit is a component of computer. The I/O unit is composed of two parts—input unit and output unit. The input unit is responsible for providing input to the computer and the output unit is for receiving output from the computer.

Input Unit

The input unit gets the data and programs from various input devices and makes them available for processing to other units of the computer.

The input data is provided through input devices, such as—keyboard, mouse, trackball and joystick. Input data can also be provided by scanning images, voice recording, video recording, etc.

Irrespective of the kind of input data provided to a computer, all input devices must translate the input data into a form that is understandable by the computer, i.e., in machine readable form. The transformation of the input data to machine readable form is done by the input interface of input device.

In brief, the input unit accepts input data from the user via input device, transforms the input data in computer acceptable form using input interface for the input device and provides the transformed input data for processing.

Output Unit

The output unit gets the processed data from the computer and sends it to output devices to make them available to the user of computer.

The output data is provided through output devices like display screen, printer, plotter and speaker.

The processed data sent to the output device is in a machine understandable form. This processed data is converted to human readable form by the output interface of output device.

In brief, the output unit accepts output data from computer via output device, transforms the output information to human readable form using the output interface of output device and provides the transformed output to user.

In addition to input devices and output devices, some devices function as both input and output devices. The I/O devices provide the input to computer as well as get output from computer. The I/O devices are used by both the input unit and the output unit. Hard disk drive, floppy disk drive, optical disk drives are examples of I/O devices.

INPUT DEVICES

Input devices allow users and other applications to input data into the computer, for processing. The data input to a computer can be in the form of text, audio, video, etc. The data is entered manually by the user or with minimal user intervention. Input devices are classified as follows—

- Human data entry devices
 - Keyboard
 - Pointing devices—mouse, trackball, joystick, digitizing tablet
 - Pick devices—light pen, touch screen
- Source data entry devices
 - Audio input—speech recognition
 - Video input—digital camera
 - Scanner—hand-held scanner, flat-bed scanner
 - Optical Scanner—OCR, OMR, MICR, barcode reader

The input is provided to the computer using an input device, and must be translated to a form that the computer can understand. The translation is done by the input interface of the input device.

In addition to the above devices, the input to a computer can also be provided from a storage device on the computer, another computer, or another piece of equipment, such as a musical instrument, thermometer or sensors.

HUMAN DATA ENTRY DEVICES

Input devices that require data to be entered manually to the computer are identified as human data entry devices. The data may be entered by typing or keying in, or by pointing a device to a particular location.

Keyboard

Features Keyboard is a common input device. It is provided along with the computer, and is easy to use. It is used for entering the text data. For inputting the data, the user types the data using the keyboard. When the data is being typed, the display monitor displays the typed data. Cursor is a vertical line, an underscore, blinking line, etc. Cursor moves with each typed character. The position of cursor indicates the location on monitor where the typed-in character will be displayed.

Description The design of a keyboard is similar to a standard typewriter. The modern keyboards are QWERTY keyboard (Q, W, E, R, T, Y are the sequence of keys in top row of letters). Standard keyboard contains 101 keys which are arranged in the same order as a typewriter. The keyboard has five sections

1. Typing keys (1, 2, 3..., A, B, C...),
2. Numeric keypad (numeric keys on right side),
3. Function keys (F1, F2... on top side),
4. Control keys (cursor keys, ctrl, alt...), and
5. Special-purpose keys (Enter, shift, spacebar...). Some keyboards have 110 keys, where the extra keys are designed to work with the Windows operating system.

Working: When a key is pressed, keyboard interacts with a keyboard controller and keyboard buffer. The keyboard controller stores the code of pressed key in keyboard buffer and informs the computer software that an action has happened on the keyboard. The computer software checks and reads the keyboard buffer and passes the code of pressed character to the system software. Due to a time gap between pressing of a key on keyboard and reading by the system software, keyboard buffer is designed to store many keystrokes together.

Pointing Devices

Pointing devices are used for providing the input to computer by moving the device to point to a location on computer monitor. The input data is not typed; instead, the data is entered by moving the pointing device. The cursor on the computer monitor moves with the moving pointing device. Operations like move, click and drag can be performed using the pointing devices.

Mouse, trackball, joystick and digitizing tablet are some of the common pointing devices.

Mouse

Features - It is the most common pointing input device. The data is entered by pointing the mouse to a location on the computer screen. The mouse may also be used to position the cursor on screen, move an object by dragging, or select an object by clicking. The key benefit of using a mouse is that the cursor moves with the mouse. So, the cursor can be positioned at any location on the screen by simply moving the mouse. Moreover, it provides an easy way to select and choose commands from menus, dialog boxes, icons, etc. Mouse is used extensively, while working with graphics elements such as line, curve, shapes, etc.

Description - Mouse is a small hand-held device having two or three buttons on its upper side. In addition to the buttons, mouse also has a small wheel between the buttons. The wheel of the mouse is used for the up and down movement, for example, scrolling a long document. A mouse is classified as physical mouse or optical mouse.

Physical Mouse has a rubber ball on the bottom side that protrudes when the mouse is moved. It requires a smooth, dust free surface, such as a mouse pad, on which it is rolled.

Optical Mouse uses a Light Emitting Diode (LED) and a sensor to detect the movement of mouse. Optical mouse requires an opaque flat surface underneath it. Optical mouse was introduced by Microsoft in 1999. Optical mouse is better than physical mouse as there is no moving part that can cause wear and tear, and dirt cannot get inside it.

Working - In a physical mouse, rollers and sensors are used to sense the direction and rate of movement of mouse. When the ball of mouse moves, the rollers sense the horizontal and vertical movement and sensors sense the speed of movement. This information is passed to computer via the mouse chord. When an optical mouse is moved, a beam of light is reflected from its underside. These pulses of light determine the direction and rate of movement. This information is sent to computer via the mouse chord.

Using the mouse - The mouse can be used in five different ways, as follows—

- Pointing points to a location or object on the computer screen. Moving the mouse by hand moves the cursor on computer screen. The cursor moves in the direction in which the mouse moves.
- Left Click or Click means pressing the left button of mouse and releasing it. Clicking is used to select a button, command or icon on the screen.
- Right Click involves pressing the right button on mouse and releasing it. Right click displays a menu that contains options like cut, copy, paste, font, paragraph, etc. for the item on which the mouse is pointing.

- Double Click means pressing the left button of mouse twice successively, without moving the mouse, and then releasing the mouse. It is used to start a program or open a folder.
- Drag and Drop drags an object and drops it at another location. Drag means pointing mouse to an object on screen, pressing the left button of mouse, keeping it pressed and moving the mouse to point to a new location. The object gets dragged to the new location along with the mouse. When the left button of mouse is released, the object gets dropped at the new location. Drag and drop is used for moving folders, files and icons to new locations on the screen.

TrackBall

Features Trackball is a device that is a variant of the mouse but has the functionality of mouse. It is easy to use and takes less space than a mouse. Trackball is generally built in laptops since there is no space for the mouse to move on the lap. Trackballs come in various sizes—small and big.

Description Trackball looks like an upside-down mouse. Instead of moving the whole device to move the cursor on computer screen, trackball requires the ball to be rotated manually with a finger. The trackball device remains stationary. The cursor on the computer screen moves in the direction in which the ball is moved. The buttons on trackball are used in the same way as mouse buttons.

Working Trackball works in the same way as a physical mouse.

Joystick

Features Joystick is a device which is commonly used for playing video games. Joystick is mainly used to control the speed of the cursor and is thus popular in games involving speed like racing and flying games. The direction of push of the stick and the amount of deflection determines the change in position and the change in speed, respectively.

Description It is a stick with its base attached to a flexible rubber sheath inside a plastic cover. The plastic cover contains the circuit that detects the movement of stick and sends the information to computer. The position of the stick movement is given by the x and y coordinates of the stick.

Digitizing Tablet

Features It is an input device used primarily to input drawings, sketches, etc. Digitizing tablet is used for Computer Aided Design (CAD) for the design of buildings, automotive designs, and designing of maps, etc. Description Digitizing tablet consists of two parts—electronic tablet and pen. The electronic tablet is a flat bed tablet. The pen looks like a ball pen but has an electronic head. The pen is moved on the tablet. Each position on the tablet corresponds to a fixed position on the screen.

Drawings can be made on the tablet using a pen, and is provided as input to computer, where, a location on the tablet corresponds to a specific location on the screen.

The tablet contains circuit that can detect the movement of pen on the tablet, convert the movements into digital signals and send the digital signal to the computer.

Pick Devices

Pick devices are used for providing input to the computer by pointing to a location on the computer monitor. The input data is not typed; the data is entered by pointing the pick device directly on the computer screen. Light pen and touch screen are some common pick devices.

Light Pen

It is a light sensitive pen-like input device and is used to select objects directly on the computer screen. It is used for making drawing, graphics and for menu selection. Figures and drawings can be made by moving the pen on computer screen.

The pen contains a photocell in a small tube. When the pen is moved on the screen, light from the screen at the location of pen causes the photocell to respond. The electric response is transmitted to the computer that can identify the position on screen at which the light pen is pointing.

Touch Screen

It is an input device that accepts input when the user places a fingertip on the computer screen. The computer selects the option from the menu of screen to which the finger points. Touch screen are generally used in applications like Automated Teller Machine (ATM), public information computers like hospitals, airline reservation, railway reservation, supermarkets, etc.

Touch screen consists of a clear glass panel that is placed over the view area of computer screen. In addition to the glass panel with sensors, it has a device driver, and a controller that translates the information captured by the glass panel sensors to a form that the computer can understand.

Touch screens have an infrared beam that criss-cross the surface of screen. When a fingertip is touched on the screen, the beam is broken, and the location is recorded. Some touch screens have ultrasonic acoustic waves that cross the surface of screen. When a fingertip is touched on the screen, the wave is interrupted, and the location is recorded. The recorded location is sent to the computer via the controller of touch screen, in a form that the computer can understand.

SOURCE DATA ENTRY DEVICES

Source data entry devices are used for audio input, video input and to enter the source document directly to the computer. Source data entry devices do not require data to be typed-in, keyed-in or pointed to a particular location.

Audio Input Device

Audio input can be provided to the computer using human voice or speech. Audio input to the computer can be used for different purposes. It can be used for making telephone calls, for audio and video conferencing over Internet, to record voice, to create audio files and embed these files to be sent over e-mail, or, to translate spoken words into text, etc.

Audio input devices like a microphone are used to input a person's voice into the computer. A sound card translates analog audio signals from microphone into digital codes that the computer can store and process. Sound card also translates back the digital sound into analog signals that can be sent to the speakers. Translating spoken words into text is also known as speech recognition or voice recognition. The audio input along with the software for voice recognition forms the speech recognition system or voice recognition system.

The computer can be operated using voice commands. The user can dictate the commands to the computer, instead of typing them. The computer has to be trained to recognize the voice of user using the speech patterns and pronunciation of words. The system thus adapts to the voice of user. Speech recognition systems are costly and difficult to develop. They are generally used by people who have difficulty in typing, people with disabilities or by corporate world for dictation. Audio input can be recorded on an mp3 recorder and provided as an input to computer. Open source software like Audacity is used for recording and editing of audio files.

Video Input Device

Video input is provided to the computer using video camera and digital camera. Video camera can capture full motion video images. The images are digitized and can be compressed and stored in the computer disk. Webcam is a common video camera device. It is placed on the computer above the screen to capture the images of the user who is working on the computer. A video capture card allows the user to connect video devices like camcorders to the computer.

Digital camera works like video camera but can capture still images. The digital camera digitizes images, compresses them and stores them on a memory card like flash memory. The information from the digital camera can be brought into the computer and stored. The video files can be edited using software like VLC media player. Computer vision is an area of computer science that deals with images. Computer vision has applications in areas like robotics and industrial processing.

OPTICAL INPUT DEVICES

Optical input devices allow computers to use light as a source of input. Scanner is an example of optical input device. Other common optical input devices are magnetic ink character reader used for Magnetic Ink Character Recognition (MICR), optical mark reader used for Optical Mark Recognition (OMR), optical character reader for Optical Character Recognition (OCR) and Barcode Reader.

Scanner

Scanner is an input device that accepts paper document as an input. Scanner is used to input data directly into the computer from the source document without copying and typing the data. The input data to be scanned can be a picture, a text or a mark on a paper. It is an optical input device and uses light as an input source to convert an image into an electronic form that can be stored on the computer. Scanner accepts the source paper document, scans the document and translates it into a bitmap image to be stored on the computer. The denser the bitmap, the higher is the resolution of the image. The quality of scan increases with the increase in resolution. Scanners come with utility software that allows the stored scanned documents to be edited, manipulated and printed. Hand-held scanner and flat-bed scanner are the two common types of scanners.

Hand-held Scanners are portable and are placed over the document to be scanned. They consist of light emitting diodes. The scanned documents are converted and stored as an image in the computer memory. Hand-held scanners have to be moved at a constant speed over the document to be scanned, to get good quality scans. They are preferably used for low volume of documents, small pictures or photos. They are difficult to use if there is a need to scan a full page document. Some of the documents that are primarily scanned using hand-held scanners are price tags, label and ISBN number on books.

Flat-bed Scanners provide high quality scan in a single pass. It is a box shaped machine similar to a photocopy machine and has a glass top and a lid that covers the glass. The document to be scanned is placed on the glass top, which activates the light beam beneath the glass top and starts the scan from left to right. They are largely used to scan full page documents.

Optical Character Recognition (OCR)

OCR is a technique for the scanning of a printed page, translating it, and then using the OCR software to recognize the image as ASCII text that is editable. OCR uses optical character reader for recognition. The optical character reader stores the scanned image as bitmap image which is a grid of dots. Thus, you cannot edit the text that has been scanned. To edit the scanned text, you need OCR software. The OCR software translates the array of dots into text that the computer can interpret as words and letters. To recognize the words and letters of text, the OCR software compares the pattern on the scanned image with the patterns stored inside the computer.

The text files generated via OCR can be stored in different formats.

Magnetic Ink Character Recognition (MICR)

MICR is used in banks to process large volumes of cheques. It is used for recognizing the magnetic encoding numbers printed at the bottom of a cheque. The numbers on the cheque are human readable, and are printed using an ink which contains iron particles. These numbers are magnetized. MICR uses magnetic ink character reader for character recognition. When a cheque is passed through Magnetic Ink Character Reader, the magnetic field causes the read head to recognize the characters or numbers of cheque. The readers are generally used in banks to process the cheques. The numbers in the bottom of the cheque include the bank number, branch number and cheque number. The reading speed of MICR is faster than OCR.

Optical Mark Recognition (OMR)

OMR is used to detect marks on a paper. The marks are recognized by their darkness. OMR uses an optical mark reader to read the marks. The OMR reader scans the forms, detects the mark that is positioned correctly on the paper and is darker than the surrounding paper, and passes this information to the computer for processing by application software. For this, it uses a beam of light that is reflected on the paper with marks, to capture presence and absence of marks. The optical mark reader detects the presence of mark by measuring the reflected light. The pattern of marks is interpreted and stored in the computer.

OMR is widely used to read answers of objective type tests, where the student marks an answer by darkening a particular circle using a pencil. OMR is also used to read forms, questionnaires, order forms, etc.

Barcode Reader

Barcodes are adjacent vertical lines of different width that are machine readable. Goods available at supermarkets use barcode for identification. Barcodes are read using reflective light by

barcode readers. This information is input to the computer which interprets the code using the spacing and thickness of bars. Hand-held barcode readers are generally used in departmental stores to read the labels, and in libraries to read labels on books.

Barcode Readers are fast and accurate. They enable faster service to the customer and are also used to determine the items being sold, number of each item sold or to retrieve the price of item.

OUTPUT DEVICES

Output devices provide output to the user, which is generated after processing the input data. The processed data, presented to the user via the output devices could be text, graphics, audio or video. The output could be on a paper or on a film in a tangible form, or, in an intangible form as audio, video and electronic form. Output devices are classified as follows—

- Hard Copy Devices
 - Printer
 - Plotter
 - Computer Output on Microfilm (microfiche)
- Soft Copy Devices
 - Monitor
 - Visual Display Terminal
 - Video Output
 - Audio Response

The output device receives information from computer in a machine readable form. The received output is translated to a human understandable form. The translation is done using the output interface of output device.

Hard Copy Devices

The output obtained in a tangible form on a paper or any surface is called hard copy output. The hard copy can be stored permanently and is portable. The hard copy output can be read or used without a computer. The devices that generate hard copy output are called hard copy devices. Printer, plotter and microfiche are common hard copy output devices.

Printer

A printer prints the output information from the computer onto a paper. Printers are generally used to print textual information, but nowadays printers also print graphical information. The print quality (sharpness and clarity of print) of the printer is determined by the resolution of the printer. Resolution is measured in dots per inch (dpi). Printers with a high resolution (more dpi) provide better quality output. Different kinds of printers are available for different types of applications. Printers are classified into two categories— impact printer and non-impact printer.

IMPACT PRINTERS use the typewriter approach of physically striking a typeface against the paper and inked ribbon. Impact printers can print a character or an entire line at a time. Impact printers are low-cost printers useful for bulk printing. Dot matrix printers, daisy wheel printers and drum printers are examples of impact printers.

Dot Matrix Printers print one character at a time. The speed of dot matrix printer lies between 200 and 600 characters per second (cps) and their resolution ranges from 72 to 360 dpi. Dot matrix printers normally come in two sizes—80 column printer and 132 column printer. Dot matrix printers can print alphanumeric characters, special characters, charts and graphs. They can print only in black and white. Some dot matrix printers can print in both directions - left to right and right to left. Dot matrix printers are commonly used for printing in applications like payroll and accounting.

Daisy Wheel Printers print one character at a time. They produce letter quality document which is better than a document printed by a dot matrix printer. The speed of daisy wheel printers is about 100 cps. The print head of the printer is like a daisy flower, hence the name. These printers are slow, can only print text (not graphics), and are costly in comparison to dot matrix printers. Daisy wheel printers are used where high quality printing is needed and no graphics is needed.

Drum Printers are line printers. They are expensive and faster than character printers but produce a low quality output. They can print 200–2500 lines per minute. Drum printers are generally used for voluminous print outputs.

NON-IMPACT PRINTERS do not hit or impact a ribbon to print. They use electro-static chemicals and ink-jet technologies. Non-impact printers are faster and quieter than impact printers. They produce high quality output and can be used for printing text and graphics both in black and white, and color. Ink-jet printers and laser printers are non-impact printers.

Ink-jet Printers spray ink drops directly on the paper like a jet. Their resolution is more than 500 dpi. They produce high quality graphics and text. Ink-jet printers are commonly found in homes and offices.

Laser Printers provide highest quality of text and graphics printing. Laser printers process and store the entire page before printing and are also known as page printers. The laser printer can print 5-24 pages of text per minute and their resolution ranges from 400 to 1200 dpi. They are faster and expensive than impact printers. Laser printers are used in applications requiring high quality voluminous printing.

Plotter

A plotter is used for vector graphics output to draw graphs, maps, blueprints of ships, buildings, etc. Plotters use pens of different colors (cyan, magenta, yellow and black) for drawing. Plotters draw continuous and accurate lines, in contrast to printers where a line is drawn as closely spaced dots. Plotter is a slow output device and is expensive. Plotters are of two kinds—drum plotter and flatbed plotter. In a drum plotter, pens mounted on the carriage are stationary and move only horizontally; for vertical movement, the drum on which the paper is fixed moves clockwise and anti-clockwise. In a flatbed plotter, the paper is fixed on a flat bed. The paper is stationary and the pens mounted on the carriage move horizontally and vertically to draw lines. Plotters are mainly used for drawings in AUTOCAD (computer assisted drafting), Computer Aided Design (CAD) and Computer Aided Manufacturing (CAM) applications.

Computer Output on Microfilm

A microfilm is in a fiche or roll format, and is used to record computer output directly from the computer tape or cartridge. Computer Output on Microfilm (COM) is a high speed and low cost process. It can produce data in microfilm form at a much faster speed from that of a paper printer. The standard roll film is 16 mm wide with a film image that is 1/24 of the original document. The copy of the image on microfilm retains its original clarity. Microfilm can be indexed to facilitate retrieving information from it.

For reading images stored on microfilm, a microfilm reader is used. A screen is used for viewing the enlarged images. COM is suited for storing large amounts of data for manuals and archive records for long periods of time that have to be referenced occasionally. COM is used for storing output in banking and insurance applications, medical X rays, etc.

SOFT COPY DEVICES

The output obtained in an intangible form on a visual display, audio unit or video unit is called soft copy output. The soft copy allows corrections to be made, can be stored, and, can be sent via e-mail to other users. The soft copy output requires a computer to be read or used. The devices that generate soft copy output are called soft copy devices. Visual output devices like computer monitor, visual display terminal, video system and audio response system are common soft copy output devices.

Monitor

Monitor is a common output device. The monitor is provided along with the computer, to view the displayed output. A monitor is of two kinds - monochrome display monitor and color display monitor. A monochrome display monitor uses only one color to display text and color display monitor can display 256 colors at one time. The number of colors displayed by a color monitor

varies with the kind of color adapter attached to it—CGA, EGA, VGA, XGA and SVGA. The CGA monitor supports four colors and SVGA supports around 16,000,000 colors. Monitors are available in various sizes like 14, 15, 17, 19 and 21 inches.

An image on the monitor is created by a configuration of dots, also known as pixels. The clarity of image on the computer screen depends on three factors —

1. Resolution of Screen—the number of pixels in horizontal and vertical direction. More the number of pixels, the sharper is the image. The common resolution of computer screen is 800x600 and 1024x768,
2. Dot Pitch—the diagonal distance between two colored pixels on a display screen, and
3. Refresh Rate—the number of times per second the pixels are recharged so that their glow remains bright.

Monitors may be Cathode Ray Tube (CRT) monitors that look like a television or Liquid Crystal Display (LCD) monitors that have a high resolution, flat screen, flat panel display. Nowadays, LCD monitors are generally used.

Visual Display Terminal

A monitor and keyboard together are known as Visual Display Terminal (VDT). A keyboard is used to input data and monitor is used to display the output from the computer. The monitor is connected to the computer by a cable. Terminals are categorized as dumb, smart and intelligent terminals. The dumb terminals do not have processing and programming capabilities. Smart terminals have built-in processing capability but do not have its own storage capacity. Intelligent terminals have both built-in processing and storage capacity.

Video Output

Screen image projector or data projector is an output device that displays information from the computer onto a large white screen. The projector is mainly used to display visual output to a large gathering of people required for the purposes of teaching, training, meetings, conference presentations, etc.

Audio Response

A complete sound system consists of sound card, microphone, speaker and the appropriate software. In addition to recording and playing the sound, the software allows editing of sound, like cutting, copy, amplification and creation of vibrant sound effects.

Audio response provides audio output from the computer. Audio output device like speakers, headset or headphone is used for audio output sound from computer. The signals are sent to the speakers via the sound card that translates the digital sound back into analog signals. The audio

response from the computer may be generated by synthesizing the input human speech to give audio output, or may be a result of a set of rules that are used to create artificial speech.

Audio output is commonly used for customer service in airlines, banks, etc. It is also used in video conferences, surveys, etc. Audio response is used by visually impaired to read information from the screen. For speech impaired people, audio response helps them to communicate with other people.

I/O PORT

The peripheral devices can be connected to computer in several ways. Devices such as network adapters and sound cards are connected to expansion slots inside the computer. Printers and scanners are connected to ports on the backside of the computer. Also in a portable computer, the PC Card connects to the PC Card slot on it.

The I/O ports are the external interfaces that are used to connect input and output devices like printer, modem and joystick to the computer. The I/O devices are connected to the computer via the serial and parallel ports, Universal Serial Bus (USB) port, Firewire port, etc.

Parallel Port A parallel port is an interface for connecting eight or more data wires. The data flows through the eight wires simultaneously. They can transmit eight bits of data in parallel. As a result, parallel ports provide high speed data transmission. Parallel port is used to connect printer to the computer.

Serial Port A serial port transmits one bit of data through a single wire. Since data is transmitted serially as single bits, serial ports provide slow speed data transmission. Serial port is used to connect external modems, plotters, barcode reader, etc.

USB Port Nowadays, USB is a common and popular external port available with computers. Normally, two to four USB ports are provided on a PC. USB allows different devices to be connected to the computer without requiring re-boot of the computer. USB also has the plug and play feature which allows devices ready to be run simply by plugging them to the USB port. A single USB port can support connection of up to 127 devices.

Firewire (IEEE 1394), Firewire is used to connect audio and video multimedia devices like video camera. It is an expensive technology and is used for large data movement. Hard disk drive and new DVD drives connect through Firewire. It has data transfer rate of up to 400 MB/sec.

In addition to the above ports, other ports also exist like Musical Instrument Digital Interface (MIDI) port to connect musical instruments like synthesizers and drum machines, PC expansion boards, and PC card and many more. **(TODO – Check the back of a computer and identify all ports by names and what is connected to each port)**

COPPERBELT UNIVERSITY

COMPUTER SCIENCE DEPARTMENT

INTRODUCTION TO COMPUTING

UNIT 5: Data Representation

5.1 Introduction

We use computer to process the data and get the desired output. The data input can be in the form of alphabets, digits, symbols, audio, video, magnetic cards, finger prints, etc. Since computer can only understand 0 and 1, the data must be represented in the computer in 0s and 1s. The purpose of this lecture is to introduce you to the data representation in the computer.

This lecture discusses the number systems that are commonly used in the computer. The number systems discussed in this lecture are— (1) Decimal number system, (2) Binary number system, (3) Octal number system, and (4) Hexadecimal number system. The number conversions described in this lecture are—

- Decimal (Integer, Fraction, Integer. Fraction) to Binary, Octal, Hexadecimal
- Binary, Octal, Hexadecimal (Integer, Fraction, Integer. Fraction) to Decimal
- Binary to Octal, Hexadecimal
- Octal, Hexadecimal to Binary

The lecture also discusses the binary arithmetic operations and the representation of signed and unsigned numbers in the computer. The representation of numbers using binary coding schemes and the logic gates used for the manipulation of data are also discussed.

5.2 Number System

A number system in base r or radix r uses unique symbols for r digits. One or more digits are combined to get a number. The base of the number decides the valid digits that are used to make a number. In a number, the position of digit starts from the right-hand side of the number. The rightmost digit has position 0, the next digit on its left has position 1, and so on. The digits of a number have two kinds of values—

- Face value, and
- Position value.

The **face value** of a digit is the digit located at that position. For example, in decimal number 52, face value at position 0 is 2 and face value at position 1 is 5.

The **position value** of a digit is $(\text{base}^{\text{position}})$. For example, in decimal number 52, the position value of digit 2 is 10^0 and the position value of digit 5 is 10^1 . Decimal numbers have a base of 10.

The **number** is calculated as the sum of, face value * base^{position}, of each of the digits. For decimal number 52, the number is $5 \cdot 10^1 + 2 \cdot 10^0 = 50 + 2 = 52$.

In computers, we are concerned with four kinds of number systems, as follows—

- Decimal Number System—Base 10
- Binary Number System—Base 2
- Octal Number System—Base 8
- Hexadecimal Number System—Base 16

The numbers given as input to computer and the numbers given as output from the computer, are generally in decimal number system, and are most easily understood by humans. However, computer understands the binary number system, i.e., numbers in terms of 0s and 1s. The binary data is also represented, internally, as octal numbers and hexadecimal numbers due to their ease of use.

A number in a particular base is written as (number)_{base of number}. For example, $(23)_{10}$ means that the number 23 is a decimal number, and $(345)_8$ shows that 345 is an octal number.

5.2.1. Decimal Number System

- It consists of 10 digits—0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.
- All numbers in this number system are represented as combination of digits 0–9. For example, 34, 5965 and 867321.
- The position value and quantity of a digit at different positions in a number are as follows—

Position:	3	2	1	0	.	–1	–2	–3
Position Value:	10^3	10^2	10^1	10^0		10^{-1}	10^{-2}	10^{-3}
Quantity:	1000	100	10	1		1/10	1/100	1/1000

5.2.2. Binary Number System

- The binary number system consists of two digits—0 and 1.
- All binary numbers are formed using combination of 0 and 1. For example, 1001, 11000011 and 10110101.
- The position value and quantity of a digit at different positions in a number are as follows—

Position:	3	2	1	0	.	-1	-2	-3
Position Value:	2^3	2^2	2^1	2^0		2^{-1}	2^{-2}	2^{-3}
Quantity:	8	4	2	1		1/2	1/4	1/8

5.2.3. Octal Number System

- The octal number system consists of eight digits—0 to 7.
- All octal numbers are represented using these eight digits. For example, 273, 103, 2375, etc.
- The position value and quantity of a digit at different positions in a number are as follows—

Position:	3	2	1	0	.	-1	-2	-3
Position Value:	8^3	8^2	8^1	8^0		8^{-1}	8^{-2}	8^{-3}
Quantity:	512	64	8	1		1/8	1/64	1/512

5.2.4. Hexadecimal Number System

- The hexadecimal number system consists of sixteen digits—0 to 9, A, B, C, D, E, F, where (A is for 10, B is for 11, C-12, D-13, E-14, F-15).
- All hexadecimal numbers are represented using these 16 digits. For example, 3FA, 87B, 113, etc.
- The position value and quantity of a digit at different positions in a number are as follows—

Position:	3	2	1	0	.	-1	-2	-3
Position Value:	16^3	16^2	16^1	16^0		16^{-1}	16^{-2}	16^{-3}
Quantity:	4096	256	16	1		1/16	1/256	1/4096

Table 5.1 summarizes the base, digits and largest digit for the above discussed number systems.

Table 5.1. Summary of number system

	Base	Digits	Largest Digit
Decimal	10	0–9	9
Binary	2	0,1	1
Octal	8	0–7	7
Hexadecimal	16	0–9, A, B, C, D, E, F	F (15)

5.3. Conversion from Decimal to Binary, Octal, Hexadecimal

A decimal number has two parts—integer part and fraction part. For example, in the decimal number 23.0786, 23 is the integer part and .0786 is the fraction part. The method used for the conversion of the integer part of a decimal number is different from the one used for the fraction part. In the following subsections, we shall discuss the conversion of decimal integer, decimal fraction and decimal integer.fraction number into binary, octal and hexadecimal number.

5.3.1. Converting Decimal Integer to Binary, Octal, Hexadecimal

A decimal integer is converted to any other base, by using the division operation.

To convert a decimal integer to—

- binary-divide by 2,
- octal-divide by 8, and,
- Hexadecimal-divide by 16.

Example 1.

Convert 25 from Base 10 to Base 2.

Solution

1. Make a table as shown below. Write the number in centre and to Base on the left side.

to Base	Number (Quotient)	Remainder
2	25	

2. Divide the number with to Base. After each division, write the remainder on right-side column and quotient in the next line in the middle column. Continue dividing till the quotient is 0.

to Base	Number (Quotient)	Remainder
2	25	
2	12	1
2	6	0
2	3	0
2	1	1
	0	1

3. Write the digits in remainder column starting from downwards to upwards,

to Base	Number (Quotient)	Remainder
2	25	
2	12	1
2	6	0
2	3	0
2	1	1
	0	1



The binary equivalent of number $(25)_{10}$ is $(11001)_2$.

The steps shown above are followed to convert a decimal integer to a number in any other base.

Example 2.

Convert 23 from Base 10 to Base 2, 8, 16.

Solution

to Base	Number (Quotient)	Remainder
2	23	
2	11	1
2	5	1
2	2	1
2	1	0
	0	1

The binary equivalent of $(23)_{10}$ is $(10111)_2$

to Base	Number (Quotient)	Remainder
8	23	
8	2	7
	0	2

The octal equivalent of $(23)_{10}$ is $(27)_8$

to Base	Number (Quotient)	Remainder
16	23	
16	1	7
	0	1

The hexadecimal equivalent of $(23)_{10}$ is $(17)_{16}$

5.3.2. Converting Decimal Fraction to Binary, Octal, Hexadecimal

A fractional number is a number less than 1. It may be .5, .00453, .564, etc. We use the multiplication operation to convert decimal fraction to any other base.

To convert a decimal fraction to—

- binary-multiply by 2,
- octal-multiply by 8, and,

- Hexadecimal-multiply by 16.

Steps for conversion of a decimal fraction to any other base are—

1. Multiply the fractional number with the Base, to get a resulting number.
2. The resulting number has two parts, non-fractional part and fractional part.
3. Record the non-fractional part of the resulting number.
4. Repeat the above steps at least four times.
5. Write the digits in the non-fractional part starting from upwards to downwards.

Example 3.

Convert 0.2345 from Base 10 to Base 2.

$$\begin{array}{r} 0.2345 \\ \underline{\times 2} \\ 0.4690 \end{array}$$

$$\begin{array}{r} .4690 \\ \underline{\times 2} \\ 0.9380 \end{array}$$

$$\begin{array}{r} .9380 \\ \underline{\times 2} \\ 1.8760 \end{array}$$

$$\begin{array}{r} .8760 \\ \underline{\times 2} \\ 1.7520 \end{array}$$

$$\begin{array}{r} .7520 \\ \underline{\times 2} \\ 1.5040 \end{array}$$

$$\begin{array}{r} .5040 \\ \underline{\times 2} \\ 1.0080 \end{array}$$



The binary equivalent of $(0.2345)_{10}$ is $(0.001111)_2$

Example 4

Convert 0.865 from Base 10 to Base 2, 8 and 16.

Solution

$$\begin{array}{r}
 0.865 \\
 \times 2 \\
 \hline
 1.730 \\
 \times 2 \\
 \hline
 1.460 \\
 \times 2 \\
 \hline
 0.920 \\
 \times 2 \\
 \hline
 1.840 \\
 \times 2 \\
 \hline
 1.680 \\
 \times 2 \\
 \hline
 1.360
 \end{array}$$

The binary equivalent of $(.865)_{10}$ is $(.110111)_2$

$$\begin{array}{r}
 0.865 \\
 \times 8 \\
 \hline
 6.920 \\
 \times 8 \\
 \hline
 7.360 \\
 \times 8 \\
 \hline
 2.880 \\
 \times 8 \\
 \hline
 7.040
 \end{array}$$

The octal equivalent of $(0.865)_{10}$ is $(.6727)_8$.

$$\begin{array}{r}
 0.865 \\
 \times 16 \\
 \hline
 \mathbf{13.840} \\
 \times 16 \\
 \hline
 5040 \\
 840 \times \\
 \hline
 \mathbf{13.440} \\
 \times 16 \\
 \hline
 2640 \\
 440 \times \\
 \hline
 \mathbf{7.040}
 \end{array}$$

The number 13 in hexadecimal is D.

The hexadecimal equivalent of $(0.865)_{10}$ is $(.DD7)_{16}$

5.3.3. Converting Decimal Integer. Fraction to Binary, Octal, Hexadecimal

A decimal integer.fraction number has both integer part and fraction part.

The steps for conversion of a decimal integer.fraction to any other base are—

1. Convert decimal integer part to the desired base following the steps shown in section 5.3.1.
2. Convert decimal fraction part to the desired base following the steps shown in section 5.3.2.
3. The integer and fraction part in the desired base is combined to get integer fraction.

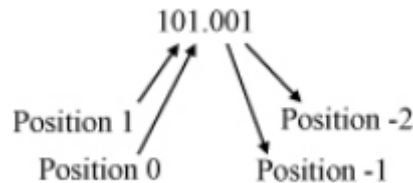
QUIZ

1. Convert 34.4674 from Base 10 to Base 2.
2. Convert 34.4674 from Base 10 to Base 8.

5.4. Conversion of Binary, Octal, Hexadecimal to Decimal

A binary, octal or hexadecimal number has two parts—integer part and fraction part. For example, a binary number could be 10011, 0.011001 or 10011.0111. The numbers 45, .362 or 245.362 are octal numbers. A hexadecimal number could be A2, .4C2 or A1.34. The method used for the conversion of integer part and fraction part of binary, octal or hexadecimal number to decimal number is the same; multiplication operation is used for the conversion. The conversion mechanism uses the face value and position value of digits. The steps for conversion are as follows—

1. Find the sum of the **Face Value * (fromBase)^{position}** for each digit in the number.
 - a) In a non-fractional number, the rightmost digit has position 0 and the position increases as we go towards the left.
 - b) In a fractional number, the first digit to the left of decimal point has position 0 and the position increases as we go towards the left. The first digit to the right of the decimal point has position -1 and it decreases as we go towards the right (-2, -3, etc.)



Example 5

Convert 1011 from Base 2 to Base 10.

Convert 62 from Base 8 to Base 10.

Convert C15 from Base 16 to Base 10.

1011 from Base 2 to Base 10

Solution

$$1011 = 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0$$

$$= 1 \cdot 8 + 0 \cdot 4 + 1 \cdot 2 + 1 \cdot 1$$

$$= 8 + 0 + 2 + 1$$

$$= 11$$

The decimal equivalent of $(1011)_2$ is 11.

62 from Base 8 to Base 10

$$62 = 6 \cdot 8^1 + 2 \cdot 8^0$$

$$= 6 \cdot 8 + 2 \cdot 1$$

$$= 48 + 2$$

$$= 50$$

The decimal equivalent of $(62)_8$ is 50.

C15 from Base 16 to Base 10

$$C15 = C \cdot 16^2 + 1 \cdot 16^1 + 5 \cdot 16^0$$

$$= 12 \cdot 256 + 1 \cdot 16 + 5 \cdot 1$$

$$= 3072 + 16 + 5$$

$$= 3093$$

The decimal equivalent of $(C15)_{16}$ is 3093

Example 6

Convert .1101 from Base 2 to Base 10.

Convert .15 from Base 16 to Base 10.

.1101 from Base 2 to Base 10

Solution

$$.1101 = 1 \cdot 2^{-1} + 1 \cdot 2^{-2} + 0 \cdot 2^{-3} + 1 \cdot 2^{-4}$$

$$= 1/2 + 1/4 + 0 + 1/16$$

$$= 13/16$$

$$= .8125$$

The decimal equivalent of $(.1101)_2$ is .8125

.15 from Base 16 to Base 10

$$.15 = 1*16^{-1} + 5*16^{-2}$$

$$= 1/16 + 5/256$$

$$= 21/256$$

$$= .082$$

The decimal equivalent of $(.15)_{16}$ is .082

Example 7

- Convert 1011.1001 from Base 2 to Base 10.
- Convert 4D.21 from Base 16 to Base 10.

Solution

1011.1001 from Base 2 to Base 10

$$1011.1001 = 1*2^3 + 0*2^2 + 1*2^1 + 1*2^0 + 1*2^{-1} + 0*2^{-2} + 0*2^{-3} + 1*2^{-4}$$

$$= 8 + 0 + 2 + 1 + 1/2 + 0 + 0 + 1/16$$

$$= 11 + 9/16$$

$$= 11.5625$$

The decimal equivalent of $(1011.1001)_2$ is 11.5625

4D.21 from Base 16 to Base 10

$$4D.21 = 4*16^1 + D*16^0 + 2*16^{-1} + 1*16^{-2}$$

$$= 64 + 13 + 2/16 + 1/256$$

$$= 77 + 33/256$$

$$= 77.1289$$

The decimal equivalent of $(4D.21)_{16}$ is 77.1289

5.5. Conversion of Binary to Octal, Hexadecimal

A binary number can be converted into octal or hexadecimal number using a shortcut method. The shortcut method is based on the following information —

An octal digit from 0 to 7 can be represented as a combination of 3 bits, since $2^3 = 8$.

A hexadecimal digit from 0 to 15 can be represented as a combination of 4 bits, since $2^4 = 16$.

The Steps for Binary to Octal Conversion are—

1. Partition the binary number in groups of three bits, starting from the right-most side.
2. For each group of three bits, find its octal number.
3. The result is the number formed by the combination of the octal numbers.

The Steps for Binary to Hexadecimal Conversion are—

1. Partition the binary number in groups of four bits, starting from the right-most side.
2. For each group of four bits, find its hexadecimal number.
3. The result is the number formed by the combination of the hexadecimal numbers.

Example 8

- Convert the binary number 1110101100110 to octal.

Solution

Given binary number 1110101100110

1. Partition binary number in groups of three bits, starting from the right-most side.

1	1 1 0	1 0 1	1 0 0	1 1 0
---	-------	-------	-------	-------

2. For each group find its octal number.

1	1 1 0	1 0 1	1 0 0	1 1 0
1	6	5	4	6

3. The octal number is 16546.

Example 9

Convert the binary number 1110101100110 to hexadecimal

Solution

1. Partition binary number in groups of four bits, starting from the right-most side.

1 1 1 0 1 0 1 1 0 0 1 1 0

2. For each group find its hexadecimal number.

1	1 1 0 1	0 1 1 0	0 1 1 0
1	D	6	6

3. The hexadecimal number is 1D66.

5.6. Conversion of Octal, Hexadecimal to Binary

The conversion of a number from octal and hexadecimal to binary uses the inverse of the steps defined for the conversion of binary to octal and hexadecimal.

The Steps for Octal to Binary Conversion are—

1. Convert each octal number into a three-digit binary number.
2. The result is the number formed by the combination of all the bits.

The Steps for Hexadecimal to Binary Conversion are—

1. Convert each hexadecimal number into a four-digit binary number.
2. The result is the number formed by the combination of all the bits.

Example 10

Convert the hexadecimal number 2BA3 to binary.

Solution

1. Given number is 2BA3
2. Convert each hexadecimal digit into four digit binary number.

2	B	A	3
0010	1011	1010	0011

3. Combine all the bits to get the result 0010101110100011.

Example 11

Convert the octal number 473 to binary.

Solution

1. Given number is 473
2. Convert each octal digit into three digit binary number.

4	7	3
100	111	011

3. Combine all the bits to get the result 100111011.

5.7 Binary Coding Schemes

The alphabetic data, numeric data, alphanumeric data, symbols, sound data and video data, are represented as combination of bits in the computer. The bits are grouped in a fixed size, such as 8 bits, 6 bits or 4 bits. A code is made by combining bits of definite size. Binary Coding schemes represent the data such as alphabets, digits 0–9, and symbols in a standard code. A combination of bits represents a unique symbol in the data. The standard code enables any programmer to use the same combination of bits to represent a symbol in the data. The binary coding schemes that are most commonly used are—

- Extended Binary Coded Decimal Interchange Code (EBCDIC),
- American Standard Code for Information Interchange (ASCII), and
- Unicode

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INTRODUCTION TO COMPUTING

UNIT 6: User-Computer Interface

Introduction

A computer system consists of hardware and software. The computer hardware cannot perform any task on its own. It needs to be instructed about the tasks to be performed. Software is a set of programs that instructs the computer about the tasks to be performed. Software tells the computer how the tasks are to be performed; hardware carries out these tasks. Different sets of software can be loaded on the same hardware to perform different kinds of tasks. For example, a user can use the same computer hardware for writing a report or for running a payroll program. The components like monitor, keyboard, processor, and mouse, constitute the hardware.

Types of Software

Software can be broadly classified in two categories:

1. System Software, and
2. Application Software.

System software provides the basic functions that are performed by the computer. It is necessary for the functioning of a computer. Application software is used by the users to perform specific tasks. The user may choose the appropriate application software, for performing a specific task, which provides the desired functionality. The system software interacts with hardware at one end and with application software at the other end. The application software interacts with the system software and the users of the computer. Figure 2 shows the hierarchy of software, hardware and users.

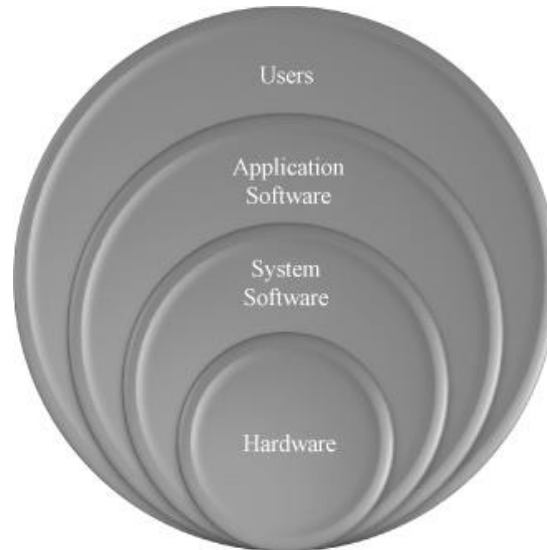


Figure 2 Software hierarchy

System Software

System software provides basic functionality to the computer. System software is required for the working of computer itself. The user of computer does not need to be aware about the functioning of system software, while using the computer. For example, when you buy a computer, the system software would also include different device drivers. When you request for using any of the devices, the corresponding device driver software interacts with the hardware device to perform the specified request. If the appropriate device driver for any device, say a particular model of a printer, is installed on the computer, the user does not need to know about the device driver, while printing on this printer.

The purposes of the system software are:

- ➡ To provide basic functionality to computer,
- ➡ To control computer hardware, and
- ➡ To act as an interface between user, application software and computer hardware.

On the basis of their functionality, system software may be broadly divided into two categories (Figure 3) as follows—

System software for the management and functionality of computer relates to the functioning of different components of the computer, like, processor, input and output devices etc. System software is required for managing the operations performed by the components of computer and the devices attached to the computer.

It provides support for various services, as requested by the application software. Operating system, device drivers, and system utilities constitute the system software for management of computer and its resources.

System software for the development of application software provides services required for the development and execution of application software. System software provides the software tools required for the development of application software. The programming language software, translator software, loader, and linker are also categorized as system software, and are required for the application software development.

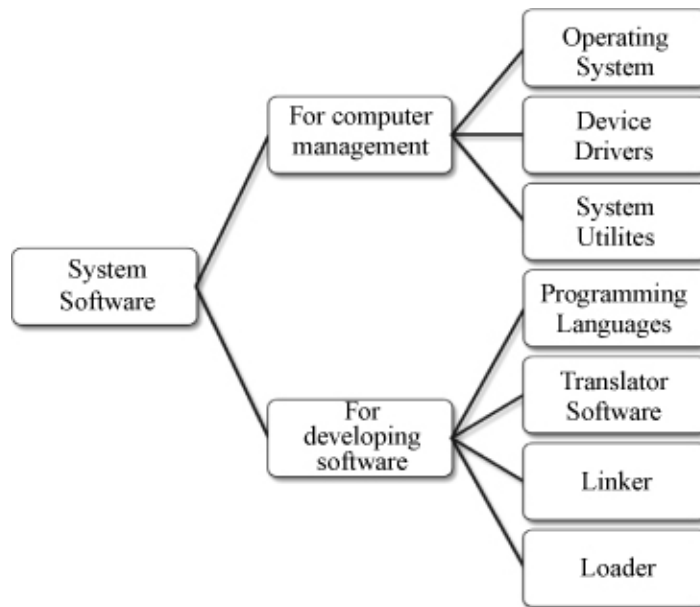


Figure 3 System software

Operating System

Operating System (OS) is an important part of a computer. OS intermediates between the user of a computer and the computer hardware. Different kinds of application software use specific hardware resources of a computer like CPU, I/O devices and memory, as needed by the application software. OS controls and coordinates the use of hardware among the different application software and the users. It provides an interface that is convenient for the user to use, and facilitates efficient operations of the computer system resources. The key functions of OS are:

- ➡ It provides an environment in which users and application software can do work.

- ➡ It manages different resources of the computer like the CPU time, memory space, file storage, I/O devices etc. During the use of computer by other programs or users, operating system manages various resources and allocates them whenever required, efficiently.
- ➡ It controls the execution of different programs to prevent occurrence of error.
- ➡ It provides a convenient interface to the user in the form of commands and graphical interface, which facilitates the use of computer.

Some available operating systems are Microsoft Disk Operating System (MS- DOS), Windows 7, Windows XP, Linux, UNIX, and Mac OS X Snow Leopard.

Device Driver

A device driver acts as a translator between the hardware and the software that uses the devices. In other words, it intermediates between the device and the software, in order to use the device.

Some devices that are commonly connected to the computer are—keyboard, mouse, hard disk, printer, speakers, microphone, joystick, webcam, scanner, digital camera, and monitor. For proper working of a device, its corresponding device driver must be installed on the computer. For example, when we give a command to read data from the hard disk, the command is sent to the hard disk driver and is translated to a form that the hard disk can understand. The device driver software is typically supplied by the respective device manufacturers.

Programmers can write the higher-level application code independently of whatever specific hardware devices it will ultimately use, because code and device can interface in a standard way, regardless of the software superstructure, or of the underlying hardware. Each version of a device, such as a printer, requires its own hardware-specific specialized commands. In contrast, most applications instruct devices (such as a file to a printer) by means of high level generic commands for the device, such as PRINTLN (print a line). The device-driver accepts these generic high-level commands and breaks them into a series of low-level, device-specific commands, as required by the device being driven.

Nowadays, the operating system comes preloaded with some commonly used device drivers, like the device driver for mouse, webcam, and keyboard. The device drivers of these devices are pre-installed on the computer, such that the operating system can automatically detect the device when it is connected to the computer. Such devices are called plug and play devices. In case the computer does not find the device driver, it prompts the user to insert the media (like a CD which contains the corresponding device driver) provided along with the device.

Most device manufacturers, host the device drivers for their devices on their companies' websites; users can download the relevant driver and install it on their computer.

Each device has its own device driver. Whenever a new device is connected to a computer, its device driver has to be loaded in the computer's memory, to enable use of the device. When you buy a new printer, you get the device driver CD with it. You must install the device driver on your computer, to use the new printer. Each printer comes with its own device driver. If you replace your old printer with a new model, you need to install the device driver for the new printer.

Device drivers can be character or block device drivers. Character device drivers are for character based devices like keyboard, which transfer data character by character. Block device driver are for devices that transfer data as a block, like in hard disk.

System Utilities

System utility software is required for the maintenance of computer. System utilities are used for supporting and enhancing the programs and the data in computer. Some system utilities may come embedded with OS and others may be added later on. Some examples of system utilities are:

- ➡ Anti-virus utility to scan computer for viruses.
- ➡ Data Compression utility to compress the files.
- ➡ Cryptographic utility to encrypt and decrypt files.
- ➡ Disk Compression utility to compress contents of a disk for increasing the capacity of a disk.
- ➡ Disk Partitioning to divide a single drive into multiple logical drives. Each drive is then treated as an individual drive and has its own file system.
- ➡ Disk Cleaners to find files that have not been used for a long time. It helps the user to decide what to delete when the hard disk is full.
- ➡ Backup Utility to make a copy of all information stored on the disk. It also restores the backed up contents in case of disk failure.
- ➡ System Profiling Utility provides detailed information about the software installed on the computer and the hardware attached to it.
- ➡ Network Managers to check the computer network and to log events.

The system utilities on a computer working on Windows XP OS and Windows 7 can be viewed by clicking <Start><All Programs><Accessories><System Tools>.

Programming Languages

A Programming Language consists of a set of vocabulary and grammatical rules, to express the computations and tasks that the computer has to perform. Programming languages are used to write a program, which controls the behavior of computer, codify the algorithms precisely, or enables the human-computer interface. Each language has a unique set of keywords (words that it understands) and a special syntax for organizing program instructions. The programming language should be understood, both by the programmer (who is writing the program) and the computer. A computer understands the language of 0's and 1's, while the programmer is more comfortable with English-like language. Programming Language usually refers to high-level languages like COBOL, BASIC, FORTRAN, C, C++, Java etc.

Programming languages fall into three categories (Figure 4):

- Machine Language is what the computer can understand but it is difficult for the programmer to understand. Machine languages consist of numbers only. Each kind of CPU has its own unique machine language.
- Assembly Language falls in between machine language and high-level language. They are similar to machine language, but easier to program in, because they allow the programmer to substitute names for numbers.
- High-level Language is easier to understand and use for the programmer but difficult for the computer.

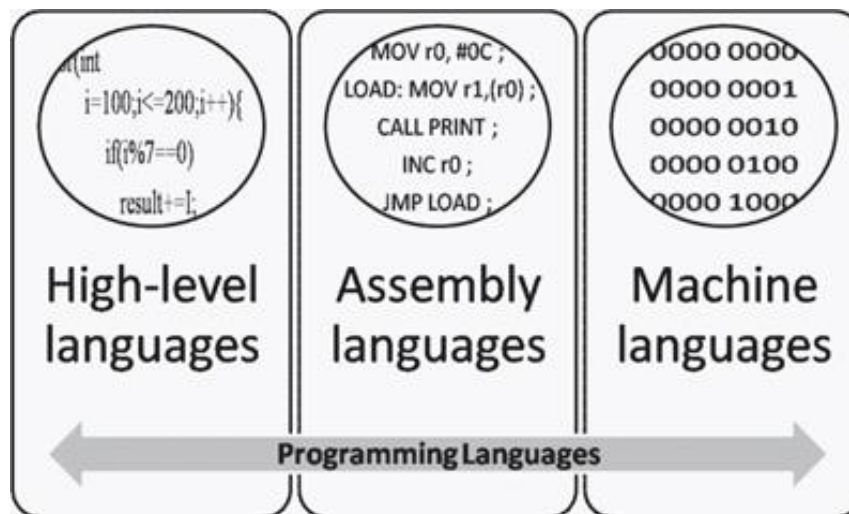


Figure 4 Programming Language Categories

Regardless of the programming language used, the program needs to be converted into machine language so that the computer can understand it. In order to do this a program is either compiled or interpreted.

The choice of programming language for writing a program depends on the functionality required from the program and the kind of program to be written. Machine languages and assembly languages are also called low-level languages, and are generally used to write the system software. Application software is usually written in high-level languages. The program written in a programming language is also called the source code.

Machine Language

A program written in machine language is a collection of binary digits or bits that the computer reads and interprets. It is a system of instructions and data executed directly by a computer's CPU. It is also referred to as machine code or object code. It is written as strings of 0's and 1's. Some of the features of a program written in machine language are as follows:

- The computer can understand the programs written in machine language directly. No translation of the program is needed. Program written in machine language can be executed very fast (Since no translation is required).
- Machine language is defined by the hardware of a computer. It depends on the type of the processor or processor family that the computer uses, and is thus machine-dependent. A machine-level program written on one computer may not work on another computer with a different processor.
- Computers may also differ in other details, such as memory arrangement, operating systems, and peripheral devices; because a program normally relies on such factors, different computer may not run the same machine language program, even when the same type of processor is used.
- Most machine-level instructions have one or more op-code fields which specify the basic instruction type (such as arithmetic, logical, jump, etc), the actual operation (such as add or compare), and some other fields.
- It is difficult to write a program in machine language as it has to be written in binary code. For e.g., 00010001 11001001. Such programs are also difficult to modify.
- Since writing programs in machine language is very difficult, programs are hardly written in machine language.

Assembly Language

A program written in assembly language uses symbolic representation of machine codes needed to program a particular processor (CPU) or processor family. This representation is usually defined by the CPU manufacturer, and is based on abbreviations (called mnemonics) that help the programmer remember individual instructions, registers, etc. Small, English-like representation is used to write the program in assembly language. Some of the features of a program written in assembly language are as follows:

- Assembly language programs are easier to write than the machine language programs, since assembly language programs use short, English-like representation of machine code. For e.g.:

ADD 2, 3

LOAD A

SUB A, B

- The program written in assembly language is the source code, which has to be converted into machine code, also called object code, using translator software, namely, assembler.
- Each line of the assembly language program is converted into one or more lines of machine code. Hence assembly language programs are also machine-dependent.
- Although assembly language programs use symbolic representation, they are still difficult to write.
- Assembly language programs are generally written where the efficiency and the speed of program are the critical issues, i.e. programs requiring high speed and efficiency.

High-level Language

A program in a high-level language is written in English-like language. Such languages hide the details of CPU operations and are easily portable across computers. A high-level language isolates the execution semantics of computer architecture from the specification of the program, making the process of developing a program simpler and more understandable with respect to assembly and machine level languages. Some of the features of a program written in high-level language are as follows:

- Programs are easier to write, read or understand in high-level languages than in machine language or assembly language. For example, a program written in C++ is easier to understand than a machine language program.
- Programs written in high-level languages is the source code which is converted into the object code (machine code) using translator software like interpreter or compiler.

- A line of code in high-level program may correspond to more than one line of machine code.
- Programs written in high-level languages are easily portable from one computer to another.

Different Generations of Programming Languages

In addition to the categorization of programming languages into machine language, assembly language, and high-level language, programming languages are also classified in terms of generations in which they have evolved. Table 6.1 shows the classification of programming languages based on generations.

Table 1 Generations of programming languages

First Generation	<i>Machine language</i>
Second Generation	<i>Assembly language</i>
Third Generation	<i>C, COBOL, Fortran, Pascal, C++, Java, ActiveX (Microsoft) etc.</i>
Fourth Generation	<i>.NET (VB.NET, C#.NET etc.)</i>
	<i>Scripting language (JavaScript, Microsoft FrontPage etc.)</i>
Fifth Generation	<i>LISP, Prolog</i>

Translator Software

Translator software is used to convert a program written in high-level language and assembly language to a form that the computer can understand. Translator software converts a program written in assembly language, and high-level language to a machine-level language program. The translated program is called the object code. There are three different kind of translator software:

- ➡ Assembler,
- ➡ Compiler, and
- ➡ Interpreter.

Assembler converts a program written in assembly language to machine language. Compiler and interpreter convert a program written in high-level language to machine language. Let's now discuss, briefly, the different kinds of translator software.

Assembler

Assembly language is also referred to as a symbolic representation of the machine code. Assembler is software that converts a program written in assembly language into machine code. There is usually a one- to-one correspondence between simple assembly statements and machine language instructions. The machine language is dependent on the processor architecture, though computers are generally able to carry out the same functionality in different ways. Thus the corresponding assembly language programs also differ for different computer architectures.

Compiler

A program written in a high-level language has to be converted to a language that the computer can understand, i.e. binary form. Compiler is the software that translates the program written in a high-level language to machine language. The program written in high-level language is referred to as the source code and compiled program is referred as the object code. The object code is the executable code, which can run as a stand-alone code. It does not require the compiler to be present during execution. Each programming language has its own compiler. Some languages that use a compiler are C++, COBOL, Pascal, and FORTRAN. In some languages, compilation using the compiler and linking using the linker are required for creating the executable object code.

The compilation process generally involves two parts—breaking down the source code into small pieces and creating an intermediate representation, and, constructing the object code for the intermediate representation. The compiler also reports syntax errors, if any, in the source code.

Interpreter

The purpose of interpreter is similar to that of a compiler. The interpreter is used to convert the high-level language program into computer- understandable form. However, the interpreter functions in a different way than a compiler. Interpreter performs line-by-line execution of the source code during program execution. Interpreter reads the source code line-by- line, converts it into machine understandable form, executes the line, and then proceeds to the next line. Some languages that use an interpreter are BASIC and Python.

Difference between a Compiler and an Interpreter

Compiler and Interpreter are used to convert a program written in high-level language to machine language; however, they work differently. The key differences between a compiler and an interpreter are as follows:

- Interpreter looks at a source code line-by-line. Compiler looks at the entire source code.
- Interpreter converts a line into machine executable form, executes the line, and proceeds with the next line. Compiler converts the entire source code into object-code and creates the object code. The object code is then executed by the user.
- For a given source code, once it is compiled, the object code is created. This object code can be executed multiple number of times by the user. However, interpreter executes line-by-line, so executing the program using an interpreter means that during each execution, the source code is first interpreted and then executed.
- During execution of an object code, the compiler is not required. However, for interpretation, both interpreter and the source code are required during execution (because source code is interpreted during execution).
- Since interpreter interprets line-by-line, the interpreted code runs slower than the compiled code.

Linker

Linker is a program that links several object modules and libraries to a single executable program. A source code of a program is often very large consisting of several hundred or more lines. The source code may also include reference to libraries. All these independent modules may not be stored in a single object file. The code is broken down into many independent modules for easy debugging and maintenance. Before execution of the program, these modules and the required libraries are linked together using the linker software. The compiled and the linked program are called the executable code.

Loader

The loader software is used to load and re-locate the executable program in the main memory. Software has to be loaded into the main memory during execution. Loader assigns storage space to the program in the main memory for execution. The java source file Sample.java is compiled and converted into a set of bytecodes and are stored in Sample.class file. At runtime, the class loader loads the bytecodes from the hard disk, checks it, and runs it in an interpreter. The interpreter executes the bytecode and makes calls to the underlying hardware.

Application Software

The software that a user uses for accomplishing a specific task is the application software. Application software may be a single program or a set of programs. A set of programs that are written for a specific purpose and provide the required functionality is called software package. Application software is written for different kinds of applications—graphics, word processors, media players, database applications, telecommunication, accounting purposes etc.

Some examples of application software packages are as follows:

- ➔ Word Processing Software: For writing letter, reports, documents etc. (e.g. MS-WORD).
- ➔ Image Processing Software: For assisting in drawing and manipulating graphics (e.g. Adobe Photoshop).
- ➔ Accounting Software: For assisting in accounting information, salary, tax returns (Tally software).
- ➔ Spreadsheet Software: Used for creating budget, tables etc. (e.g. MS- Excel).
- ➔ Presentation Software: To make presentations, slide shows (e.g. MS- PowerPoint)
- ➔ Suite of Software having Word Processor, Spreadsheet and Presentation Software: Some examples are MS-Office, Google Docs, Oracle LibreOffice, Apple iWork.
- ➔ CAD/CAM Software: To assist in architectural design. (E.g. AutoCAD, Autodesk)
- ➔ Geographic Information Systems: It captures, stores, analyzes, manages, and presents data, images and maps that are linked to different locations. (e.g. ArcGIS)
- ➔ Web Browser Software: To access the World Wide Web to search documents, sounds, images etc. (e.g. Internet Explorer, Firefox, Chrome).

Software Acquisition

Different kinds of software are made available for use to users in different ways. The user may have to purchase the software, can download for free from the Internet, or can get it bundled along with the hardware. Nowadays with the advent of Cloud computing, many application software are also available on the cloud for use through the Internet, e.g. Google Docs. The different ways in which the software are made available to users are:

- Retail Software is off-the-shelf software sold in retail stores. It comes with printed manuals and installation instructions. For example, Microsoft Windows operating system.
- OEM Software stands for “Original Equipment Manufacturer” software. It refers to software which is sold, and bundled with hardware. Microsoft sells its operating system as OEM software to hardware dealers. OEM software is sold at reduced price, without the manuals, packaging and installation instructions. For example, Dell computers are sold with the “Windows 7” OS pre-loaded on them.
- Demo Software is designed to demonstrate what a purchased version of the software is capable of doing and provides a restricted set of features. To use the software, the user must buy a fully-functional version.
- Shareware is a program that the user is allowed to try for free, for a specified period of time, as defined in the license. It is downloadable from the Internet. When the trial period ends, the software must be purchased or uninstalled.

- Freeware is software that is free for personal use. It is downloadable from the Internet. The commercial use of this software may require a paid license. The author of the freeware software is the owner of the software, though others may use it for free. The users abide by the license terms, where the user cannot make changes to it, or sell it to someone else.
- Public Domain Software is free software. Unlike freeware, public domain software does not have a copyright owner or license restrictions. The source code is publicly available for anyone to use. Public domain software can be modified by the user.
- Open-Source Software is software whose source code is available and can be customized and altered within the specified guidelines laid down by the creator. Unlike public domain software, open-source software has restrictions on their use and modification, redistribution limitations, and copyrights. Linux, Apache, Firefox, OpenOffice are some examples of open-source software.

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INTRODUCTION TO COMPUTING

UNIT 7: OPERATING SYSTEM

INTRODUCTION

The computer system comprises of a functional set of hardware, software, user and data. Hardware consists of the components of computer like memory, processor, storage devices, and Input/Output devices. The software may be of different kinds—application software and system software. A computer system may be a single stand-alone system or may consist of several interconnected systems. The user uses the application software to perform various tasks, for example, the user uses word processing software for document preparation. While using the application software, the user uses the storage of a computer—to store a document on the hard disk, to execute a command on the CPU, to retrieve a document from a peripheral device or to print document on printer. For using the hardware, there is a need for software that interacts with both the hardware and the application software. Operating system (OS) is the software that provides an interface between the computer hardware, and the application programs or users (Figure 1).

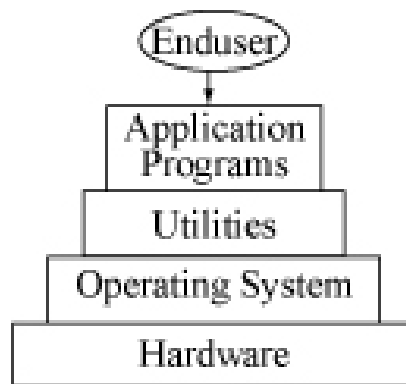


Figure 1 View of components of computer system

In this unit, we discuss about the components of operating system, the different types of operating system and the functions of operating system. A brief description of some operating systems is also given.

OBJECTIVES OF OPERATING SYSTEM

Operating system is system software that controls and coordinates the use of hardware among the different application software and users. OS intermediates between the user of computer and the computer hardware. The user gives a command and the OS translates the command into a form that the machine can understand and execute. OS has two main objectives—

- i. To make the computer system convenient and easy to use, for the user, and
- ii. To use the computer hardware in an efficient way, by handling the details of the operations of the hardware.

OS hides the working of the hardware from the user and makes it convenient for the user to use the machine. The application program used by the user requires the use of the hardware during processing. Some examples are—display of application's user interface, loading a program into memory, using I/O devices, allocating CPU to different processes during execution, and store or load data from hard disk. When using the machine, the user gives the command to perform the required actions to the OS and the OS handles all the operational steps. The user is not bothered about how these actions will be performed. This is the job of OS. OS provides an interface to the application programs to interact with the hardware. The user need not get into the details of this interaction.

At the other end, the different resources of computer hardware have to be managed and controlled. This includes managing the communication between different devices, controlling the sequence and execution of processes, allocating space on hard disk, providing error handling procedures etc. OS supervises and manages the hardware of the computer.

Some of the commonly used operating systems are Microsoft Disk Operating System (MS-DOS), Windows 7, Windows XP, Linux, UNIX, and Mac OS X Snow Leopard.

TYPES OF OS

OS are classified into different types depending on their capability of processing

- 1) Single user,
- 2) Multiuser,
- 3) Multitasking,
- 4) Multiprocessing,
- 5) Real time, and
- 6) Embedded.

Single User and Single Task OS is for use by a single user for a standalone single computer for performing a single task. Operating system for Personal Computers (PC) are single-user OS. For example, if the user is editing a document, then a document cannot be printed on the printer simultaneously. Single user OS are simple operating system designed to manage one task at a time. MS-DOS is an example of single user OS.

Single User and Multitasking OS allows execution of more than one task or process concurrently. For this, the processor time is divided amongst different tasks. This division of time is also called time sharing. The processor switches rapidly between processes. For example, the user can listen to music on the computer while writing an article using word processor software. The user can switch between the applications and also transfer data between them. Windows 95 and all later versions of Windows are examples of multitasking OS.

Multiuser OS is used in computer networks that allow same data and applications to be accessed by multiple users at the same time. The users can also communicate with each other. Linux, UNIX, and Windows 7 are examples of multiuser OS.

Multiprocessing OS have two or more processors for a single running process. Processing takes place in parallel and is also called parallel processing. Each processor works on different parts of the same task, or, on two or more different tasks. Since execution takes place in parallel, they are used for high speed execution, and to increase the power of computer. Linux, UNIX and Windows 7 are examples of multiprocessing OS.

Real Time OS are designed to respond to an event within a predetermined time. These operating systems are used to control processes. Processing is done within a time constraint. OS monitors the events that affect the execution of process and respond accordingly. They are used to respond to queries in areas like medical imaging system, industrial control systems etc.

Embedded OS is embedded in a device in the ROM. They are specific to a device and are less resource intensive. They are used in appliances like microwaves, washing machines, traffic control systems etc.

FUNCTIONS OF OS

Operating system is large and complex software consisting of several components. Each component of the operating system has its own set of defined inputs and outputs. Different components of OS perform specific tasks to provide the overall functionality of the operating system. Main functions of the operating system are as follows:

Process Management

The process management activities handled by the OS are—(1) control access to shared resources like file, memory, I/O and CPU, (2) control execution of applications, (3) create, execute and delete a process (system process or user process), (4) cancel or resume a process (5) schedule a process, and (6) synchronization, communication and deadlock handling for processes.

Memory Management

The activities of memory management handled by OS are—(1) allocate memory, (2) free memory, (3) re-allocate memory to a program when a used block is freed, and (4) keep track of memory usage.

File Management

The file management tasks include—(1) create and delete both files and directories, (2) provide access to files, (3) allocate space for files, (4) keep back-up of files, and (5) secure files.

Device Management

The device management tasks handled by OS are—(1) open, close and write device drivers, and (2) communicate, control and monitor the device driver.

Protection and Security

OS protects the resources of system. User authentication, file attributes like read, write, encryption, and back-up of data are used by OS to provide basic protection.

User Interface or Command Interpreter

Operating system provides an interface between the computer user and the computer hardware. The user interface is a set of commands or a graphical user interface via which the user interacts with the applications and the hardware.

Below is a detailed explanation of each of the above points.

PROCESS MANAGEMENT

A process is a program in a state of execution. It is a unit of work for the operating system. A process can be created, executed, and stopped. In contrast, a program is always static and does not have any state. A program may have two or more processes running. A process and a program are, thus, two different entities.

To accomplish a task, a process needs to have access to different system resources like I/O devices, CPU, memory etc. The process management function of an operating system handles allocation of resources to the processes in an efficient manner. The allocation of resources required by a process is made during process creation and process execution.

A process changes its state as it is executed. The various states that a process changes during execution are as follows:

- New—process is in a new state when it is created,
- Ready—process is in ready state when it is waiting for a processor,
- Running—process is in running state if processor is executing the process,
- Waiting—process is in waiting state when it waits for some event to happen (I/O etc), and
- Terminated—process that has finished execution is in terminated state.

A system consists of collection of processes—(1) system process that executes system code, and (2) user process that execute user code. OS mainly handles the execution of user code, though it may also handle various system processes.

The concurrent execution of the process requires process synchronization and CPU scheduling. The CPU scheduling, process synchronization, communication, and deadlock situations are described in the following subsections.

CPU Scheduling

CPU or processor is one of the primary computer resources. All computer resources like I/O, memory, and CPU are scheduled for use.

CPU scheduling is important for the operating system. In a multiprogramming and time sharing system, the processor executes multiple processes by switching the CPU among the processes, so that no user has to wait for long for a program to execute. To enable running of several concurrent processes, the processor time has to be distributed amongst all the processes efficiently.

Scheduler is a component of the operating system that is responsible for scheduling transition of processes. At any one time, only one process can be in running state and the rest are in ready or waiting state. The scheduler assigns the processor to different processes in a manner so that no one process is kept waiting for long.

Scheduling can be non-pre-emptive scheduling or pre-emptive scheduling. In non-pre-emptive scheduling, the processor executes a process till termination without any interruption. Hence the system resources are not used efficiently. In pre-emptive scheduling, a running process may be interrupted by another process that needs to execute. Pre-emption allows the operating system to interrupt the executing task and handle any important task that requires immediate action. In pre-emptive scheduling, the system resources are used efficiently.

There are many different CPU scheduling algorithms that are used to schedule the processes. Some of the common CPU scheduling algorithms are as follows—

- i. **First Come First Served (FCFS) Scheduling:** As the name says, the process that requests for the CPU first, gets the CPU first. A queue is maintained for the processes requesting the CPU. The process first in the queue is allocated the CPU first. FCFS scheduling is non-pre-emptive. The drawback of this scheduling algorithm is that the process that is assigned to the CPU may take long time to complete, keeping all other processes waiting in the queue, even if they require less CPU time.
- ii. **Shortest Job First (SJF) Scheduling:** The process that requires the least CPU time is allocated the CPU first. SJF scheduling is non-pre-emptive. The drawback of this scheduling is that a process that requires more CPU time may have to wait for long time, since processes requiring less CPU time will be assigned the CPU first.
- iii. **Round Robin (RR) Scheduling:** It is designed for time-sharing systems. RR scheduling is pre-emptive. In this scheduling, a small quantum of time (10–100 ms) is defined, and each process in the queue is assigned the CPU for this quantum of time circularly. New processes are added at the tail of the queue and the process that has finished execution is removed from the queue. RR scheduling overcomes the disadvantage of FCFS and SJF scheduling. A process does not have to wait for long, if it is not the first one in the queue, or, if it requires CPU for a long period of time.

Process Synchronization

In a computer, multiple processes are executing at the same time. The processes that share the resources have to communicate with one another to prevent a situation where one process disrupts another process.

When two or more processes execute at the same time, independent of each other, they are called concurrent processes.

A situation where multiple processes access and manipulate the same data concurrently, in which the final result depends on the order of process execution, is called a race condition. To handle such situations, synchronization and coordination of the processes is required.

Deadlock

In a multiprogramming environment, multiple processes may try to access a resource. A deadlock is a situation when a process waits endlessly for a resource and the requested resource is being used by another process that is waiting for some other resource.

A deadlock arises when the four necessary conditions hold true simultaneously in a system. These conditions are as follows:

- **Mutual Exclusion:** Only one process at a time can use the resource. Any other process requesting the resource has to wait until the resource is released.
- **No Pre-emption:** A process releases the resource by itself. A process cannot remove the resource from another process.
- **Hold and Wait:** A process holds a resource while requesting another resource, which may be currently held by another process.
- **Circular Wait:** In this situation, a process P1 waits for a resource held by another process P2, and the process P2 waits for a resource held by process P1.

Deadlock handling can be done by deadlock avoidance and deadlock prevention.

- **Deadlock Prevention** is a set of method that ensures that at least one of the above four necessary conditions required for deadlock, does not hold true.
- **Deadlock Avoidance** requires that the operating system be given information in advance regarding the resources a process will request and use. This information is used by the operating system to schedule the allocation of resources so that no process waits for a resource.

MEMORY MANAGEMENT

In a computer, there may be multiple processes executing at the same time. Every process that needs to execute, requires a certain amount of memory. Memory management is one of the tasks handled by the operating system. Memory management schemes handle the allocation of memory to different processes. On completion of process execution, the memory is de-allocated and made available to another process. Additionally, different processes that have been allocated memory should not interfere into each other's memory space. This requires some memory protection and sharing mechanism.

Virtual Memory

In the memory management, the whole process is kept in memory before the execution starts. However, for some applications, large memory is required to run the applications, and the whole program cannot be loaded into the memory.

Virtual memory allows the execution of those processes that are not completely in memory. When a process executes and a page is required, that page is swapped-in into the memory. This allows execution of large-sized programs without loading them completely into the memory.

FILE MANAGEMENT

The file management function of the operating system involves handling the file system which consists of two parts—a set of files, and a directory structure.

- File is a collection of related information, has a name, and is stored on a secondary storage. It is the smallest named unit that can be written to a secondary storage device. Data cannot be stored on the secondary storage if it is not in the form of a file. A file has attributes like its name, location, size, type, time, and date of creation etc. The information stored in a file can be accessed in different ways—sequential access (access is in a sequential order from start to end) and direct access (the file can be accessed in any order).
- Directory structure provides information about the files stored on the secondary storage. Directory contains information about all the files within it. The information about the files is kept as entries in the directory of device. A directory further may have a sub-directory defined within it. Directory contains the name, location, size, and type of all the files defined on the device. The tree-structured directory is the commonly used directory structure.

The operating system manages the storage media like the disk and implements the abstract concept of the file. System calls are an interface between the process and the operating system. Operating system provides system calls for creating, reading, writing, deleting, repositioning, and truncating a file. Some of the operations that can be performed on a directory are—search for a file, create, delete and rename a file, list a directory, and traverse the file system within the directory. The user simply uses the system calls like “dir”, “list” to perform operation on a file or directory, without going into the details of its working.

DEVICE MANAGEMENT

Several peripheral devices like mouse, hard disk, printer, plotter etc. are connected to the computer. The peripheral devices have varying characteristics like character or block device, sequential or random access device, and dedicated or shared device.

OS manages and controls the devices attached to the computer. OS provides appropriate functionality to the application programs for controlling different aspects of the devices.

OS handles the devices by combining both hardware and software techniques. The I/O hardware includes the ports, buses, and device controllers for the devices. The OS communicates with the I/O hardware via the device driver software. The device driver software comes along with each device.

A device communicates with the computer hardware via a port (for example, a serial port or a parallel port). Bus is a common set of wires used by one or more devices. The computer uses different kinds of buses like PCI bus for connecting processor or memory to the fast devices, expansion bus to connect to slow I/O devices and SCSI bus to connect disks. A device controller operates a port, bus, and a device. Device controller is just like a bridge between the device and the operating system. The device controller receives the data from a connected device, stores it temporarily, and then communicates the data to the device’s device driver. Device driver is the software with which the device controller communicates with the operating system.

Operating system enables handling of the different I/O devices in a uniform way. The complexity of handling the different devices is abstracted and hidden in the device drivers of the devices. The device drivers hide the differences among the different device controllers and present a uniform interface to the operating system.

In addition to managing the peripheral devices, OS also provides various services related to I/O like I/O scheduling, buffering, spooling, and error handling.

Scheduling of I/O requests involves ordering the requests to improve performance of the system and provide fair access to all processes. For this, a queue of request is maintained for each device. The I/O scheduler re-arranges the queue to improve the efficiency of the overall system.

Buffer is a memory area that stores the data, while it is being transferred between two devices or between a device and an application. The speed at which the I/O device can transfer data is different from the speed at which the data is processed. Buffering handles the speed mismatch by storing the data in a buffer till the complete data has arrived and then writing it in a single write operation.

Spool (Simultaneous Peripheral Operation On-Line) is a buffer in memory area or disk. Spooling stores the jobs in a spool where the device can access it when it is ready. Spooling is commonly used for printers. Users may give several print commands, and continue working with other operations. However, the printer can print only one job at a time. The rest of the jobs are stored in the spool in a queue, and the printer accesses the spool when it is ready to print the next job.

PROTECTION AND SECURITY

The access of programs, processes, and users, to the resources defined by the computer are controlled by the protection mechanism.

Protection ensures that the resources of the computer are used in a consistent way.

Security mechanism prevents unauthorized access to the computer. Security concerns include—security of software, security of data stored in the computer, and security of physical resources of the computer.

In a personal computer, security can be ensured using—(1) user accounts—individual accounts for each user, (2) user authentication— using password protection, (3) access rights—define rights for access of different kind of information for different people, (4) data encryption—store data in computer in encrypted form, and (5) data backup—storing data on a peripheral device other than the hard disk. In a networked environment, only trusted computers should be able to share data. Some of the common security threats occur due to hacking, viruses etc.

USER INTERFACE

The primary goal of operating system is to make the computer convenient for use by its user. It should allow users to easily access and communicate with the applications and the hardware.

The users can interact with the computer by using mainly two kinds of interfaces—(1) Command Line Interface (CLI), and (2) Graphical User Interface (GUI).

CLI requires the user to interact with operating system in the form of text keyed in from the keyboard. In this, the user has to learn and remember the different commands required for copying, deleting, opening a file or folder etc.

GUI use graphics to display the various commands. The interface consists of icons, menus, windows, and pointers. The user need not learn the commands, instead, the user can give instructions by moving the pointer on the screen using a mouse and pressing the mouse button.

EXAMPLES OF OPERATING SYSTEMS

MS-DOS, Windows family of operating systems, Unix OS, Linux OS, and Mac OS X are some of examples of commonly used OSs. Each operating system has specific characteristics. Here, we will discuss the features of the MS-DOS, Windows family of operating systems and Linux operating system.

MS-DOS

MS-DOS was the first widely-installed operating system for PCs in 1980s.

MS-DOS is easy to load and install. It neither requires much memory for the operating system, nor a very powerful computer to run on.

MS-DOS is a command line user interface operating system. This means that the user has to type single line commands through the command interface. So, user has to remember the different commands and their syntax.

It is a single-user and single-tasking operating system for the PC. Only one user can use it and only one task can be executed, at a given point of time. Also, it does not have a built-in support for networking.

MS-DOS is a 16-bit OS, meaning thereby that it can send or receive 16 bits of data at a time and can process 16 bits of data. It is not able to take the advantage of 32-bit processors.

To use MS-DOS, user must know where the programs and data are stored and how to interact with it. In the MS-DOS command mode, command.com routine interprets the typed in command from the keyboard.

Command prompt is the command line interpreter on the current Windows-based OS. It is similar to command.com in MS-DOS. cmd.exe is a Windows program that acts as a DOS-like command line interpreter.

Windows Family of OS

Windows is a personal computer operating system from Microsoft.

The Windows family of OS which is currently in use includes the Windows 9x family (Windows 95, Windows 98 and Windows 2000), Windows XP, Windows Vista, and Windows 7 operating systems.

Windows family of OS is GUI-based operating system. Since GUI interfaces are easy to use and are user-friendly, these have become very popular.

Windows support multi-tasking. It means Windows OS allows simultaneous execution of multiple tasks.

Windows contains built-in networking, which allows users to share files and applications with each other, if their PCs are connected to a network.

Windows 7 comes in six different editions, Starter, Home Basic, Home Premium, Professional, Enterprise and Ultimate.

With each new version of the Windows OS, the user interface undergoes some changes and the user has to learn to use the new interface. This becomes troublesome for the user.

Windows 8 is the current release of the Windows operating system, produced by Microsoft for use on personal computers, including home and business desktops, laptops, tablets, and home theater PCs. Development of Windows 8 started before the release of its predecessor in 2009. Its existence was first announced at CES 2011, and followed by the release of three pre-release versions from September 2011 to May 2012. The operating system was released to manufacturing on August 1, 2012, and was released for general availability on October 26, 2012.

Windows 8 introduces significant changes to the operating system's platform, primarily focused towards improving its user experience on mobile devices such as tablets to rival other mobile operating systems like Android and iOS taking advantage of new or emerging technologies like USB 3.0, UEFI firmware, near field communications, cloud computing and the low-power ARM architecture, new security features such as malware filtering, built-in antivirus software, and support for secure boot (a UEFI feature which allows operating systems to be digitally signed to prevent malware from altering the boot process), along with other changes and performance improvements.

Windows 8 also introduces a new shell and user interface based on Microsoft's "Metro" design language, featuring a new Start screen with a grid of dynamically updating tiles to represent applications, a new app platform with an emphasis on touch screen input, the new Windows Store to obtain and purchase applications for the system, and the ability to synchronize programs and settings between multiple devices.

Linux OS

Linux is a Unix-like OS. Unix OS has a user interface called shell. The kernel provides interface for the programs to interact with the hardware, and provides services like process management and memory management. The shell interacts with the kernel through the system calls.

Linux was developed by Linus Torvalds in 1992. Linux is copyright under the GNU Public License. Linux is a “free” operating system that is easily available. Since Linux follows the open development model, it is being constantly upgraded by programmers across the globe.

Some organizations offer Linux with add-on features and capabilities. Red Hat, Mandrake, Debian and Novell are the popular vendors of Linux OS.

Tux, the Linux penguin is the official mascot of Linux.

Linux is a command line user interface OS. Linux has GUI interfaces called desktop environments like GNOME and K Desktop Environment (KDE). The GUI interface is convenient for the user to use.

Linux is a 32-bit, multi-tasking OS. It supports multiple users and multiple processors.

Linux is a reliable and secure OS, and is available almost for free. So, Linux is fast becoming very popular and powerful OS.

Linux OS is easily available, such as Redhat Linux ver. 9, and, Debian’s— Ubuntu, Kubuntu, and Edubuntu.

COPPERBELT UNIVERSITY

COMPUTER SCIENCE DEPARTMENT

INTRODUCTION TO COMPUTING

UNIT 8: COMPUTER NETWORKS

INTRODUCTION

The communication process involves—sender of information, receiver of information, language used for communication, and medium used to establish the communication. Communication between computers also follows a similar process.

This unit discusses the data communication and the computer networks. The section on data communication discusses the media used for transmission of data, how data can be transferred across the communication media and the relationship between data transmission and data networking.

The section on computer network discusses different network types, network topologies, communication protocol and network communicating devices. A brief explanation of wireless networks is also provided.

Importance of Networking

Networking of computers provides a communication link between the users, and provides access to information. Networking of computers has several uses, described as follows:

Resource Sharing In an organization, resources such as printers, fax machines and scanners are generally not required by each person at all times. Moreover, for small organizations it may not be feasible to provide such resources to each individual. Such resources can be made available to different users of the organization on the network. It results in availability of the resource to different users regardless of the physical location of the resource or the user, enhances optimal use of the resource, leads to easy maintenance, and saves cost too.

Sharing of Information In addition to the sharing of physical resources, networking facilitates the sharing of information. Information stored on networked computers located at same or different physical locations, becomes accessible to the computers connected to the network.

As a Communication Medium Networking helps in sending and receiving of electronic-mail (e-mail) messages from anywhere in the world. Data in the form of text, audio, video and pictures can be sent via e-mail. This allows the users to communicate online in a faster and cost effective manner. Video conferencing is another form of communication made possible via networking. People in distant locations can hold a meeting, and they can hear and see each other simultaneously.

For Back-up and Support Networked computers can be used to take back-up of critical data. In situations where there is a requirement of always-on computer, another computer on the network can take over in case of failure of one computer.

DATA TRANSMISSION MEDIA

The data is sent from one computer to another over a transmission medium. The transmission media can be grouped into guided media, and unguided media.

In the guided media, the data signals are sent along a specific path, through a wire or a cable. Copper wire and optical fibers are the most commonly used guided media. Copper wire transmits data as electric signals. Copper wires offer low resistance to current signal, facilitating signals to travel longer distances. To minimize the effect of external disturbance on the copper wire, two types of wiring is used—(1) Twisted Pair, and (2) Coaxial Pair. Optical fibers transmit data as light signals.

In the unguided media, the data signals are not bounded by a fixed channel to follow. The data signals are transmitted by air. Radio, microwave, and satellite transmissions fall into this category.

Now let's discuss both the guided and the unguided data transmission media.

Twisted Pair

A twisted pair cable consists of four pairs of copper wires coated with an insulating material like plastic or Teflon, twisted together. The twisting of wires reduces electromagnetic interference from external sources.

Twisted pair cabling is often used in data networks for short and medium length connections because of its relatively lower costs compared to optical fiber and coaxial cable.

Twisted pair is of two kinds—Shielded Twisted Pair (STP), and Unshielded Twisted Pair (UTP).

STP cable has an extra layer of metal foil between the twisted pair of copper wires and the outer covering. The metal foil covering provides additional protection from external disturbances.

However, the covering increases the resistance to the signal and thus decreases the length of the cable. STP is costly and is generally used in networks where cables pass closer to devices that cause external disturbances.

UTP is the most commonly used medium for transmission over short distances up to 100m. Out of the four pairs of wires in a UTP cable, only two pairs are used for communication. Figure 1 shows the cross- section of STP and UTP cables.

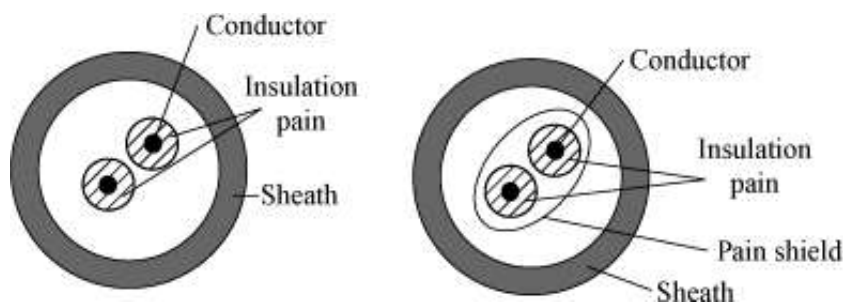


Figure 1 Cross section of (a) UTP (b) STP

UTP cables are defined in different categories. The commonly used UTP cable is the Cat-5 cable which is used with fast Ethernet.

Coaxial Cable

A coaxial cable (Figure 2) has a single inner conductor that transmits electric signals; the outer conductor acts as a ground. The two conductors are separated by insulation. The inner conductor, insulator, and the outer conductor are wrapped in a sheath of Teflon or PVC.

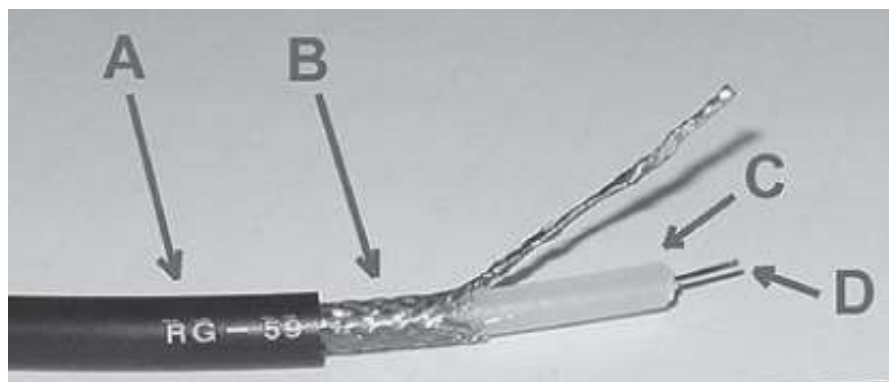


Figure 2 Coaxial cable (A: outer plastic sheath, B: woven copper shield, C: inner dielectric insulator, D: copper core)

The copper wire is used for both inner and outer conductor. The signal is transmitted over the surface of the inner conductor.

In an ideal coaxial cable the electromagnetic field carrying the signal exists only in the space between the inner and outer conductors. This allows coaxial cable runs to be installed next to metal objects such as gutters without the power losses that occur in other transmission lines, and provides protection of the signal from external electromagnetic interference.

A thicker coaxial cable can transmit more data than a thinner one. The commonly used coaxial cable is 10 base 2 that transmits over a distance of 185 m, and 10 base 5 that transmits over a distance of 500 m.

Optical Fiber

Optical fibers are being used for transmission of information over large distances more cost-effectively than the copper wire connection. Communication systems are now unthinkable without fiber optics. Optical fiber transmits data as light signals instead of electric signals.

An optical fiber cable (Figure 3) consists of (1) core—optical fiber conductor (glass) that transmits light, (2) cladding—an optical material that surrounds the core to prevent any light from escaping the core, and (3) jacket—outer covering made of plastic to protect the fiber from damage.



Figure 3 (a) Optical fiber (b) Cross section of optical fiber

Modern optical fiber cables can contain up to a thousand fibers in a single cable, so the performance of optical networks easily accommodate large demands for bandwidth on a point-to-point basis.

Optical fibers come in two types: (a) Single-mode fibers, and (b) Multi-mode fibers

Single-mode fibers have small cores (about 3.5×10^{-4} inches or 9 microns in diameter) and transmit infrared laser light (wavelength = 1,300 to 1,550 nanometers).

Multi-mode fibers have larger cores (about 2.5×10^{-3} inches or 62.5 microns in diameter) and transmit infrared light (wavelength = 850 to 1,300 nm) from Light Emitting Diodes (LEDs).

The Advantages of Optical Fibers over wires are:

- Optical fibers do not cause electrical interference in other cables, since they use light signals.
- Due to much lower attenuation and interference, optical fiber has large advantages over existing copper wire in long-distance and high-demand applications.
- A fiber can carry a pulse of light much farther than a copper wire carrying a signal.
- Optical fiber can carry more information than a wire (light can encode more information than electrical signal).
- A single optical fiber is required for light to travel from one computer to another (two wires are required for electric connection).
- Because signals in optical fibers degrade less, lower-power transmitters can be used instead of the high-voltage electrical transmitters needed for copper wires. Again, this saves your provider and you, money.
- No amplification of the optical signal is needed over distances of hundreds of kilometers. This has greatly reduced the cost of optical networking, particularly over undersea spans where the cost reliability of amplifiers is one of the key factors determining the performance of the whole cable system.
- Optical fibers are ideally suited for carrying digital information, which is especially useful in computer networks.
- They are highly secure as they cannot be tapped and for lack of signal radiation.

The Disadvantages of Optical Fiber are:

- Installing an optical fiber requires special equipment.
- If a fiber breaks, finding the broken location is difficult.
- Repairing a broken optical fiber is difficult and requires special equipment.
- Due to its high installation costs, they are economical when the bandwidth utilization is high.

Radio Transmission

The electromagnetic radio waves that operate at the radio frequency are also used to transmit computer data. This transmission is also known as Radio Frequency (RF) transmission. The computers using RF transmission do not require a direct physical connection like wires or cable. Each computer attaches to an antenna that can both send and receive radio transmission.

Microwave Transmission

Microwave transmission refers to the technique of transmitting information over a microwave link. Microwaves have a higher frequency than radio waves. Microwave transmission can be aimed at a single direction, instead of broadcasting in all directions (like in radio waves). Microwaves can carry more information than radio waves but cannot penetrate metals. Microwaves are used where there is a clear path between the transmitter and the receiver.

Microwave transmission has the advantage of not requiring access to all contiguous land along the path of the system, since it does not need cables. They suffer from the disadvantages: a) needing expensive towers and repeaters, and b) are subject to interference from passing airplanes and rain. Because microwave systems are line-of-sight media, radio towers must be spaced approximately every 42 km along the route.

Satellite Transmission

The communication across longer distances can be provided by combining radio frequency transmission with satellites. Geosynchronous satellites are placed in an orbit synchronized with the rotation of the earth at a distance of 36,000 km above the surface of the earth. Geosynchronous satellites appear to be stationary when viewed from the earth. The satellite consists of transponder that can receive RF signals and transmit them back to the ground at a different angle. A ground station on one side of the ocean transmits signal to the satellite which in turn sends the signal to the ground station on the other side of the ocean.

COMPUTER NETWORK

A computer network is an interconnection of two or more computers that are able to exchange information. The computers may be connected via any data communication link, like copper wires, optical fibers, communication satellites, or radio links. The computers connected to the network may be personal computers or large main frames. The computers in a network may be located in a room, building, city, country, or anywhere in the world.

NETWORK TYPES

Computer network is broadly classified into three types—(1) Local Area Network (LAN), (2) Metropolitan Area Network (MAN), and (3) Wide Area Network (WAN). The different network types are distinguished from each other based on the following characteristics:

- Size of the network
- Transmission Technology
- Networking Topology

The size of the network refers to the area over which the network is spread. Transmission technology refers to the transmission media used to connect computers on the network and the transmission protocols used for connecting. Network topology refers to the arrangement of computers on the network or the shape of the network. The following subsections discuss the three types of networks and their characteristics.

Local Area Network

LAN is a computer network widely used for local communication. LAN connects computers in a small area like a room, building, office or a campus spread up to a few kilometers. They are privately owned networks, with a purpose to share resources and to exchange information.

The computers in a LAN are generally connected using cables. LAN is different from other types of network since they share the network. The different computers connected to a LAN take turns to send data packets over the cables connecting them. This requires coordination of the use of the network. Some of the transmission protocols used in LAN are Ethernet, Token bus, and FDDI ring.

Star, Bus, and Ring are some of the common LAN networking topologies. LAN runs at a speed of 10 Mbps to 100 Mbps and has low delays. A LAN based on WiFi wireless network technology is called Wireless Local Area Network (WLAN).

Metropolitan Area Network

MAN is a computer network spread over a city. Cable television network is an example of MAN. The computers in a MAN are connected using coaxial cables or fiber optic cables. MAN also connects several LAN spread over a city.

Wide Area Network

WAN is a network that connects computers over long distances like cities, countries, continents, or worldwide. WAN uses public, leased, or private communication links to spread over long distances. WAN uses telephone lines, satellite link, and radio link to connect. The need to be able

to connect any number of computers at any number of sites, results in WAN technologies to be different from the LAN technologies. WAN network must be able to grow itself. Internet is a common example of WAN.

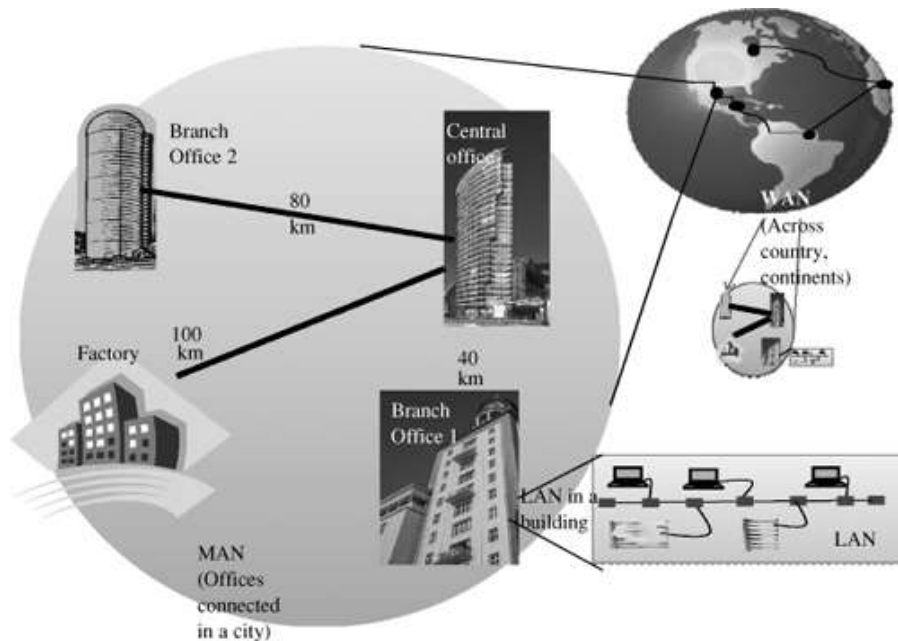


Figure 5 LAN, MAN and WAN

LAN TOPOLOGIES

There are different types of network topologies that are used in a network. The network topologies differ in the structure or the layout of the different devices and computers connected to the network. The three topologies commonly used in LAN are—Bus topology, Star topology, and Ring topology.

Bus Topology

All devices on the network are connected through a central cable called a Bus (Figure 5).

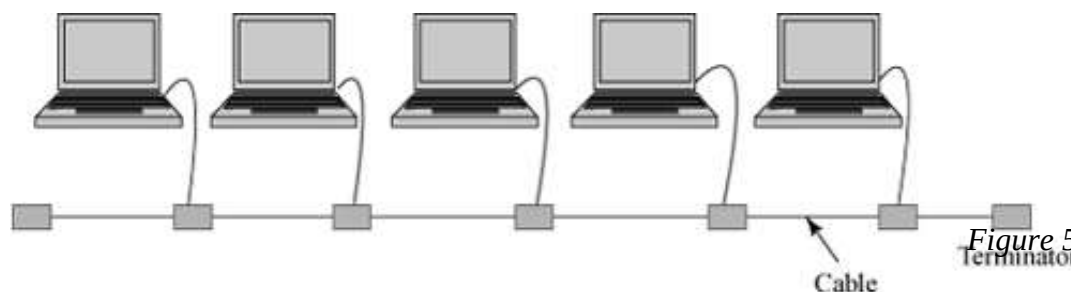


Figure 5 Bus topology

The data signal is available to all computers connected to the bus.

The data signal carries the address of the destination computer.

Each computer on the network checks the destination address as the data signal travels through the bus. The computer whose address matches makes a copy of the signal and converts it into data. The data signal on the bus does not get destroyed and still transmits along the bus, and is finally absorbed by the terminator attached to the end of the network.

It is good for connecting 15–20 computers.

A single coaxial cable is generally used in bus topology, to which the computers or devices are connected.

Ethernet is a commonly used protocol in networks connected by bus topology.

Ring Topology

All devices in the network are connected in the form of a ring.

Each device has a receiver and transmitter to receive the data signals and to send them to the next computer, respectively.

Ring network does not have terminated ends, thus data signals travel in a circle.

Ring topology (Figure 6) uses token passing method to provide access to the devices in the network.

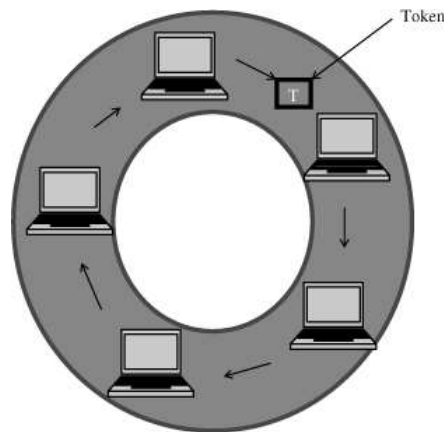


Figure 6 Ring topology

The computers or devices are connected to the ring using twisted pair cables, coaxial cables or optic fibers.

The protocols used to implement ring topology are Token Ring and Fiber Distributed Data Interface (FDDI).

Star Topology

All devices are connected through a central link forming a star-like structure.

The central link is a hub or switch. The computers are connected to the hub or switch using twisted pair cables, coaxial cables or optic fibers.

Star topology (Figure 7) is the most popular topology to connect computer and devices in network.

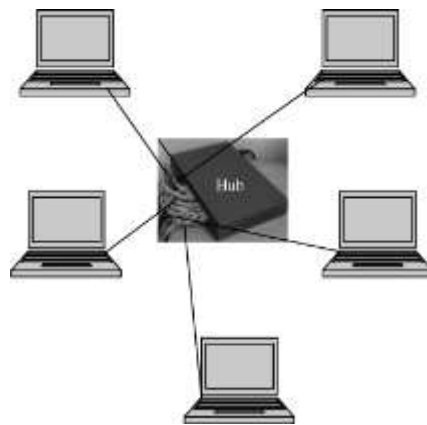


Figure 7 Star topology

The data signal is transmitted from the source computer to the destination computer via the hub or switch.

The common protocols used in star topology are Ethernet, Token Ring, and LocalTalk.

In addition to the bus, ring, and star topologies, there are complex topologies like the tree topology, and the mesh topology used for networking in LAN. Table 1 lists the advantages and disadvantages of the different LAN network topologies.

	BUS	RING	STAR
<u>ADVANTAGES</u>	Easy to implement (computers connected linearly through cable) Easily extendable (new devices can be easily added) Not very expensive	All computers in the ring have equal access to the ring. Each computer in the ring gets an opportunity to transmit data.	Failure of a device attached to the network does not halt the complete network; only that device is down. Easily extendable—by attaching a new device to the hub or switch.

			No disturbance when a new device is added or removed. Easy to troubleshoot the network.
<u>DISADVANTAGES</u>	If the cable gets damaged, the whole network collapses A computer can transmit data only if network is not being utilized Network slows down if additional computers are connected to the network.	Adding or removing devices is difficult and affects the complete network Failure in a node or the cable breaks down the ring and thus the network The length of the ring and the number of nodes are limited	It is costly, since each device on the network is attached by a single cable to the central link. Failure of the hub or switch breaks the complete network

NETWORK DEVICES

The cables are used to transmit data in the form of signals from one computer to another. But cables cannot transmit signals beyond a particular distance. Moreover there is a need to connect multiple computers and devices. A concentrator is a device having two or more ports to which the computers and other devices can be connected. A concentrator has two main functions—(1) it amplifies the signal to restore the original strength of the signal, and (2) it provides an interface to connect multiple computers and devices in a network. Repeater, hub, switch, bridge, and gateway are examples of network connecting devices.

Two or more LANs using different protocols may not be able to communicate with the computers attached to their network. For example, a LAN connected using Ethernet may not be able to communicate with a LAN connected using Token Ring. Bridge, Router, and Gateway are devices used to interconnect LANs.

Network Interface Card

A Network Interface Card (NIC) is a hardware device through which the computer connects to a network.

NIC is an expansion card, it can be either ISA or PCI, or can be on-board integrated on a chipset. NIC has an appropriate connector to connect the cable to it. NIC for different LAN are different (NIC for token ring is different from NIC for Ethernet).

NIC depends upon the configuration of the computer, unlike hub or switches that perform independently.

Repeater

Repeaters are used to extend LAN. It has only two ports and can connect only two segments of a network. Multiple repeaters can be used to connect more segments. (Segment is a logical section of the same network).

They are useful when computers in a network are located far away from each other.

Repeaters amplify the signal so that the signal is as strong as the original signal. They can thus extend the reach of a network.

Repeaters cannot be used if multiple computers need to be interconnected or multiple segments need to be interconnected.

Repeaters cannot identify complete frames. Thus, in addition to the valid transmissions from one segment to another, repeater also propagates any electrical interference occurring on a segment to other segment.

Bridge

Bridge is used to connect two LAN segments like a repeater; it forwards complete and correct frames to the other segment. It does not forward any electrical interference signals to the other segment.

Bridge forwards a copy of the frame to the other segment, only if necessary. If a frame is meant for a computer on the same segment, then bridge does not forward a copy of the frame to other segment.

A bridge is also used to divide a network into separate broadcast domains to reduce network traffic while maintaining connectivity between the computers.

Hub

It is like a repeater with multiple ports. But, hub does not amplify the incoming signal.

Hubs are used to connect multiple segments of the same network.

Hubs are also used to connect computers to network that use Star topology.

The port on the hubs can also be used to connect another hub, switch, bridge or router.

Hubs increase the network traffic because they broadcast data to all the devices connected to all the ports of the hub.

It is preferable to use a hub in a small LAN having about 8-10 computers connected to it.

Switch

Like hub, switch also connects multiple computers in a network or different segments of the same network. A hub simulates a single segment that is shared by all computers attached to it (hub transmits the data to all computers attached to it). In a hub, at most two computers can interact with each other at a given point of time.

However, in a switch each computer attached to a switch has a simulated LAN segment.

A data frame contains the MAC address of the destination computer. A switch receives a signal as a data frame from a source computer on a port, checks the MAC address of the frame, forwards the frame to the port connected to the destination computer having the same MAC addresses, reconverts the frame back into signal and sends to the destination computer. (Switching is a technique that reads the MAC address of the data frame and forwards the data to the appropriate port). Switches, thus, regenerate the signals.

Since a switch does not broadcast data, but sends the data from the source computer to the destination computer, a half of the computers attached to the switch can send data at the same time.

Switch is also referred to as a multi-port bridge. In general, bridges are used to extend the distance of the network, and switches are primarily used for their filtering capabilities to create a multiple and smaller virtual LAN (a LAN segment can be connected to each port of the switch) from a single large LAN.

Router

Router is used to connect heterogeneous networks.

A router has a processor, memory, and I/O interface for each network to which it connects.

A router connects networks that use different technologies, different media, and physical addressing schemes or frame formats.

A router can connect two LANs, a LAN and a WAN, or two WANs.

A router is used to interconnect the networks in the Internet.

Physically, a router resembles a bridge, but is different from a bridge. A router determines which way is the shortest or fastest in a network, and routes packets accordingly.

Gateway

Gateway is a generic term used to represent devices that connect two dissimilar networks.

It can accept a packet formatted for one protocol and convert it to a packet formatted for another protocol, before forwarding it. An application gateway can translate messages from one format to the other.

A gateway can be implemented in hardware, software, or in both hardware and software. Generally, gateway is implemented by software installed within a router.

WIRELESS NETWORKING

Wireless technology, as the name suggests, is used to establish a wire-free connection or communication between two or more devices. In contrast to the wired technology where data is encoded as electric current and signals travel through wires, in wireless technology data is encoded on electromagnetic waves that travel through air. The wireless technology is used for broadcasting in radio and television communication, for communication using mobile phones and pagers, for connecting components of computers using Bluetooth technology, for Internet connection using Wi-Fi, Wireless LAN, PDA, and in remote controls for television, doors etc.

Wireless network is a computer network connected wirelessly. The communication is done through a wireless media like radio waves, infrared or Bluetooth.

The wireless networks have two main components—the wireless access points that include the transmitter along with the area it can cover, and the wireless clients like mobile handsets, laptops with Ethernet cards etc.

The access point receives data frames from the computers attached to it wirelessly, checks the frames, and transmits them to their destination. The coverage area of a transmitter depends on the output power of the transmitter, its location, and the frequency used to transmit the data. Higher frequencies require a clear line of sight as compared to lower frequencies.

The speed of wireless connection is determined by the distance of the wireless client device from the access point, the obstruction-free path (walls, trees etc.), interference, and the number of users using the network at a given time.

Wireless networks can be divided into three categories based on their use:

- Bluetooth technology to connect the different components of the computer in a room, a small office or home.
- Wireless LAN is used to connect computers and devices wirelessly in a LAN, for example, different computers or devices in an office or campus.
- Wireless WAN is used to connect wide area systems, for example access to Internet via mobile devices like cell phone, PDAs and laptops.

COPPERBELUT UNIVERSITY

COMPUTER SCIENCE DEPARTMENT

INTRODUCTION TO COMPUTING (CS 110)

UNIT 9: INTERNET BASICS

INTRODUCTION

In the past two decades, computers have played a significant role in our everyday life. With the growth in popularity and networking of computers, intercommunication between different computers became easier, which led to evolution of Internet. It is the most talked about computer technology in recent years. For a majority of people, who are unaware of computer technology, Internet depicts some enigmatic computer gizmo, which has something to do with games, music and generally entertainment. However, Internet is a much broader concept than mere entertainment and recreational activities. It is revolutionizing and enhancing the ways of human interaction and communication.

The term 'Internet' is derived from two words—interconnection and networks. Also referred to as 'the Net', Internet is a worldwide system of computer networks, that is, a network of networks, which allows the participants (users) to share information. It consists of thousands of separately administered networks of various sizes and types. Each of these networks comprises tens of thousands of computers. Moreover, the total number of users of the Internet is known to be in billions (approximately 2.5 Billion in 2013 - Wikipedia). This high level of connectivity encourages an unparalleled degree of communication, resource sharing and information access. It is probably the most powerful and important technological advancement since the introduction of the desktop computer.

To benefit from what the net has to offer, a basic understanding of what it is and how it works is helpful.

EVOLUTION OF INTERNET

The foundation of Internet was laid in 1969 by the Department of Defense (DOD) of United States of America. They wanted to create a computer network that could continue to function in the event of a disaster such as a nuclear war. Even if a part of the network was damaged or destroyed, the rest of the system would continue to work. That network was known as ARPANET (Advanced Research Projects Agency Network), which linked US scientific and academic researchers. It was the forerunner of today's Internet.

Later in 1980, another agency, the National Science Foundation (NSF) created a new network of computers based on ARPANET, called NSFNET, which turned out to be more efficient and capable. Initially, NSFNET was designed to link five super computers situated at the major universities of NSF and allowed only academic research. Over the time, this network expanded to include sites for business, universities, government and so on, and finally becoming a network consisting of millions of computers, now known as the Internet.

In the 1980s, Usenet newsgroups and electronic mail (e-mail) came into picture. To keep track of the information on the Internet, indices such as Archie and the wide area information server (WAIS) were also created in this decade. To give users an easy-to-use interface to work with, the University of Minnesota created Gopher, a simple menu-driven system for accessing files. The Internet really became popular in the 1990s after the development of the World Wide Web (WWW). Before that, it was open for a handful of sites only. It is amazing that as recently as June 1993, there were only 130 websites, but now there are millions.

With the growth of the Internet, the quality, quantity and variety of information also grew. The Internet today is a repository of every type of information. Nowadays, an Internet user can get all sorts of information ranging from how to add to the design of a functional spaceship to choosing a product for personal use.

Internet Organizations and Committees

Now that you are aware of how the Internet evolved, you may ask that who owns the Internet. Well the answer is 'nobody'. The Internet is a loosely organized international collaboration of autonomous, interconnected networks, which supports host-to-host communication through voluntary adherence to open protocols and procedures defined by Internet standards. Nobody really owns or controls the Internet. However, there are some governing bodies overseeing the technical aspects of the Internet. They define how the Internet works and how new technology is to be implemented. Some of the important governing bodies of Internet are as follows:

- ✧ Internet Society (ISOC): It is concerned with the growth and evolution of the Internet and the way in which the Internet is and can be used.
- ✧ Internet Architecture Board (IAB): It is a technical advisory group of the ISOC and is chartered to provide oversight of the architecture of the Internet and its protocols.
- ✧ Internet Engineering Task Force (IETF): It is a self-organized group of people who provides technical and other contributions to the engineering and evolution of the Internet and its technologies.
- ✧ Internet Engineering Steering Group (IESG): This group is responsible for the technical management of IETF activities and the Internet standards' development process.
- ✧ Internet Research Task Force (IRTF): Its main purpose is to create research groups that focus on Internet protocols, applications, architecture and technology.

- ✎ Internet Assigned Number Authority (IANA): The task of IANA is to assign protocol parameters such as Internet addresses, domain names and protocol numbers for the Internet.
- ✎ Internet Network Information Center (InterNIC): It is a collaborative activity of AT&T, Network Solutions Inc. and National Science Foundation (NSF). It provides directory and database services, the Internet white pages and domain name registration.
- ✎ World Wide Web Consortium (W3C): It was founded in 1994 to develop common protocols for the evolution of the World Wide Web.

BASIC INTERNET TERMS

World Wide Web (WWW)

The World Wide Web (abbreviated as the Web or WWW) is a collection of linked documents or pages, stored on millions of computers and distributed across the world. The concept of the Web began at the European Center for Nuclear Research (CERN), Geneva, Switzerland, in the year 1989. Since then, WWW is the most popularly used Internet subnetwork. One of the main reasons that led to its popularity is that it provides information in multimedia form, that is, in more than one medium such as, text, graphics, video and audio. Further, it provides a simple and consistent way of accessing the information available on the Internet by using hypertext system. In hypertext system, the documents are connected to other related documents on the Internet through links. The Web uses a specific Internet protocol called HTTP to support hypertext documents.

NOTE: Although the WWW is often referred to as the Internet, they are actually two different concepts. The WWW is one of the most popular services available on the Internet.

Web Page

The hypertext documents on the World Wide Web are known as web pages. A web page is written in a language called HTML (Hypertext Markup Language), which enables to embed hypertext links (discussed later in this section) in the document. Using these hyperlinks, user can jump from one web page to another.

NOTE: Web pages are also known as HTML documents.

Website

A website is a set of related web pages (linked through hypertext links), published by an organization or an individual. Typically, a website contains a home page along with other additional web pages. Each website is accessed by its own address known as URL (Uniform Resource Locator). All the websites on the Internet constitute the World Wide Web.

Home Page

When you browse the World Wide Web, you will often see the term 'Home Page'. A home page (also called index page) is the starting point or a doorway to the website; it refers to the web page that identifies a website and contains the hyperlinks to other web pages in the website. It may contain links to other websites also. Like the table of contents of a book, the home page usually provides an overview of what could be found at the website. If there is not much information, the home page may be the only page of the website. However, usually you will find at least a few other pages.

Browser

A browser (short for web browser) is a special software that enables the users to read/view web pages and jump from one web page to another. It is the software that is needed to find, retrieve, view and send information over the Internet. The most popular browsers are Microsoft Internet Explorer, Mozilla Firefox, Google Chrome, Safari and Opera.

Uniform Resource Locator

Each web page has a unique address, called a URL that identifies its location on the Internet. Usually, the format of a URL consists of four parts—protocol, name of the web server (or domain name), path and filename, for example, <http://www.xyz.com/tutor/start/main.htm>. The structure of this URL is as follows:

- Protocol: http
- Web Server Name/Domain Name: www.xyz.com
- Path: tutor/start/
- File Name: main.htm

The first part of the address, the part before the colon, is the protocol. Most of the time, you see http (hypertext transfer protocol) for accessing a web page. Apart from http, you could also find other protocols such as ftp (file transfer protocol), news (news server) and telnet (for accessing remote computers). After the protocol, comes the domain name. The protocol and the domain name are separated by colons and slashes (<http://www.xyz.com>)

A domain name is a unique name that identifies a particular website and represents the name of the server where the web pages reside. They are the human-readable addresses and hence are easier to remember. Domain names have two or more parts separated by dots. The letters occurring after the dot (.) identify the domain type. A domain type represents the type of organization or countries to which the web server (sometimes called host computer) belongs. Table 14.1 describes some of the domain types that are typical on the Internet.

In addition, dozens of domain types have been assigned to identify and locate files stored on the host computers in countries across the world. These are referred to as two-letter Internet country codes, and have been standardized by the International Standards Organisation (ISO). For example, uk for the United Kingdom, de for Germany, au for Australia, and in for India. The last part of a URL consists of the path and the file name. Path name specifies the hierarchic location of the said file on the computer. For instance, in <http://www.xyz.com/tutor/start/main.htm>, the file main.htm is located in start, which is a subdirectory of tutor.

Hypertext

The hypertext is a system that provides a simple and consistent way of organizing large data (that includes text, images, pictures and videos) available on the Internet. In hypertext systems, the documents contain links to other related documents on the Internet. These links are called hyperlinks, hotlink or simply links. A hyperlink is used to jump to another part of the same page or to load a different web page. Usually, hypertext links appear in a different colour (typically in blue and underlined) from the rest of the text. When the cursor is moved over a text link or over a graphic link, it will change from an arrow to a hand. The hypertext words often provide a hint about the connected page. When the user returns to a page with a link he/she had already visited, the hypertext words will often be in a different colour (usually magenta), indicating that the link has already been used. However, the user can certainly use that link again.

Domain Types

- Type Description
- com Commercial and for profit organizations
- edu Colleges and universities (education providers)
- gov Federal government agencies
- mil US military sites
- net Internet infrastructure and service providers
- org Miscellaneous and nonprofit organizations

Figure 14.4. Hypertext Links

Hyperlinks are the foundation of the Web and the operation of the Web relies primarily on them as it is a means of information retrieval. Producing hypertext for the Web is accomplished by creating documents with HTML.

14.3.8 Internet Service Provider

An Internet Service Provider (ISP) is an organization that connects its subscribers' computer using modem to the Internet. The connections can be provided by telephone lines, cable lines or wireless connections. To use the Internet connection, the user has to first log on to the host computer using the username and password provided by the ISP. Some of the well-known ISPs are America Online (AOL), Microsoft Network (MSN), Yahoo! and so on.

Figure 14.5. Data Transfer from Web Server to Browser

While choosing an ISP, many factors should be kept in mind for subscribing. First, the user should check the speed and consistency of the Internet access. One should also check the reliability of ISP, that is, whether the ISP provides consistent access to the Internet or does it have frequent down times. Apart from these two criteria, the price of the ISP and other additional services provided by it should also be considered. For example, many ISPs provide

free web space so that users can create their own website, free e-mail

address and much more. Last but not the least, the user should also check whether the ISP provides a round-the-clock technical assistance.

14.3.9 Web Server

A server is a computer equipped with server software, which provides a specific kind of services to client software running on other computers. It can refer to a particular piece of software, for example, a web server, or to a machine on which the server software is running. A web server receives request from the clients, processes this request and sends the result back to the client. In other words, web server fulfils the requests of the client (browser).

To view a website, the browser sends a request to the server. On receiving the request, the server sends the appropriate web page to the client's machine. The client's machine (browser) receives the information in the form of HTML commands. The browser interprets the HTML, finds all the pictures (or other types of media) and displays the information onto the user's screen (Figure 14.5).

14.3.10 Download and Upload

Download refers to the act of transmitting data from a remote computer on the Internet or other network to one's own computer. When the user downloads a file, he/she is actually copying a particular file from the remote computer and placing it on the hard drive. Downloading allows a user to save file for later use thereby reduces the time spent online and the corresponding charges.

Upload is just the opposite of download. Upload refers to the act of transmitting data from local computer to any other computer on the Internet or network. For example, a software company may upload a demonstration of its new software onto the web so that the users can then download the software.

14.3.11 Online and Offline

The term online is commonly referred to the state of being connected to the networked computer system or the Internet. For example, if you are browsing the WWW, your computer is said to be online. Being online also refers to any peripheral device (like printer) that is connected to the system and ready to use.

Offline is just the opposite of online; it refers to the state of not being

connected to the remote computer or the Internet. A printer or other peripheral that is not ready to use is also considered offline.

GETTING CONNECTED TO INTERNET

Now, you are aware of the basic Internet terms, and let us shift our focus on how to connect to the Internet. The basic requirements for getting connected to the Internet are discussed in this section.

The Computer

The computer must have at least 386-microprocessor chip with a minimum of 16 MB of RAM. For a good browsing experience, use a faster chip (Pentium IV, Intel Core i3 etc) with more RAM (1GB or greater). In addition, a color monitor, with at least 640×480 resolution and a capability of displaying a minimum of 256 colors, is also vital. The system should also possess a hard disk, with at least 200 MB of free space to store Internet software and temporary Internet files. Since the web is a multimedia, you can also enjoy sound on your system with a sound card and a pair of speakers.

Modem

Apart from the previously mentioned configuration, a computer system must have a device called modem, which enables a computer to transmit data over telephone lines. A modem (i.e. modulator-demodulator) is a hardware, which converts digital data into analog signals (i.e. modulation) that can be sent over an analog telephone line and convert the analog signal back into digital data (i.e. demodulation). Thus, when the modem receives the data from a web server via phone system, it reconstitutes the analog signal into digital form so that the computer can understand it. The most critical aspect of the modem is its speed of operation. The speed of modem is measured in kilobits per second (Kbps). Modern modem supports 56 - 128 Kbps speeds. Modems are of two types:

- ✎ Internal Modem is a card, which is fitted inside the computer with a lead running directly from the computer to the phone socket.
- ✎ External Modem is a small external box wired between the computer and the phone socket.

Internet Connection

Before connecting to the Internet, you need to buy an Internet connection from an authorized Internet Service Provider (ISP). The speed of accessing the Internet entirely depends upon the type of connection you bought from the ISP. Some of connections available for the users these days are briefed as follows:

- ✎ Dial-up: A dial-up connection is the access method that uses telephone lines to connect to the Internet. It is the most common way for the individuals who use home computers to connect to the Internet. To connect to the Internet by using dial up, the user needs to specify a username, a password and a telephone number. After the connection is established, the user can start browsing sites on the Internet.
- ✎ ISDN: It stands for Integrated Services Digital Network and is more common in business and commercial use. It was the first high-speed alternative to regular analog phone modems. ISDN involves the digitization of telephone network so that voice, graphics, text and other data can be provided to users from a single terminal over existing telephone wiring. Digital connections have fewer errors in transmission, which means

speed of downloading graphics, web pages, sound and so on is increased to four times faster than with dial-up modems.

- ✧ Cable Modem: A cable modem connects the user to the Internet through a cable television line. Nowadays, many cable companies offer Internet along with television accessibility. A cable modem will typically have two connections, one to the television outlet and the other to the computer. Cable modems not only provide a faster Internet access (10–100 times as fast as the dial-up modem), but interactivity to the television also.
- ✧ Leased Line: This facility provides reliable, high-speed Internet access ranging from 2.4 Kbps to 45 Mbps. A leased line connection is an affordable way to link two or more sites for a fixed monthly charge. Usually, leased lines facility can be provided via fibre optics or copper lines. Leased line service provides a consistent amount of bandwidth and is an excellent way to provide data, voice and video links between the sites.
- ✧ DSL: Digital Subscriber Line (DSL) service is provided through an existing phone line, but it works differently than regular analog modem dial-up access. DSL operates over normal telephone lines and it can be used simultaneously with the telephone. DSL can increase the connection speed by as much as tenfold from a standard dial-up modem.
- ✧ Broadband: This type of access is good for remote locations, where ISDN, cable or DSL are not available. It gives a decent download speed, but to upload data, the user still needs a regular analog modem to dial in, via a phone line. Satellite connection can be either a two-way service or a one-way service. In case of two-way satellite service, the data are transmitted via satellite to a dish antenna at the user's house. In one-way system, the user needs a conventional modem and telephone link to an ISP. Satellite connection is expensive, but sometimes is the only fast option for people who are beyond the service area of cable and DSL providers.

Before deciding the type of Internet connectivity, one should first decide the requirements. If the user wants the Internet mainly for sending e-mail, occasional chats infrequent browsing, then one should work with regular modem dial-up access. In this case, user should invest in a good quality 56 Kbps modem and a good ISP.

If the user is using the Internet frequently for research, downloading or uploading a fair amount of data, to play multiplayer games or to use the multimedia capabilities of the Internet like live audio or video streaming, then one should look into other high-speed accesses such as cable modem or ISDN.

Internet Software

To connect to the Internet, a computer requires mainly three different kinds of software.

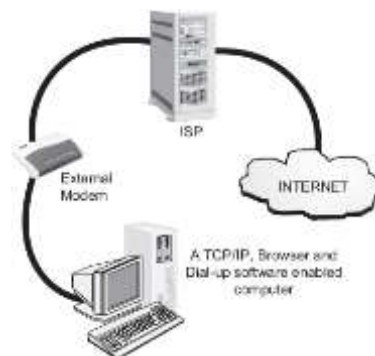
- ✧ TCP/IP: TCP/IP, or Transmission Control Protocol/Internet Protocol, is the basic communication protocol of the Internet. It allows programs on the user's computer to communicate over the Internet. Usually, when you are set up with direct access to the

Internet, your computer is provided with a copy of the TCP/IP. Similarly, every other computer that you may send messages to or get information from also has a copy of TCP/IP.

- ✎ **Dialer Software:** This software is provided by the ISP to instruct the modem to dial the phone number and to identify the user's machine to the access provider's system for access to the network.
- ✎ **Browser:** To use the Internet, a web browser is essential, which allows the user to view the information available on the World Wide Web.

INTERNET APPLICATIONS

Today, the Internet has grown from a small network to biggest network of computers. Due to this colossal growth, the Internet has become the biggest domain of information. Nowadays, an Internet user has access to a wide variety of services such as electronic mail, file transfer, vast information resources, interest group membership, interactive collaboration, multimedia displays, real-time broadcasting, shopping opportunities and much more. Some of the important services provided by the Internet are briefed in the following few sections.



Electronic Mail

Electronic mail (e-mail) is a fast, easy and inexpensive way of communicating with other Internet users around the world. It is one of the basic and earliest services of the Internet and the most used application on the Internet too. E-Mail overcomes most of the problems and delays of getting a physical document from one person to another. Rather, it has the dual advantage of not only being faster, but also cheaper.

Each user of e-mail has a mailbox address to which messages are sent and the mailbox can be accessed from anywhere and at anytime. It does not matter what type of connectivity the user is having, he will always be able to use this service. One can receive and send e-mails from all systems whether it is a Windows PC or a UNIX machine. Besides, e-mail can also be used to send documents, images, audio, and video and so on as an attachment along with the mail.

File Transfer Protocol

The file transfer protocol FTP is a set of rules that enables a user to transfer files from one system to another. It was the first service developed for the Internet so that government and

educational institutions could easily exchange files. It allows the user to get access to the files stored in the directory of a remote computer that is connected to the Internet. Using FTP, one can upload and download files from the remote computer (known as FTP servers), if one has access permission on the remote machine. Although files can be transferred using e-mail, this is not a good choice if the file size is large or when there are many files. FTP sites consists of hundreds to thousands of files with information on books, music, software, games, images, sounds, multimedia, course ware and so on.

Telnet

The term 'telnet' is derived from 'telecommunications and network' and is a protocol that allows a user to log on to a remote computer. Telnet is also known as remote login, which means connecting one machine to another in such a way that a person may interact with another machine as if it is being used locally. It means that someone in New Delhi could connect to a computer in the CBU Library and search the card catalogue the same way as someone located at a terminal in the library. Once connected, the user's computer emulates the remote computer.

When the user types in commands, they are executed on the remote computer. The user's monitor displays what is taking place on the remote computer during the telnet session. The user's computer, which initiates the connection, is referred to as the local computer or telnet client, and the machine being connected to, which accepts the connection, is referred to as the remote computer or telnet server. Note that the telnet client computer must run a telnet client application and the telnet server must run a telnet server application. Moreover, the user's machine has to be connected to the Internet or to a network to use telnet. In addition, to transmit information between the telnet client and the telnet server, the TCP/IP protocol is used.

Internet Relay Chat

Internet Relay Chat (IRC) is a service on the Internet that allows people to communicate in real time and carry on conversations via the computer with one or more people. It provides the user with the facility to engage in simultaneous (synchronous) online 'conversations' with other users from anywhere in the world. Unlike other forms of group communication on the Internet, chat is an instant communication. However, both parties must be connected to the Internet and that too at the same time, running the right software and actively participating to chat.

Generally, the user runs a program (called 'client') to connect to an IRC server. The server relays information to and from other servers on the same network. Popular IRC client applications are mIRC, Virc and LeafChat. Once connected to an IRC server on an IRC network, the user will usually join one or more 'channels' (also known as chat rooms) and converse with others there. Each channel is devoted to a different topic. Conversations may be public (where everyone in a channel can see what the user types) or private (messages between only two people, who may or may not be on the same channel).

Chatting and Instant Messaging

Chat programs allow users on the Internet to communicate with each other by typing in real time. They are sometimes included as a feature of a website, where the users can log into chat rooms to exchange comments and information about the topics addressed on the site. For example, America Online is well known for sponsoring a number of topical chat rooms. Chatting has become one of the 'killer applications' of the Internet and has become quite popular with all kinds of Internet users. It is quite cheap to chat online. E-mail is also a cheap source of communication, but it is not done on 'real time' and the response to the message solely depends on the recipient. Note that even though chatting is based on standardized IRC system, but unlike IRC, the user does not need to have any special software to connect to any chat rooms.

A variation of chat is instant messaging where a user on the Web can contact another user currently logged in and type a conversation. To avail this Internet service, the user must have software called instant messenger installed on the system. Instant messaging is like making a text-only phone call over the Internet, that is, the user is 'talking' to the other person (possibly more) instantly. There are various instant messaging products such as MSN Messenger and Yahoo Messenger.

Internet Telephony

Internet telephony is the use of the Internet rather than the traditional telephone company infrastructure, to exchange spoken or other telephonic information. It consists of hardware and software that enable people to use the Internet as a transmission medium for telephone calls. There are many Internet telephony applications available. Some applications such as CoolTalk and NetMeeting come bundled with popular web browsers.

The required hardware for the Internet telephony generally consists of end devices (either traditional telephones or audio-equipped personal computers) and gatekeepers that provide call admission control, bandwidth management, addresses translation, authentication and user location. They communicate with each other using the voice over Internet Protocol (VoIP). For the cost of a local connection to their service providers, users can call anywhere in the world. Note that there is no single Internet phone standard at this time. Therefore, participants need to use the same software at each end of the connection.

Video Conferencing

Video conferencing uses the same technology as IRC, but also provides sound and video pictures. It enables direct face-to-face communication across the networks. A video conferencing system has two or more parties in different locations, which have the ability to communicate using a combination of video, audio and data. A video conference can be person to person (referred to as point-to-point) or can involve more than two people (referred to as multipoint) and the video conferencing terminals are often referred to as endpoints.

In this form of meeting, participants in remote locations can view each other and carry on discussions via web cameras, microphones and other communication tools. The following five elements are common to all video conferencing endpoints:

- ✎ Web Camera: It captures live images to send across the network.
- ✎ Visual Display: It displays the images of the people taking part in the video conference.
- ✎ Audio System: It includes microphones to capture audio from the endpoint and loudspeakers to play back the audio received from other endpoints across the network connection.
- ✎ Compression: Videos are highly bandwidth-intensive and they take a long time to load. Therefore, video systems include technologies, often referred to as codecs, to compress and decompress video and audio data, allowing transmission across a network connection in near-real time.
- ✎ User Interface and Control System: The user interface allows the users to control interactions, for example, placing calls, storing and locating numbers, and adjusting environment settings such as volume. The control system handles the underlying communication that takes place between the endpoints.

Video conferencing has many benefits, as a tool for both teaching and learning. A key factor is that it provides real-time, visual communication, unlike other communication methods such as e-mail. Video conferencing technology is still in its infancy, and one of its major limitations is the bandwidth (the volume of information per unit time that a computer or transmission medium can handle) available on the Internet. As the protocols and applications for video conferencing develop higher resolution and improved speed, participation will increase.

Commerce through Internet

Today, business is taking place through electronic telecommunication media. Nowadays, various organizations offer the facility of shopping online. This type of business model is known as electronic commerce or E-commerce. E-commerce refers to buying and selling goods and services online. It integrates communication, data management and security capabilities, which allow organizations to exchange information about the sales of goods and services. Essentially, it is the end-to-end digital exchange of information needed to conduct business including electronic data interchange (EDI) and electronic funds transfer (EFT). EDI is the computer-to-computer exchange of routine business transactions including payments, information exchange and purchase-order requests. EFT is a technology that allows the transfer of funds from the bank account of one person or organization to that of another.

While discussing about e-commerce, we normally talk about virtual shopping malls, up-front stores where the user virtually visits and selects the product(s) of his/her choice. After selecting the desired product(s), the payment for the purchase is done. Online payment can be done through credit card. Techniques for cash transactions have also been developed in the form of Internet money such as Cyber cash and Digital money. After transfer of money, the items are

received at the doorstep, which means delivery of the product is done at the address desired by the buyer.

Through the development of e-commerce, the Internet has established an electronic global worldwide market that operates 24 hours and 365 days a year. Organizations and individuals are able to market goods and services, while purchasing is supported by secure facilities such as digital signatures (the process that operates on a message to assure message source authenticity and integrity) and encryption (a method of ensuring data secrecy). Developments in software mean that an increased number of facilities are becoming available all the time. Some of the well-known e-commerce sites are www.amazon.com and www.ebay.com.

A Simple E-commerce Structure

The increased use of mobile devices such as personal digital assistants (PDAs) and mobile phones has opened a new horizon of proliferation in the electronic market and has coined a new term 'mobile commerce'. Mobile commerce or M-commerce refers to transactions through a mobile phone network and data connection that result in the transfer of value (monetary or otherwise) in exchange for goods and services. Mobile commerce has evolved as a new and simpler concept to operate financial transaction through mobile phones and it includes services such as banking, payment and ticketing. M-commerce users may view and interact with information that appears through SMS (Short Message Service), WAP (Wireless Application Protocol) or a standard HTML browser on a PDA, web-enabled mobile phone or laptop computer connected to a mobile phone on a wireless network.

Newsgroups (Usenet)

Newsgroups are international discussion groups that focus on a particular topic and helps in gathering information about that topic. The topics discussed here cover all the fields such as politics, computers, technology and many more. The information or articles that make up the 'news' are written by people interested in a specific topic. These articles are posted to the newsgroup so that others can read, reply and comment on them. To read and post articles in a newsgroup, newsreader software such as Microsoft Outlook

Express or Netscape News is required. Some newsgroups are devoted to current events and headlines. However, newsgroups usually describe topical discussion groups, not 'the news' in the traditional sense. Most newsgroups exist on a network known as Usenet, which contains thousands of newsgroups.

Reading News with Newsreader

Newsgroups provide a source of information and a medium through which the users can ask questions from the Internet community. They provide a forum for users to discuss topics of mutual interest. It is normal to post the rules, restrictions and subject matter for the newsgroup in a message called frequently asked questions (FAQ). To simplify the selection of suitable

newsgroups, they are divided into subject classifications known as top-level categories. These top-level categories are then further subdivided and this continues down the hierarchy.

Mailing Lists (List server)

The Internet is home to a large community who carry out active discussions, organized around topic-oriented forums that are distributed via e-mail. This method of Internet communication is known as mailing list and it enables people with similar interests from all over the world to communicate and share information with each other. The mailing list groups are administered by a software program known as listserver. There are varieties of programs available such as Listserv and Majordomo. A listserver watches for incoming mail on a certain mailbox continuously and forwards any message to a list of other addresses. Such a mailing list can bring people together with a common interest. There are hundreds of mailing lists, each for a different subject. Subscribers can send special messages to a separate address to retrieve files from the archive or to get other information about the mailing list.

When the user subscribes to a listserver, messages from the other subscribers are automatically sent to their electronic mailbox (e-mail address). Note that to participate in a listserver discussion, the user must have an e-mail account. To subscribe to a listserver, the user has to send an E-mail message to the desired listserver located on the computer networks throughout the world.

ELECTRONIC MAIL: AN INTRODUCTION

The Internet was developed mainly to communicate and interchange knowledge irrespective of the distance limitation. Hence, from the very beginning, users used the Internet primarily for sending and receiving messages. Even today, communicating through the Internet is the main application of the Internet. Since these messages are communicated electronically, they are known as electronic mail or E-mail. Thus, electronic mail can be defined as the process of exchanging messages electronically, via a communications network, using the computer. E-mails allow users to communicate with each other in less time and at nominal cost as compared to traditional phone or mail services. Apart from a textual message, e-mails can also consist of other data formats such as pictures, sound and video. E-mails can be sent anywhere in the world using a computer and a modem. Its delivery is almost instant and is very economical to use especially for international messages, costing only the price of the telephone connection to the ISP. You may send many messages at one time or just one to a designated location.

E-mail Address Structure

To use e-mail, one must have access to the Internet and an e-mail account. An e-mail account is a service that allows the user to send and receive e-mails through the Internet. Usually, e-mail accounts are provided as part of ISP's monthly packages. One can also obtain free e-mail accounts such as Yahoo and Hotmail on the Web. An e-mail account provides a unique e-mail address and a mailbox where the user can save all the mails.

Generally, there are two parts of an e-mail address: the logon identity and the identity of the e-mail server. These are separated by the symbol @ (pronounced as ‘at’). For example, a typical e-mail address would look like:

username@yahoo.com
└──┬──┘ ↑ └──┬──┘
Login at Mail Server

The first part of the address indicates the identifiable name of the user. It is just like a home address (and it is unique) so that the mails could be sent to that address. The user name is a name, which is assigned to or desired by the user, while signing up for e-mail address. For example, mayuka@yahoo.com. The @ symbol in the address is used to separate the user name from the rest of the address. Next comes the host name (yahoo), also called the domain name. This refers to the mail server, the computer where the recipient has an electronic mailbox. It is usually the name of a company or organization. The end of the domain name consists of a dot (.) followed by three or more letters (such as .com or .net). This part of the domain name indicates the type of organization or the country where the host server is located. Note that you may find variations in e-mail addresses, but these common elements will always remain the same.

NOTE: There are no spaces in an e-mail address and it is usually (but not always) in lower case letters.

Sending and Receiving E-mail

To send and receive an e-mail, the user must have an e-mail account. In case, you do not have a mail account, you must sign up with e-mail provider. Let us assume that you have an e-mail account. Now, follow the steps given below to check the received e-mails.

1. Open Internet Explorer and type the address of the website on which you have an e-mail account. For example, type www.rediff.com in the address bar and press the Enter key. The home page of the website appears.
2. Click Sign In. A dialog box, which allows the existing users to log on, appears on the home page.
3. Type your email ID and password in the specified fields and click the Go button. By default, the Inbox page of your e-mail account appears which shows the received e-mails. Note that you can view the Inbox page any time by clicking the Check mails.
4. Click on the mail you want to read.
5. After reading the mail, you can reply to the mail or forward the same mail to other persons by clicking the Reply or Forward button, respectively. In case, you do not want to keep the mail in the mailbox, click the Delete button to delete it.

One of the important functions of an e-mail service is to provide a platform to send e-mails. A user can reply to the received e-mail or send a new e-mail.

To send an e-mail, follow the steps given below.

1. Click on the Write Mail link so that you can create and send e-mail to the desired address. A webpage will be displayed. The key elements of the Write Mail page are listed as follows:
 - ✉ To: It denotes to whom the mail is to be sent. The user has to write the recipient's e-mail address.
 - ✉ Cc: Cc or carbon copy is used to specify the addresses of all the recipients who will also receive copies of the same mail. It is used to send the same message to several people (all the addresses are separated with commas).
 - ✉ Bcc: Bcc or blind carbon copy is used to send message to several addresses without showing everyone all the addresses.
 - ✉ Subject: It denotes the subject of the message as specified by the sender.
 - ✉ Attachments: Clicking this link opens another page where the user can attach files (created with programs like MS Word or Zip files) that are to be sent with the e-mail.
2. Type the addresses of the recipients and subject of mail in the corresponding fields and the contents in the text area.
3. Click the Send button.

How E-mail Works

The electronic mailing system works in a similar fashion as the postal service. Just as the postal service uses letters in envelopes with addresses on them, mailboxes and post offices, electronic mailing system completes the same operation with the help of computers (clients and servers).

To send e-mail, one needs a connection to the Internet and access to a mail server, which forwards the mail. The standard protocol used for sending e-mail is called SMTP (Simple Mail Transfer Protocol). It works in conjunction with POP (Post Office Protocol) and IMAP (Internet Mail Access Protocol) servers.

When an e-mail is sent to a person, who has an e-mail address like someone@theirdomain.com, it is broken down into two parts: someone (the recipient's account name) and theirdomain.com (the recipient's domain name). The SMTP server contacts a domain name service (DNS) server and asks for the location of theirdomain.com. The DNS server sends the address back to the SMTP server. The SMTP server then sends the e-mail message to the SMTP server where theirdomain.com is located. This SMTP server delivers the e-mail message to 'someone's' account on the POP or IMAP server. Finally, when 'someone' logs on to the computer and opens the e-mail client, the e-mail client requests the POP or IMAP server to send all mails from the account to 'someone's' computer. Note that if you are sending mail to somebody whose account resides on the same mail server, the SMTP server will simply direct the mail to the local POP or IMAP server, where it will be delivered to the appropriate account.

Advantages and Disadvantages of E-mail

- ✧ E-mail can be sent anywhere in the world to anyone who has an e-mail address. It can take days to send a letter across the country and weeks to go around the world. To save time and money, more and more people are using e-mail. However, the slightest error in the address can deliver the mail to the wrong address. Hence, like there are two faces of a coin, e-mail also comes with its own share of benefits and limitations.
- ✧ Multiple copies of the same message can be sent to a group of people at the same time and can be sent as easily to a single person. Since e-mail passes through a network, it may be intercepted in between.
- ✧ Pictures, documents and other files can also be attached to messages.
- ✧ Electronic mailing system depends on the electricity and telephone system. Thus, failure of any one of them can prevent the user from sending or receiving e-mails.

Netiquette

Communicating online, whether by e-mail, chat rooms or message boards, is informal, quick and easy. However, there are certain unsaid rules that one should follow. On the Web, you are what you write, so take some pride in your work and do it correctly. The standard rules of a polite Internet behavior are called netiquette. Like e-mail and other online communication, the rules of netiquette are short and to the point. Some of these rules are briefed below:

- ✧ Never do anything online that you would not do in real life. Even while communicating online, you are dealing with real people. Therefore, do not say anything over the Internet that you would not say on their faces.
- ✧ Good discussion should always be welcomed, but do not waste other's time with useless queries. For example, if you respond to a listserver to send a private message to one person, you have wasted a lot of other's time. Moreover, before participating in a newsgroup discussion, one should read the FAQs (frequently asked questions).
- ✧ Do not use all upper case letters because using all caps equates to yelling. A single word in uppercase (to emphasize) is acceptable, but use upper case words sparingly.
- ✧ Never send unsolicited junk mail and do not forward junk mails or chain letters.
- ✧ Respect other people's privacy and copyrights. Treat the mails that you receive as confidential unless the sender permits you to share it with others. Moreover, if you are quoting something from an online book or paper, always mention the source.
- ✧ When posting a message to a public bulletin board, forum or newsgroup, stick to the topic. Posting unrelated comments to a newsgroup is known as spamming, which leads to another unpleasant Internet practice, known as flaming. Flaming is the practice of attacking people on a personal level.
- ✧ Respect other people's time and bandwidth. While mailing or posting to a discussion group, you are taking up other's time. So, it is your responsibility to ensure that the time they spend reading your posting is not wasted. Hence, be concise and brief while mailing and posting information online.

- ☞ When someone makes a mistake, whether it is a spelling error, an unnecessarily long answer, be kind about it. If it is a minor error, try to avoid it and even if you feel strongly about it, be polite and point it preferably by private e-mail rather than in public.

SEARCHING THE WEB (SEARCH ENGINES)

The Internet provides access to a wealth of information on countless topics contributed by people throughout the world. However, the Internet is not a library in which all its available items are identified and can be retrieved by a single catalogue. In fact, no one knows how many individual files (could be in billions) reside on the Internet. Hence, to conduct a search on the Internet, a special search tool known as search engines is used.

A search engine searches a database of Internet files collected by a computer program called a wanderer, crawler or spider. It allows the user to enter keywords relating to particular topics and retrieve information about the Internet sites containing those keywords. As such, a search engine consists of four components:

- ☞ Spider: Program that traverses the Web from link to link, identifying and reading pages.
- ☞ Indexing Software: Program that analyses web pages that are downloaded by spiders.
- ☞ Database: Warehouse of the web pages downloaded and processed.
- ☞ Search Engine Mechanism: Software that enables the users to query the index and that usually returns results in relevancy ranked order.

With most search engines, the user fills out a form with the search terms and then asks the search engine to find web pages relevant to those search terms. Some of the well-known search engines are www.google.com, www.altavista.com, www.lycos.com and www.hotbot.com.

Let us assume that you want to search the Web to get information regarding World Wide Web. To do this, first open the search engine's website (such as www.google.com). Now type your keyword(s) in the search box and click the Google Search button or press Enter.

The engine searches its index and generates a page with links to those resources containing some or all of the search terms.

Clicking on any link opens a website or web page in which the searched keyword appears.

Refining the Search

The major search engines allow the user to choose whether to search for the exact typed phrase, all the words in the phrase, any of the words in a phrase and so on. However, a few search engines do not provide such options. In this case, the user can refine the search by adding one or more words or symbols to the search topic.

- ☞ AND: It is used to search for two or more terms on the same page. Type the word AND between the terms (e.g. poverty AND crime) or put a plus sign right before the second term (e.g. poverty+crime).
- ☞ OR: It is used to search for either of two (or more) terms on the same page. Type the word OR between the terms (e.g. college OR university).
- ☞ NOT: It is used to search for pages that include the first term but not the second. Type the word NOT between the terms (e.g. cats NOT dogs) or put a minus sign right before the second term (e.g. cats–dogs).
- ☞ "'": Double quotes are used to search for an exact phrase, for example, 'desktop wallpapers'.
- ☞ "()": Parentheses are used to group parts of the search phrase, for example, type desktop AND (themes OR wallpapers). This returns pages with the words desktop and themes or both the words themes and wallpapers.
- ☞ "*": Asterisk is used to search for various forms of a word, for example, prior* returns pages with the words prior, prioritize, priority and so on.

DATA OVER INTERNET

The data, in terms of Internet, primarily comprises the hypertext documents containing links that connects to other documents or files. These documents may even include animations and multimedia components. Depending upon how information is displayed in the browser, the displayed data can be classified into two categories—static component and dynamic component.

Static Component

Static component includes the information displayed on the browser that does not interact with the user. For example, the text written in the form of a paragraph on the web page is a static component. Producing static component for the web page is accomplished by creating documents in HTML.

Dynamic Component

Dynamic component imparts life to a web page. This component refers to a specification by which a user can interact with a web server. For example, when a user fills a form on the web page, the texts filled in various fields are sent back to the originating point of the web page. This way a user is able to interact with the server hosting that web page. To impart dynamism, a special program (often called scripts) is written in any programming language, including Perl, VBScript, JavaScript and so on. Some of the programming languages used to provide dynamism to a web page include

- ☞ Perl
- ☞ JavaScript and Jscript
- ☞ VBScript
- ☞ Active Server Pages

- ☞ PHP
- ☞ eXtensible Markup Language (XML)
- ☞ Java

INTERNET AND VIRUSES

With the use of the Internet and communications between the computers, viruses are spreading faster than ever. Virus (stands for Vital Information Resources Under Seize) is a program or small code of segment, which is written with a malicious intent. It has the capability to attach itself to existing programs or files and infect them as well as replicate itself without the user's knowledge or permission. The virus instructs the computer to perform a specific task, which is often destructive, for example, deleting important information or crashing the hard drive. However, other kinds of viruses also exist that may only be slightly irritating.

It enters into the computer system from external sources like CD, pen drive or e-mail and executes when an infected program is executed. Further, as an infected computer gets in contact with an uninfected computer (e.g. through computer networks), virus may pass on to the uninfected system and destroy the files.

Types of Computer Viruses

There are many kinds of viruses. The level of destructiveness of viruses varies widely. Some viruses display pictures or messages on the screen periodically. Others erase or destroy programs and data. They can also inflict havoc at different speeds. Some viruses spread through the computer system quickly, but some may take days or even months to activate. Various categories of viruses are:

- ☞ **Boot Sector Virus:** Boot sector virus infects the master boot record of a computer system. This virus either moves the boot record to another sector on the disk or replaces it with the infected one. It then marks that sector as a bad spot on the disk. This type of virus is very difficult to detect since the boot sector is the first program that is loaded when a computer starts. In effect, the boot sector virus takes full control of the infected computer.
- ☞ **File Infecting Virus:** File infecting virus infects files with extensions .com and .exe. This type of virus usually resides inside the memory and infects most of the executable files on a system. The virus replicates by attaching a copy of itself to an uninfected executable program. It then modifies the host programs and subsequently, when the program is executed, it executes along with it. File-infecting virus can only gain control of the computer if the user or the operating system executes a file infected with the virus.
- ☞ **Stealth Virus:** Stealth virus attempts to conceal its presence from the user. It reads system files or system sectors and when some other program requests for information from portions of the disk, it changes back into the correct (unchanged) form. Use of stealth virus is the major reason why most antivirus programs operate best when the system is started (booted) from a known-clean floppy disk. When this happens, the virus does not

gain control over the system and is immediately available to be seen and dealt with. The Stoned Monkey virus is an example of stealth virus. This virus uses 'read stealth' capability and if a user executes a disk-editing utility to examine the main boot record, the user would not find any evidence of infection.

- ✎ **Multipartite Virus:** Multipartite virus infects both boot sectors and executable files, and uses both mechanisms to spread. It is the worst virus of all because it can combine some or all of the stealth techniques along with polymorphism to prevent detection. For example, if a user runs an application infected with a multipartite virus, the virus activates and infects the master boot record of the hard disk. Moreover, the next time the computer starts; the virus gets activated again and starts infecting every program that the user runs. One-half virus is an example of a multipartite virus, which exhibits both stealth and polymorphic behavior.
- ✎ **Polymorphic Virus:** Polymorphic virus, unlike other viruses, consists of static virus program that gets copied from file to file as it propagates. Such virus is difficult to detect because each copy it generates, appears different from the other one. It uses encryption algorithm to multiply new copies of the program. For an encrypted virus to execute, it must decrypt the encrypted portion of itself. When an infected program launches, the virus decryption routine gains control of the computer and decrypts the rest of the virus body so that it can execute normally.
- ✎ **Worm:** It is by far the most quickly spreading and sophisticated viruses. Internet worms exploit e-mail programs and network software to multiply their numbers. Once a system is infected, the worm may look for an address book and mail itself to everyone listed in the book.
- ✎ **Trojan horse:** It is a kind of virus that hides its true intention. It was named after the famous hollow wooden horse used by the Greek hero Ulysses to sneak soldiers into the ancient city of Troy. It can be a threat to the computer software. They are often encountered as e-mail attachments with tempting or intriguing names that compel people to open them.

III-effects of Viruses

As already discussed, viruses are computer programs, which can virtually do anything like any normal program. This includes deleting files, formatting a user's hard drive, overwriting the system BIOS and completely disabling the computer. Some ill-effects that can be caused by viruses are as follows:

- i. Corrupting the data
- ii. Increasing the file size
- iii. Formatting the hard disk
- iv. Slowing down the system
- v. Manipulating the directory contents
- vi. Renaming all files with different names

Remedy of Viruses

Different organizations have different styles of operation. This fact extends to the way they set up their computer networks and operating procedures. If computer operations consist of one or two personal computers used by fewer people, then the need for an elaborate defence system is not of utmost importance. However, if system is large enough to include the worldwide networks used by large corporations, then a detailed and systematic defense system is required.

Using Antivirus Software

In the early days of computer networking, computers were not networked very well, and computer viruses spread extremely slowly. Files were transmitted by means of bulletin board systems (BBSs) or on diskette. As a result, the transmission of infected files was not that fast and easy. However, as the connectivity improved, mostly by the use of computers in the workplace, the scope of virus threat widened. First, there was LAN, and then there was WAN and now the Internet. The extensive use of e-mail has also contributed to the significant rise in the number of virus incidents. As a result, probability of getting infected by a virus today is more than it was a few years ago.

Until the preventive measures for virus were not invented, the only option to get rid of them was to get rid of that valuable file, which was infected. This proved very costly to the companies whose valuable work was destroyed due to viruses. As the computer technology gained heights, these viruses also advanced in causing destruction. Then came a software utility called Antivirus to the rescue. Antivirus is a software utility, which (upon installing on a computer) scans the hard disk for viruses and tries to remove them, if found. Most Antivirus programs include an auto-update feature that enables the program to download profiles of new viruses so that it can check for the new viruses as soon as they are discovered. The most popular Antivirus software available include Norton Antivirus, McAfee Antivirus and Avira AntiVir.

Antivirus software has normally a built-in scanner, which scans all files on the computer hard disk. It looks for changes and activities in computers typical in case of a virus attack. Scanner look for particular types of codes within the programs. The software generally relies on having prior knowledge of the virus. As a result, frequent update to the tools is necessary. The important thing is to be aware of the possibility of an attack to possess a good virus checking software and to have data backups.

COPPERBELT UNIVERSITY

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

CS120

Data Control and Security

Introduction

Security and control have become a critical, although perhaps unappreciated, area of information systems investment. When computer systems fail to run or work as required, firms that depend heavily on computers experience a serious loss of business function. The longer computer systems are down, the more serious the consequences for the firm. Some firms relying on computers to process their critical business transactions might experience a total loss of business function if they lose computer capability for more than a few days. And with so much business now dependent on the Internet and networked systems, firms are more vulnerable than ever to disruption and harm. Security incidents have been growing at a phenomenal rate.

For example, in 2003 corporate networks and home computer systems were overwhelmed by attacks from the SoBig.F worm. SoBig.F disguises itself in e-mail, which, once opened, scans a computer for e-mail addresses and then sends scores of messages to the addresses it has collected using its own built-in sending program. SoBig.F caused millions of dollars in damage in the United States, temporarily disabling freight and computer traffic in Washington, D.C., and overwhelming computer systems with the sheer volume of e-mail.

Companies have very valuable information assets to protect. Systems often house confidential information about individuals' taxes, financial assets, medical records, and job performance reviews. They also may contain information on corporate operations, including trade secrets, new product development plans, and marketing strategies. Government systems may store information on weapons systems, intelligence operations, and military targets. These information assets have tremendous value, and the repercussions can be devastating if they are lost, destroyed, or placed in the wrong hands.

Protection of information resources requires a well-designed set of controls. Computer systems are controlled by a combination of *general controls* and *application controls*. **General controls** govern the design, security, and use of computer programs and the security of data files in general throughout the organization's information technology infrastructure. On the whole, general controls apply to all computerized applications and consist of a combination of hardware, software, and manual procedures that create an

overall control environment. **Application controls** are specific controls unique to each computerized application, such as payroll or order processing. They consist of controls applied from the business functional area of a particular system and from programmed procedures.

GENERAL CONTROLS

General controls include software controls, physical hardware controls, computer operation controls, data security controls, controls over the systems implementation process. The following table describes each type of control:

Type of General Control	Description
Software controls	Monitor the use of system software and prevent unauthorized access of software programs, system software, and computer programs. System software is an important control area because it performs overall control functions for the programs that directly process data and data files.
Hardware controls	Ensure that computer hardware is physically secure and check for equipment malfunction. Computer equipment should be specially protected against fires and extremes of temperature and humidity. Organizations that are dependent on their computers also must make provisions for backup or continued operation to maintain constant service.
Computer operations controls	Oversee the work of the computer department to ensure that programmed procedures are consistently and correctly applied to the storage and processing of data. They include controls over the setup of computer processing jobs and computer operations and backup and recovery procedures for processing that ends abnormally.
Data security controls	Ensure that valuable business data files on either disk or tape are not subject to unauthorized access, change, or destruction while they are in use or in storage.
Implementation controls	Audit the systems development process at various points to ensure that the process is properly controlled and managed. The systems development audit looks for the presence of formal reviews by users and management at various stages of development; the level of user involvement at each stage of implementation; and the use of a formal cost-benefit methodology in establishing system feasibility. The audit should look for the use of controls and quality assurance techniques for program development,

	conversion, and testing and for complete and thorough system, user, and operations documentation.
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APPLICATION CONTROLS

Application controls include both automated and manual procedures that ensure that only authorized data are completely and accurately processed by that application. Application controls can be classified as

1. input controls
2. processing controls, and
3. output controls.

Input controls check data for accuracy and completeness when they enter the system. There are specific input controls for input authorization, data conversion, data editing, and error handling. **Processing controls** establish that data are complete and accurate during updating. Run control totals, computer matching, and programmed edit checks are used as processing controls. **Output controls** ensure that the results of computer processing are accurate, complete, and properly distributed. The following table provides more detailed examples of each type of application control:

Name of Control	Type of Application Control	Description
Control totals	Input, processing	Totals established beforehand for input and processing transactions. These totals can range from a simple document count to totals for quantity fields, such as total sales amount (for a batch of transactions). Computer programs count the totals from transactions input or processed.
Edit checks	Input	Programmed routines that can be performed to edit input data for errors before they are processed. Transactions that do not meet edit criteria are rejected. For example, data might be checked to make sure they are in the right format (for instance, a nine-digit social security number should not contain any alphanumeric characters).
Computer matching	Input, processing	Matches input data with information held on master or suspense files and notes unmatched items for investigation. For example, a matching program might match employee time cards with a payroll master file and report missing or duplicate time cards.
Run control totals	Processing, output	Balance the total of transactions processed with total number of transactions input or output.
Report distribution logs	Output	Documentation specifying that authorized recipients have received their reports, cheques, or other critical documents.

NOTE

Not all of the application controls discussed here are used in every information system. Some systems require more of these controls than others, depending on the importance of the data and the nature of the application.

Control, therefore, applies throughout the process of data processing. One important aspect of control is the control of data to prevent its loss, misuse or disclosure. This type of control is called “data security”.

DATA CONTROL

Data to be processed by the computer must be presented to it in *machine-sensible* form. Therein lies the basic problem since much data originates in a form which is far from machine sensible. Thus a painful error-prone process of transcription must be undergone before the data is suitable for input to the computer.

INPUT CONTROLS

Murphy's Law: *'If something can go wrong, it will go wrong.'*

Input designs are aimed at reducing the chance of mistakes or errors during data entry. However, the analyst must always assume that errors will occur.

The term given to methods aimed at detecting errors in input is input validation. Three main categories of these methods are concerned with:

1. Checking the transaction;
2. Checking the transaction data; and
3. Changing or modifying the transaction data.

CHECKING THE TRANSACTION

First and foremost, it is essential to identify any transactions that are not valid, that is, not acceptable. Transactions can be invalid because they are **incomplete**, **unauthorized**, or even **out of order**.

TRANSACTION VALIDATION

Transaction validation is the examination of input from, especially, a remote site to determine if it is acceptable for processing on the system. A transaction can be new data to be stored in the system, data to update an existing record, or a request to retrieve data (print a report or display answers to an inquiry) from the system. Users may sometimes submit invalid or unacceptable transactions for processing. It is the responsibility of the analyst to specify validation procedures that test the acceptability of a transaction.

The steps the system takes to ensure that the transaction is acceptable are called transaction validation. The system must verify that it is capable of processing the request. For example, if a user enters a request to retrieve sales data while using a personnel system, the transaction is invalid. However, unless the system is designed to detect that this is an invalid request, refuse to process it, and alert the user through an error message, the system could crash. Crashing occurs when the program attempts to process a request that it has not anticipated, that is, an undefined operation. Unless the design specifies how to handle undefined operations, all processing will stop and the

system will have to be restarted (with the likelihood that there will be loss of data in memory at the time the system crashed). Transaction validation is performed in conjunction with user identification procedures. Several different levels of user identification are needed to fully protect a system from accidental loss of data and unauthorized use.

As the user signs onto the system, the first level of identification takes place. This is done by provision of an individual password that uniquely identifies them or a general password that all authorized users should know. (The general password is changed periodically by supervisors to protect against unauthorized users who accidentally learn it.) In some systems, it is also necessary to provide an approved account number against which usage costs are to be charged.

Even if an authorized user is signed onto the system and attempting to submit a transaction the system is able to process, the combination of user and transaction may not be valid. In other words, systems analysts can specify additional levels of protection that require individuals to prove they have authorization. Analysts sometimes design systems so that the user is automatically signed off if an invalid request is made. However, this is an extreme action, since honest mistakes in keying (typing errors) will occur. A better approach is allowing individuals to reenter the password a second or third time and, if still incorrect, to then terminate processing.

The process of checking the transaction has the following parts:

1. Ensuring the transaction is valid;
2. Ensuring the data are valid; and
3. Ensuring the data are not invalid due to error.

SEQUENCE TEST

Sequence tests use codes in the data (serial numbers) to test for either of two different conditions depending on the characteristics of the application. In some systems, the order of transactions is important. For example, when processing bank cheque deposits and withdrawals, it is important to ensure that each is processed in the order in which it arrived. If a series of withdrawals is mistakenly processed before a deposit that actually occurred first, the customer could be penalized for overdrawing the account when in fact that did not actually happen.

Sequence tests also point out missing items. Cheque numbers are used to identify every cheque. When reconciling an account at the end of a month or quarter, it is usual to sort the cheques into order based on cheque numbers.

COMPLETENESS TEST

Point-of-sale (POS) systems in retail centers are oriented to automatically performing completeness tests. Many POS terminals use systems of lights that instruct the operators

what to do next. The light under the key for transaction type lights up first. Until a valid transaction code (such as sale, void, or credit) is entered, nothing else can happen; the keyboard locks. Following entry of a valid transaction code, the operator is guided through the transaction by a system of lights under the keys that shows which key to depress next.

Completeness tests are one more way of validating the transaction to ensure that it is accurate and acceptable.

CHECKING THE TRANSACTION DATA

Even valid transactions can contain invalid data. Therefore, analysts should be sure to specify methods for validating the data when developing input procedures. There are four data validation methods. These are:

1. Existence tests;
2. Limit and Range tests;
3. Combination test; and
4. Duplicate processing.

EXISTENCE TEST

Some data fields in transactions are designed to not be left empty or blank. Existence tests examine those essential fields to determine that they contain data. In a file maintenance processing program, for example, some items, for instance a key field, should always exist, as should a code indicating the type of transaction (add, delete, or change).

It is the responsibility of the analyst, working with the users, to learn when data items must be present and when their absence is acceptable. This information belongs in the design specification and should be passed along to the programmers.

LIMIT AND RANGE TESTS

These tests verify the reasonableness of the transaction. They can also be used to verify the result of processing. Limit tests validate either the minimum or maximum amount acceptable for an item. Range tests validate both minimum and maximum values.

In most banks, tellers must notify a bank officer before completing a cash transaction in excess of a certain amount. Analysts working on the design of banking systems must build the maximum amount figure for cash transactions into their specifications. When unauthorized transactions occur, the system should note the exception and reject the invalid transactions.

Sometimes both upper and lower limits are important.

COMBINATION TEST

Combination tests validate that several data items jointly have acceptable values; that is, the value for one element of data determines whether other data values are correct.

Example

An ordering system for the motor car industry should note that a customer order for a car with factory air-conditioning must also specify other options, such as heavy-duty battery, an oversize radiator, etc. All must be ordered in combination with each other.

DUPLICATE PROCESSING

In especially sensitive areas, it may be necessary to process data more than once, either on different equipment or in different ways. The results are then compared for agreement and accuracy.

In the United States Space programme, which relies heavily on computers for course and direction control, multiple computers process the same data and compare results. Duplicate processing ensures utmost accuracy. (If there is disagreement, there are also specific procedures used to resolve the difference).

CHANGING OR MODIFYING THE TRANSACTION DATA

A third way of validating data involves modifying the data themselves. Two methods are:

1. Automatic correction of errors; and
2. Self-checking digits on key fields.

AUTOMATIC CORRECTION OF ERRORS

Sometimes analysts specify that programs be written to correct errors in the data. This input validation method is used to minimize the number of separate error correction steps or rejections of transactions during processing. This method simply requires the program to detect an error and make the correction automatically.

Example

Data entry personnel keying data into, say, a six-digit numeric field can enter only three digits. Yet the entire field must contain numbers. (A blank is not a number). Rather than have the program reject the transaction because of missing leading zeros (i.e. zeros in the unused positions at the beginning of a data field), it can be designed to insert them automatically. There is no need to insert trailing zeros automatically.

CHECK DIGITS ON KEY FIELDS

Two of the most common errors in handling data occur with data that are captured correctly but entered incorrectly into processing. These are:

1. Transcription errors; and
2. Transposition errors.

Transcription errors occur if data are inadvertently copied incorrectly by the data entry person. E.g. A customer number of 24589 is transcribed incorrectly if it is entered as 24587. With transposition errors two or more digits are reversed so that their positions in the data are incorrect. E.g. entering 24598 instead of 24589.

A special method devised to help detect these errors during computer processing is called the check digit method. This method adds an additional digit to a data element being used for identification purposes. The check digit is added to the original number before that number goes into use. To use check digits with four-digit customer numbers, the check digit is computed and added to the customer number, making it a five-digit number, before it is assigned to any customer.

ILLUSTRATION

Suppose we have a system of four-digit customer numbers and want to assign check digits. The original customer number is 2458. Let's use modular division to develop the check digit ($10 \bmod 5 = 0$, $11 \bmod 5 = 1$, etc.). We select 11 as the divisor, a commonly used one for this method, hence the name Modulus 11 check method.

First we assign weights to each digit, starting from the low order (right) position in the data, using values 2 through 10 (starting over again with 2 if we run out of digits), and moving to the left. The following shows weights of 2, 3, 4, and 5 under each digit in our customer number 2458.

Customer Number:	2	4	5	8
Weights:				<u>5</u>
			<u>4</u>	<u>3</u>
Multiply numbers by weights	10	16	15	16
Sum results:	$10+16+15+16 = 57$			
Divide by modulus number:	$57 \bmod 11 = 2$			
Subtract remainder from modulus number:	$11 - 2 = 9$			
Add check digit to original number:	24589			

Thus the number 9 is the check digit and becomes a permanent part of the customer number when the number is assigned.

TRANSPOSITION ERRORS

When the data are entered for processing (batch or directly in an online system), the program reads the first four digits of the customer number, 2453. The modulus 11 process is carried out. The result of multiplying and adding the digits is 47. $47 \bmod 11 = 3$ which when subtracted from 11 produces a check digit of 8. When the program compares the value 8 with the original digit of 9, it will find that they do not match and can produce an error message.

Although check digits add an additional number to the data they improve the quality of data entering the system by helping to eliminate transposition and transcription errors.

AUDITING

Users often have a tendency to trust systems more than they should, to the extent that they frequently believe the results produced through a computer-based information system without sufficient skepticism. Therefore, the need to ensure that adequate controls are included in the system is an essential step in the selection of software. Auditors must have the ability to validate reports and output and to test the authenticity and accuracy of data and information.

Specific audit and control procedures of interest include these abilities:

1. Trace a transaction through each processing step by having the capability to examine intermediate data values produced during processing.
2. Print selected records and transactions in the system that meet certain criteria (such as a highly active account or a high-balance account) to validate the accuracy and authenticity of both transactions and results.
3. Maintain a constant balance in the system when financial matters are involved and report whether the system is in balance.
4. Produce a detailed journal of all transactions and the effect of the transactions on account balances or master file records.
5. Provide sufficient controls on input, such as batch and transaction controls and counts.

AUDIT TRAIL

In online systems, unlike batch environments, there may not be copies of input source documents to fall back on if the system fails during processing. It is also possible for online users to sign onto a system, alter data stored in the files, and sign off again, without leaving a visible clue as to what happened. Unless the systems analyst develops an audit trail, no such protection exists in online and distributed systems.

An *audit trail* is designed to permit tracing of any input record or process performed on a system back to its original source. One way of accomplishing this is by automatically maintaining a transaction log. The details of each transaction are recorded in a separate transaction file on the system. Before and after images can provide information on how the record was changed. The storage of these details is automatic and invisible to the user, about whom information should also be stored so that it is clear who conducted the transaction. If the system has an internal clock, each transaction is also time-stamped to tell when it occurred. If the need arises to audit a particular record in a file, it is relatively easy to determine who submitted the transaction, when it occurred, what data the transaction contained, and how the database or master file record was modified. In other words, there is a complete trail of the entire transaction and its effect on the system.

Another form of audit trail presumes that storing transaction data on magnetic disk is not fully reliable. For example, in some small business systems, if the system is turned off, perhaps through a power failure, before the data captured on disk during an editing session has been backed up, it will be lost completely. (Some computer systems use disk systems in which the read-write heads drop down to the disk surface when the power is turned off. In these systems, users must remove the disk *before* powering down or they will lose data.)

Printing a copy of the transaction before processing it is one of the best ways to protect against data loss. Then, if anything happens during the online session, a backup copy of the master file can be mounted and the transaction reentered by using the printed transaction list.

Users a long distance away have no way of knowing when a malfunction occurs. As long as they can continue to enter transactions, they have every reason to believe that the system is operating correctly. Therefore, the systems analyst must anticipate these problems for safeguarding the integrity of a system by providing ways to audit its use.

FILE CONTROL AND ON-LINE SECURITY

File control is to ensure suitable storage and to limit access to the files. Files are labeled and stored in a library when not in use. Entry to this library is controlled so that unauthorized persons cannot gain access to the files held.

In any configuration involving terminals, procedures are needed to ensure that access to the system is gained for legitimate purposes only. Passwords or some equivalent should be used to restrict access to the system or parts of it. Dedicated terminals may be used where only certain facilities are needed by the persons who use these terminals. Further control is provided by logging all terminal usage. Possible attempts to use the system illegally can be highlighted by a review of the log together with an exception report on invalid use. At the data file level, write-protect facilities provide yet another layer of control.

OUTPUT CONTROLS

The major concerns in this area are dissemination of output and the verification of the necessary relationships between input and output. Systems should exist, therefore, to ensure that output is distributed to authorized persons only, and that the data contained in that output have been reconciled with the various controls established over input and files.

FILE CONTROLS

As the whole file is passed in each processing run in sequential-file processing, it is comparatively simple to compute file controls by counting the number of records on the file and summing key fields. These totals can then be compared against totals calculated

from the total of transactions input plus the total of the file at the end of the previous processing run. These 'run-to-run' controls afford a very high degree of file integrity, particularly when used in conjunction with a grandfather-father-son method of file version retention.

With direct access files it is uneconomical to read through the entire file after each updating run to sum the key fields. So only the records updated are included in the addition, and the assumption is made that the other records have not changed. Every so often the file must be checked by a complete pass through it to ensure the actual sum of the fields equals the calculated sum. This is normally done each time the file is backed up, when it must be read in its entirety.

SECURITY

Security may be defined simply as **“protection against attack or failure”**.

Computer-based systems are designed to perform particular functions or provide particular services: any loss of security may result in the inability or failure to do these things in the way intended.

Data security is the protection of data. In some situations data security will be concerned with preventing the loss of data eg. in the file security method used during updates. In other situations data security will be concerned with preventing the misuse or unwanted modification of data eg. due to access by unauthorized persons. A third situation is the prevention of disclosure of data to unauthorized persons eg. where the data is important to national security.

Various measures can be taken to ensure all three types of security. The following are some common methods.

1. The use of backup copies of tapes or disks eg. in conjunction with generations of files.
2. Physical prevention eg.
 - (a) Write permit rings (security tag holes on diskettes and magnetic tape which can be covered by tape which are used when there is a need to stop recorded data being accidentally lost).
 - (b) Restricting the access of personnel.
 - (c) Keeping data under lock and key.
3. The use of passwords to prevent unauthorized use of computer terminals or unauthorized access to online files.
4. Constant checks of security.

These methods are considered in more detail in the following sections.

SYSTEM AND FILE ACCESS CONTROLS

CONTROLLED ACCESS TO THE SYSTEM

Most computer crimes can be prevented by denying access to the system to all who should not have access. There are two widely used mechanisms for controlling access to computer systems:

- Identification codes and
- Passwords

The first way is to have each user appropriately authorized to access the computer or network in question. This involves assigning users unique ID, or *identification*, codes that they must use to access the network. The second control mechanism is to issue each user a *password*.

With these controls in place, in order to access a computer system a user must have a valid ID and a password. The password can be suitably encrypted within the computer system so that the user alone creates his/her unique password. As an additional measure of security, users should change their passwords frequently and make sure that the password is not something that can easily be guessed.

To use such a system, you are first prompted for a user ID. If the ID is accepted, you are then prompted for a password. If it matches the recorded password for the ID, then you are allowed access to the system. If the password does not match, you are either disconnected or prompted again for the correct password. Usually, after a limited number of unsuccessful tries, the unauthorized user will be locked off the system.

CONTROLLED ACCESS TO FILES

The second step in creating a secure system is to control access to the files of data themselves. Combinations of passwords and *access rights* are usually used. Access rights provide that only the creator (owner) of a given file or certain restricted classes of users may read it or write to it.

DATA ENCRYPTION

A final security technique is to use data encryption, in which data is processed with a secret key to render it unintelligible except to the receiver of the file who holds the necessary key to de-encrypt the data. Such a scheme is regularly used in military or banking telecommunications environments. This encryption/de-encryption process can be done with either hardware or software, but it can substantially slow the rate of data transfer due to the additional encrypting and de-encrypting task. This technique not only shields the data from unauthorized users, but also carries with it a measure of control of the integrity of the data: if encrypted data has been tampered with, then the de-encrypting process will fail.

SECURITY BACK-UP FACILITIES

Back-up facilities are provided for both data files and programs to ensure that:

- (a) Data files can be reconstructed in the event of master file loss or corruption;
- (b) Application and system software can be reinstated in the event of loss or corruption.

These objectives are achieved by keeping copies of data files and programs in a safe place (which may be an alternative remote site) and in such a way that they can be reconstructed if necessary.

SECURITY POLICY

Firms must develop a coherent corporate policy that takes into account the nature of the risks, the information assets that need protecting, and the procedures and technologies required to address the risks, as well as implementation and auditing mechanisms.

A security policy consists of statements ranking information risks, identifying acceptable security goals, and identifying the mechanisms for achieving these goals. What are the firm's most important information assets? Who generates and controls this information in the firm? What existing security policies are in place to protect the information? What level of risk is management willing to accept for each of these assets? Is it willing, for instance, to lose customer credit data once every 10 years? Or will it build a security edifice for credit card data that can withstand the once-in-a-hundred-year disaster? Management must estimate how much it will cost to achieve this level of acceptable risk.

MALICIOUS SOFTWARE: COMPUTER VIRUSES, TROJAN HORSES, AND SPYWARE

Malicious software programs referred to as **malware** include a variety of threats such as computer viruses, worms, and Trojan horses.

Virus

Virus is a software program that is destructive in nature. Virus programs have the following properties:

- ☐ It can attach itself to other healthy programs.
- ☐ It can replicate itself and thus can spread across a network.
- ☐ It is difficult to trace a virus after it has spread across a network.

Viruses harm the computer in many ways—

- ☐ Corrupt or delete data or files on the computer,
- ☐ Change the functionality of software applications,
- ☐ Use e-mail program to spread itself to other computers,
- ☐ Erase everything on the hard disk, or,
- ☐ Degrade performance of the system by utilizing resources such as memory or disk space.

Virus infects an executable file or program. The virus executes when a program infected with virus is executed or you start a computer from a disk that has infected system files. Once a virus is active, it loads into the computer's memory and may save itself to the hard drive or copies itself to applications or system files on the disk.

However, viruses cannot infect write protected disks or infect written documents. Viruses do not infect an already compressed file. Viruses also do not infect computer hardware; they only infect software.

Viruses are most easily spread by attachments in e-mail messages. Viruses also spread through download on the Internet.

Worms

Worm is self-replicating software that uses network and security holes to replicate itself. A copy of the worm scans the network for another machine that has a specific security hole. It copies itself to the new machine using the security hole, and then starts replicating from there, as well. A worm is however different from a virus. A worm does not modify a program like a virus; it however replicates so much that it consumes the resources of the computer and makes it slow.

Trojan horse

Trojan horse is destructive programs that masquerade as useful programs. The name "Trojan horse" is given because of the Greek soldiers who reached the city of Troy by hiding themselves inside a large wooden horse. The people of the city of Troy themselves pulled the horse inside their city, unaware of the fact that the Greek soldiers were hiding

inside the horse. Similarly, users install Trojan horses thinking that it will serve a useful purpose such as a game or provide entertainment. However, Trojan horses contain programs that corrupt the data or damage the files. Trojan horses can corrupt software applications. They can also damage files and can contain viruses that destroy and corrupt data and programs. Trojan horse does not replicate themselves like viruses.

COPPERBELT UNIVERSITY

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

CS 120

Unit 11: DATA PROCESSING STANDARDS

INTRODUCTION

One of the most important roles of the Quality Assurance (QA) team in a large software project is the development of appropriate product and process standards. **Product standards** set out attributes that all product components should exhibit. An example of a product standard is a programming language standard that sets out how a language should be used. **Process standards** set out how the software process should be conducted. An example of a process standard is a standard setting out how design reviews should be conducted and documented.

Standards are one of the key means by which management are assured of levels of quality, maintaining project schedules and security. Standards minimize the likelihood of errors and misunderstandings and have been developed (for example, by IBM and other major computer manufacturers) and may be applied to the following:

- (a) Documentation;
- (b) Operations; and
- (c) Authority to approve activities.

SOURCES OF STANDARDS

Other sources of standards, apart from the major computer manufacturers, are:

- (a) Standards organizations, such as The International Organization for Standardization (ISO), ITU Telecommunications Standardization Sector (ITU-T) and the Internet Engineering Task Force (IETF);
- (b) User groups and other individual users; and
- (c) Informal standards that are already in use.

Standards on operating procedures should be applied to the following:

- (a) Computer and other machine operators;
- (b) Each item of equipment (use, periodic checks, etc.);
- (c) The recording of breakdowns, usage time, etc.;
- (d) Filing, Storage, Security;
- (e) Obtaining (ordering) fresh supplies.

Standards on authority to approve activities should be applied to the following:

- (a) Approving the feasibility study;
- (b) Approving the system study and design;
- (c) Approving the system specification and program specifications;
- (d) Purchasing equipment; and
- (e) Amendments to a system or program.

The standards used should be formulated into a *STANDARDS MANUAL* and distributed to relevant staff, which includes management, analysts, programmers, operators, user departments, and auditors. The standards need to be periodically revised to ensure that they are up to date and in line with company/organization policy. The standards manual should cover all areas of systems development. It should clearly indicate which standards are mandatory and which are of the 'guideline' type.

IMPORTANCE OF STANDARDS

Standards are important for a number of reasons:

- (a) They provide an encapsulation of best, or at least most appropriate, practice. This knowledge is often only acquired after a great deal of trial and error and building it into a standard avoids the repetition of past mistakes. In essence, the standard captures some wisdom that is of value to an organization.
- (b) They provide a framework around which the quality assurance (QA) process may be implemented. Given that standards encapsulate best practice, quality assurance becomes the activity of ensuring that standards have been properly followed.
- (c) They assist in continuity where work carried out by one person is taken up by and continued by another. Standards ensure that all developers within an organization adopt the same practices so that the learning effort when starting new work is reduced.

The primary benefit of adhering to software standards is *efficiency*. Standards enable each member of a software team to work anywhere in the source code without needing to recognize and adopt a different programming style.

Documentation of a Computer System

Murphy's law: "When all else fails, read the instructions"

Like standards, documentation is one of the key means by which management are assured of levels of quality, maintaining project schedules and security.

Documentation is an integral part of the software development process. Unfortunately, it is all too often treated as an afterthought, sloppily prepared or outright omitted. This is unacceptable for software projects that are being developed jointly by several individuals or those that may be used or maintained by someone other than the original developer. The complex and abstract ideas embodied in most programs must be communicated clearly if there is to be any continuity and efficiency in their development as well as end use. This is certainly true of the user interface which will be operated without the benefit of instruction by the original developers.

Documentation of a software project is organized into several categories:

Project

The highest level of software documentation is a description of the purpose and functionality of the entire software system. At this level, the reader should be introduced to the overall problem statement, requirements for an acceptable solution, assumptions made, and important characteristics of the design and implementation of the software such as module interface specifications or resource and timing requirements.

Module

Many software systems are designed as a collection of interacting modules (or objects), each of which should be documented individually. In addition, a more detailed description of methods, data structures and algorithms should be included. A software requirements specification serves this purpose well.

Source-Code

Each file of source code should contain comments to aid the reader as well as maintainer in understanding the code.

User-Level

Documentation intended for the users of a software system must be produced for all but the most trivial programs. The format and contents of user-level documentation depends largely upon the application.

TYPES OF DOCUMENTS

USER DOCUMENTATION

The documentation provided for system users is usually the first contact they have with the system. It should provide an accurate initial impression of the system. The user should not have to read most of the documentation to find out how to make simple use of the system. The documentation should be structured in such a way that the user may read it to the level of detail appropriate to his or her needs.

There are at least five (5) documents (or perhaps chapters in a single document) which are an essential part of *user documentation*. These are:

- 1) A *Functional description*, which explains what the system can do.
- 2) An *Installation document*, which explains how to install the system and tailor it for particular hardware configurations.
- 3) An *Introductory manual*, which explains, in simple terms, how to get started with the system.
- 4) A *Reference manual*, which describes in detail all of the system facilities available to the user and how these facilities can be used.
- 5) A *System Administrator's guide* (if necessary), explaining how to react to situations which arise while the system is in use and how to carry out system housekeeping tasks such as making a system backup.

Commonly used user documentation are:

- (a) User manuals; and
- (b) Operators manuals.

USER MANUALS

Features of user manuals include:

- 1) Details of how to install the system in a particular environment;
- 2) A detailed description to the system's users of how to use the system;
- 3) An informal introduction to the system, describing its 'normal' usage.
- 4) A description of how to get started on the system and how the user might make use of the common system facilities;
- 5) Being liberally illustrated with examples; and
- 6) Instructions to the system user on how to get out of trouble when things go wrong.

OPERATORS MANUALS

These manuals give details to the computer operators of how to run the programs. They may include:

- 1) Details of the procedure for starting the program;
- 2) Details of disks or tapes required;

- 3) Special stationery to be used;
- 4) The number of copies of each report, and who is to receive the output.
- 5) Backup procedures to be followed; and
- 6) Recovery procedures to be followed in the event of hardware failure.

SYSTEM DOCUMENTATION

System documentation encompasses all of the documents describing the implementation of the system from the requirements specification to the final acceptance test plan. Documents describing the design, implementation and testing of a system are essential if the program is to be understood and maintained. Like user documentation, it is important that system documentation is structured, with overviews leading the reader into more formal and detailed descriptions of each aspect of the system.

The documents making up the system documentation should include:

- 1) The requirements definition and specification and an associated rationale.
- 2) An overall system specification showing how the requirements are decomposed into a set of interacting programs. This document is not required when the system is implemented using only a single program.
- 3) For each program in the system, a description of how that program is decomposed into components and a statement of the specification of each component.
- 4) For each unit, a description of its operation. This need not extend to describing program actions as these should be documented using intra-program comments.
- 5) A comprehensive test plan describing how each program unit is tested.
- 6) A test plan showing how integration testing, that is, the testing of all units/programs together is carried out.
- 7) An acceptance test plan, devised in conjunction with the system user. This should describe the tests that must be satisfied before the system is accepted.

DOCUMENTATION STANDARDS

Documentation standards consist of two parts: *in-line* and *external*. In-line documentation aids in module debugging and maintenance. External documentation of support routines is typically used to assist in determining which subroutines to call and how to do so; whereas, external documentation of application modules is used to assist the user in operating the software in the correct manner.

In-Line Program Documentation

In-line documentation includes two particular areas:

- (1) The prologue is text describing the program unit that is placed at the beginning of the file before any code. This implies that there is only one prologue per file. Local

subroutines do not need a full prologue since they will not be called by any routines except for those in the same file, but subroutines do need an algorithm description. The routines should be simple enough so that a few comments can explain how and what they do.

- (2) All code should contain text which identifies and describes each major block and any complicated code sequence.

External Support Module Documentation

External support module documentation is documentation of a support module that is kept external to the program. This documentation would be useful to a programmer trying to understand the function of a piece of code or an entire package. It would also be useful to a programmer trying to use an entry point out of a library. It would not be useful to a typical user trying to run a program or package.

CS 120

UNIT 13: INFORMATION SYSTEMS

Introduction

In the previous units, the focus was on learning about parts of the computer, networking of computers and, the Internet. In this unit, the focus is on the use of the computer and the telecommunication technology to generate information for businesses and organizations. Information system is an organized combination of people, hardware, software, communication network, and data resources that perform collection of data, process it and transform it to provide information, in an organization. The information required from the information systems at different levels of the organization hierarchy is different in nature, like operational information, management information, or information that supports the decision making process.

Information systems are vital for the functioning of businesses and organizations. This unit discusses the information systems, and their categorization to provide support at different levels of the organization.

Data, Information and Knowledge

Data are raw facts. Data is isolated values which by themselves do not have much significance. Data may be numbers, letters or characters, image represented as graphic images and pictures, audio or video. It represents something in the real world. Data forms the raw material for the production of information. Data is not organized and does not convey a meaning. For example, the data elements like ‘24th’, ‘Oct’, 2012’, ‘Kitwe’, and ‘Independence Day’ just represent values.

Information is processed, manipulated, and interpreted data. It is the collection of data organized in such a way that they have value beyond the data itself. Information is created by defining and organizing relationships among data. For example, ‘The Independence Day celebrations will be held in Kitwe on the 24th October, 2012’. The processed data gives us information which can be used for different purposes.

Knowledge is an inference or conclusion drawn from the information. For example, a person from Mongu who wishes to go to Kitwe can use the information ‘The Independence Day celebrations will be held in Kitwe on the 24th October, 2012’ to enhance his knowledge so that this person can plan his trip to Kitwe accordingly.

Characteristics of Information

Information that is meaningful for the people and the organization must have certain characteristics. The characteristics of the information are listed as follows—

- **Accurate:** Information must not contain any errors. It must be accurate.
- **Accessible:** Authorized users should be able to access the information whenever required.
- **Complete:** Information must contain all important and related data.
- **Economical:** Information should be economical to produce both in terms of time and cost.
- **Format:** Information should be available in the desired format.
- **Flexible:** Information should be flexible enough to be used for different purposes.
- **Reliable:** Information is dependable and is generated using correct data.
- **Relevant:** Information must be relevant so that it can be used by the organization.
- **Secure:** Unauthorized users should not be able to access the information. Access is allowed only to authorized individuals.
- **Simple:** Information must be easily understandable and usable. Complex information is difficult to use and may not serve its purpose.
- **Timely:** Information must be available when it is needed. Late or outdated information is of no use.
- **Verifiable:** There should be a means to cross check the available information.

The information having the above characteristics is valuable and is used by the organization for achieving its goals. The organization may use the information for different purposes like decision making, managing the organization etc.

Information System (IS)

A system is a set of components that work together to achieve a common goal. A system accepts input, processes it, and produces an output (Figure below). A system is composed of one or more subsystems. A system may be a closed system or an open system. A closed system is a stand-alone system that does not interact with other systems. An open system interacts with other systems.

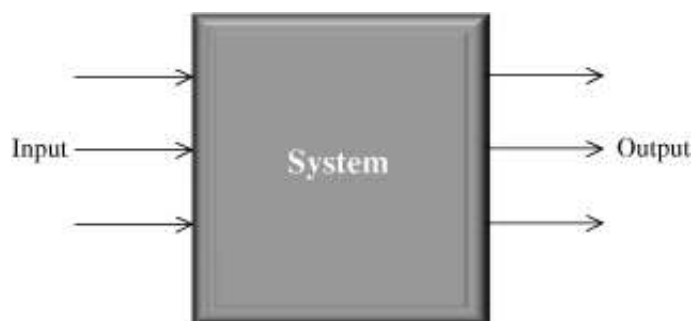


Figure – System

Information System (IS) is a set of inter-related components that (1) collect data, (2) manipulate data, (3) disseminate data and information, and (4) provide feedback to meet an objective. An information system uses resources like people, hardware, software, data, and networks to perform input, processing, output, storage, and control activities. Some examples of IS are ATMs, railway reservation system, and movie ticket reservation systems.

Components of Information System: An IS consists of four main components—Input, Process, Output, and Feedback (Figure below). Input is the activity of gathering and capturing raw data. A process converts or transforms data into useful output. During processing, the input data is manipulated into information using mathematical, statistical, and other tools. Output is the generation and presentation of useful information, usually in the form of documents and reports. Feedback is an output that is used to make changes to the input, or the processing activities.

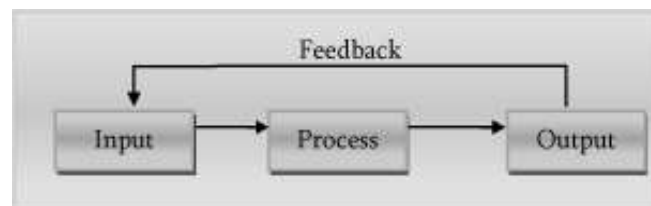
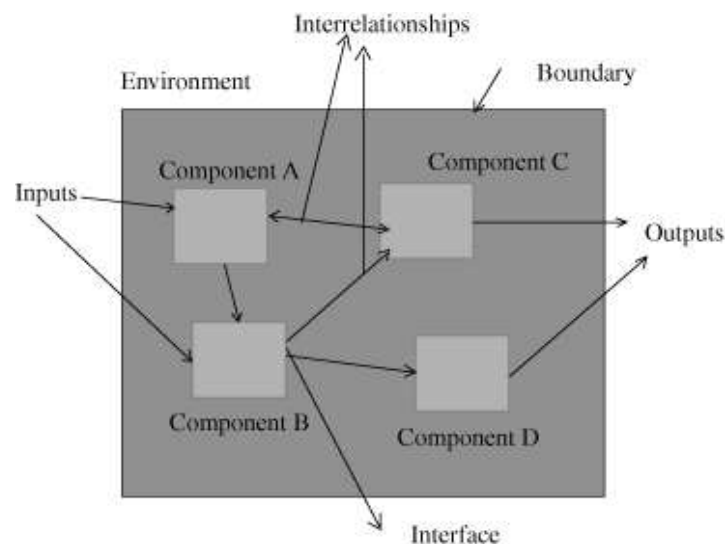


Figure – Components of Information System

Information systems are needed both by the people and the business organizations. People need information system for entertainment and enlightenment. Organizations require information systems to support decision making, coordination, problem solving and control. Information systems may also help to analyze problems, and to create new products.

Characteristics of IS: Like any other system, ISs have seven characteristics



Characteristics of Information System

- Environment is anything outside the system.
- Boundary separates the system from environment. Any part within boundary is the system and outside boundary is the environment.
- Input is the data from environment to the system. The system uses the input.
- Output is the information from the system to the environment. The system uses input to generate output.
- Components are the subparts of the system.
- Interrelationships are the associations between components of a system.
- Interface is a mechanism for interaction between different components of the system.

The performance of Information System is measured by its efficiency and effectiveness. Efficiency is a measure of what is produced divided by what is consumed. Effectiveness is the extent to which a system attains its goals.

The role of the Information System is expanding and evolving continuously. In 1950s–1960s, the focus was on data processing. Management reporting and decision support was provided during 1960s–1970s and 1970s–1980s, respectively. 1980–1990s saw it being also used for strategic and end user support. Lately in 1990s–2000s, there is a shift towards global internetworking.

An information system can be manual and computerized. Here, we shall focus on the computerized information system or the Computer-based Information System.

Computer-Based Information System (CBIS)

Computer-Based Information System (CBIS) use computer and telecommunication technology for the generation of information. CBIS uses hardware, software, databases, telecommunications, people, and procedures for the collection, manipulation, storage, and processing of data into information. They take raw data as input, process it, and produce information as output.

The components of a CBIS (Figure below) include the following:

- Hardware—computer devices like keyboard, monitor, processor, and printer, used for performing the input, processing, and output activities.
- Software—computer programs that govern the operation of the computer.
- Database—an organized collection of data, consisting of two or more related data files.
- People—those who work with the CBIS. People are the most important element in most CBIS.
- Procedures—strategies, policies, methods, and rules for using the CBIS.
- Telecommunication, Network, and Internet—Telecommunication and network is used for connecting the computers and computer devices in a building, city, country, or across the world to enable electronic communication.

Internet is the world's largest computer network, which is an interconnection of networks. World Wide Web (WWW) is a system with universally accepted standards for storing, retrieving, formatting, and displaying information in a networked environment. The Internet and WWW have had a tremendous impact on the role IS play in organizations. They are responsible for the increased connectivity and collaboration within and outside the organization. They have helped in transforming the structure of an organization, scope of operations, reporting and control mechanisms, work practices, and services.

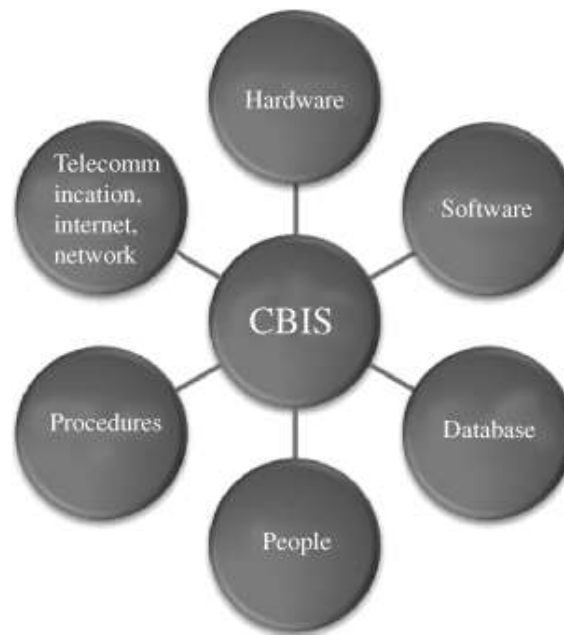


Figure – Computer Based Information System

IS or CBIS is transforming the way businesses are done nowadays. Communication tools like cell phones, e-mail, online conferencing, and international teleconferencing over the Internet have all become essential tools of a business. E-commerce, e-banking, e-governance, e-learning, Internet advertising, and social media marketing have all seen an increasing usage.

In this unit, when we refer to Information Systems, we are talking about the Computer-Based Information System.

Need for Efficient Information System

In today's world, information systems are indispensable to businesses, industries, and organizations, to meet the future challenges. Organizations need ISs to meet the information needs of its employees. More and more business organizations are using information systems to improve the way they conduct business, as discussed below:

- IS, for many businesses, provide an opportunity to do business in a new way and thus gain huge profits and market shares.
- ISs facilitate the acquisition, transformation, and distribution of information. ISs can improve decision making, enhance organizational performance, and help increase profitability.
- The use of information systems to add value to the organization can also give an organization a competitive advantage. It provides significant, long-term benefit to a company over its competition. ISs are used for strategic planning in a competitive world. It allows changing the structure, creating of new services, and improving upon the existing services.
- Damaged or lost data can cause disruptions in normal business activities leading to financial losses, law suits, etc. ISs help an organization to better manage and secure its critical data.
- ISs also improve integration and the work processes.

For an organization, ISs impact the way business is carried out. Organizations need to have an efficient IS to improve and manage their function, performance, and productivity. Briefly summarized, ISs are important for business today due to the need for operational excellence, new products, services and business models, improved decision making, competitive advantage, and the most important—survival.

Categories of Information System

An organization structure can be shown as a pyramid which is divided both horizontally and vertically. Vertically, the divisions represent the different functional areas of the organization, like sales, marketing, accounting, human resource, and manufacturing. Horizontally, the divisions are made according to the operational level of the organization, hierarchically from the base of the pyramid to the top, i.e. operations at low level, middle level, and top level (Figure below). Information systems provide support at all the levels of the organization and in the different functional areas.

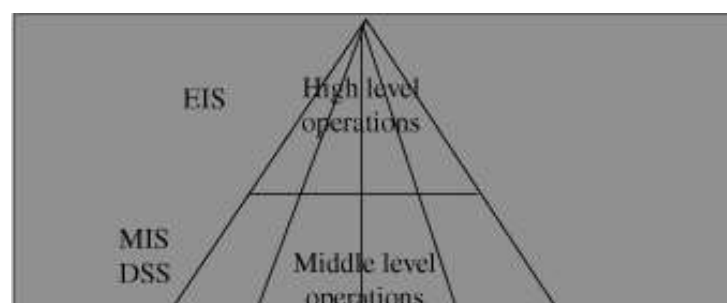


Figure – Information System Categories

Major application categories of the information systems on the basis of the support provided by them at the different levels of the operations hierarchy are as follows:

- Operations Support System processes data generated by business operations, and is required for the low-level operations. The categories of operations support system are:
 - Transaction Processing System (TPS), and
 - Office Automation System (OAS).
- Management Support System provides information and support needed for decision making by the managers at middle level and top level of the organization. Major categories of management support systems are:
 - Management Information System (MIS),
 - Decision Support System (DSS), and
 - Executive Information System (EIS).

In addition to the above information systems, there is specialized information systems like—

- Expert Systems,
- Enterprise Resource Planning (ERP), and
- Electronic commerce (E-commerce) system.

Operations Support System

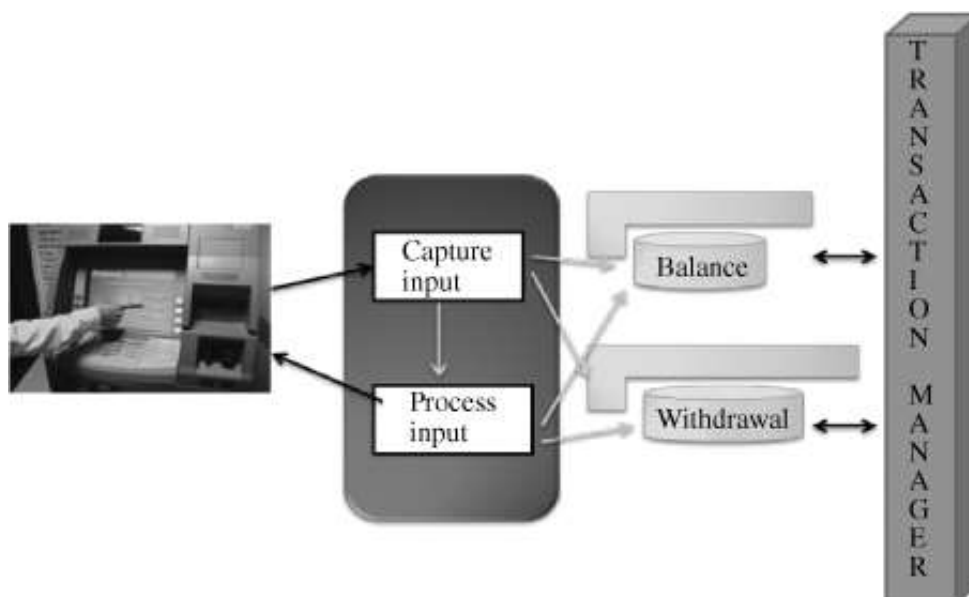
The operations support system provides information about the day-to-day activities of the organization. They support the operations of the organization, by keeping track of the resources and the transactions. For example, in a banking operation, the operations support system are used to keep track of current balance of the customer; in a manufacturing unit, it helps to keep track of the inventory of the items; and, in a sales unit, it keeps track of the number of units of each item sold. Thus, transaction processing is the main function of the operations support system.

Transaction Processing System (TPS)

Transaction is a basic level of activity that is performed during the business operations. It is the business-related exchange of payments to employees, sales to customers, and payments to suppliers. Examples of transactions are—withdrawal of money from bank, booking ticket for an airline, selling of goods to the customer etc.

TPS is an organized collection of people, procedures, software, databases, and devices used for recording the business transactions. TPSs are essential parts of any business organization.

TPSs are used for routine, repetitive, and ordinary business transactions, and transactions that are critical to the daily function of the business. TPS records the business transactions and generates the documents related to the transaction. For example, daily routine transactions for sales orders from customers, bank deposits and bank withdrawals.



TPS automates handling of the data for the business activities and transactions.

The goal of TPS is to increase transaction processing speed, enhance productivity, and simplify the processes.

TPS gathers all the input necessary for other types of systems. It provides the basic input to the company's database. A transaction generates data, which is recorded by the transaction processing system. This data forms the input for the generation of the other information systems, like MIS, DSS and EIS. For example, based on the daily transactions, a monthly sales report for middle management and marketing information for senior managers is generated.

In an organization, there are different functional areas like manufacturing, human resource, accounts, finance, administration, sales, and marketing. The TPS information systems are developed for and used by people in the different functional areas. Examples of TPS are Payroll TPS, Sales TPS, and Marketing TPS.

Office Automation System (OAS)

OAS supports automation of the tasks in an office. It supports general office work for handling and managing documents. It facilitates communication, both within the office, and, between the office and its environment. It provides support for the traditional office operations like managing the documents, sending messages for interaction, arranging and conducting meetings, scheduling of the projects etc.



OAS allows use of text processing systems like word processors and desktop publishing system, to enable the creation of professional documents with graphics and special layout features.

Spreadsheets, presentation packages like PowerPoint, personal database systems, and, note-taking systems like appointment book, notepad and calendar, are parts of an OAS.

OAS includes the communication systems required for transmitting the messages, documents, images, video or voice. For this, it uses the network and the Internet. It incorporates the e-mail facility, voice mail, teleconferencing capabilities, accessing web sites etc.

Management Support System

The purpose of the management support system is to assist the managers at different levels in an organization, in the controlling and the management of the business. Management support systems provide different kinds of reports, such as summary report, sales report, and customer balance report. They also assist the managers in the decision making process by generating projected reports like projected sales report etc. Management support systems also provide information to the executives related to the overall operations of the organization, the overall growth, profits, its performance etc. Moreover, for any of the reports, it also provides a detailed report specifying its internal details, if required.

MIS is a collection of people, procedures, software, databases, and devices that provides information to managers and decision makers.

MIS provides information to the managers to help them make informed decisions. MIS supports the managers in the decision making process and enhances their productivity and performance. For example, MIS may generate information related to the sales of a product in a month and the inventory status during that month. It can also monitor the performance and maintain coordination between the different functional areas, like purchase and accounts payable, in an organization.

MIS works with the transaction processing system. MIS uses the data generated from TPS as input, and converts them into meaningful aggregate form.

MIS generates different kinds of reports like summary report, exception report, and detailed report. Summaries are for the high-level management and give statistical information to the managers. Exception reports provide information about the activities that are different from the normal functioning. Detailed reports generally provide operational level reports.

The different kinds of MIS reports are either scheduled to be generated at fixed intervals of time, may be event-driven and generated when an event occurs, or may be generated in an ad-hoc manner, for a particular query. Scheduled reports are periodic reports that are generated weekly, monthly, or quarterly. Ad-hoc reports are generated on-demand, whenever information is required by the managers.

MIS can also generate customized reports, where managers can select from different data, sorting options, and display choices.

The focus of MIS is operational efficiency. The reports generated by the MIS helps the managers in controlling the operations of the organization and in the decision making process.

Decision Support System (DSS)

DSS is a collection of people, procedures, software, databases, and devices that supports problem-specific decision making.

DSS support the middle level managers in the decision making process. They supplement the decision making power of the manager with the information technology.

DSS facilitate decision making by applying mathematical or logical models and interactions to solve problems. For examples, a large number of mathematical and logical calculations are required to be performed on data which may be more than million in number, in order to generate information that helps in the decision making process.

DSSs are user-friendly and highly interactive.

DSS focuses on the decision-making effectiveness.

DSS use data from the TPS and MIS, and, also any new data from external sources, such as current share prices or prices of competitors.

DSS provides comparison of alternatives and recommendations of a preferred option. DSS provide powerful models and tools for simulation, optimization etc. to the managers so that they can evaluate from several alternatives before making a decision. For a particular problem, the managers can use DSS to get outcome of several different scenarios, and then compare them to arrive at the right decision.

Executive Information System (EIS)

EIS provides a generalized computing and communication environment to senior managers to support strategic decisions. It provides summarized information in a convenient form to the top level managers.

Unlike DSS, they are not designed to use analytical models for specific problem solving. EIS is designed to facilitate access to information quickly and effectively, for the senior managers.

EIS is used by the senior managers to identify a problem, an opportunity, or a trend which affects the organization's future growth. The middle level managers use DSS to suggest solution to the problem or analyze the opportunity.

EIS takes data from the TPS, MIS and allows communication with external sources of information.

EIS has menu-driven user-friendly interfaces, interactive graphics to help visualization of the situation, and telecommunication links to connect to the external information, if required.

Specialized Information System

Specialized information systems are both for organizational and individual use. The specialized systems help organizations and individuals achieve their goals.

Expert Systems

Artificial Intelligence (AI) system is the people, procedures, hardware, software, data, and knowledge needed to develop computer that display the characteristics of intelligence. For certain types of well-defined problems, AI systems can replicate the human decision making. AI systems mimic the functions of the human brain.

Expert system is an applied area of AI. An expert system is a knowledge- based system having two main components—knowledge base and inference engine. It uses the knowledge base to store relevant information, data, rules, cases, and relationships, and, the inference engine gets information and relationships from the knowledge base, and, provides answers and predictions the way a human expert does.

Expert systems are used in areas where there is a need to capture and preserve irreplaceable human expertise, provide expertise that is expensive or rare, and to provide expertise needed for training and development to share the wisdom of human experts with a large number of people.

The expert systems are used to explore new business possibilities, increase overall profitability, and reduce costs.

Expert systems are used to provide information for problems related to classification, diagnosis, process control, scheduling and planning etc. They are used in products like antilock braking system, television, plant layout and manufacturing, credit granting and loan analysis, and stock picking.

Enterprise Resource Planning (ERP)

ERP system is a computer software system that manages and coordinates all the resources, information, and functions of a business from shared data stores.

ERP systems are integrated programs that manage all business operations. For example, they coordinate the planning, inventory control, production, and ordering.

ERP systems integrate functions such as human resources, supply chain management, customer relations management, financial, manufacturing functions, and warehouse management functions.

Since ERP systems are integrated systems, they help to streamline different processes and workflows, allow data to be easily shared across various departments in an organization, improve the efficiency and productivity levels, and improves customer service.

SAP is an example of ERP system. SAP stands for Systems, Applications and Products.

Electronic-Commerce (E-Commerce)

E-commerce involves any business transaction executed electronically between parties. It uses Internet and Web for doing the business. It uses services like e-mail, workflow software tools, Intranet, and, the e- payment services.

E-commerce involves buying and selling of products and services, electronically.

The parties involved in e-commerce may be of the following kinds:

- Companies and Companies (B2B). A data processing company handling data services for a company.
- Companies and Consumers (B2C).
- Consumers and Consumers (C2C). A customer selling goods to another customer, like in e-bay.com.
- Business and the public sector, and, consumers and the public sector.

E-commerce web sites are like on-line market places where you can sell and buy items, and facilitate it by advertising your product, establishing newsgroups and blogs, posting job-oriented resumes etc.

The on-line shopping is a fast growing segment as consumers are becoming more confident to use it, with the widespread use of the Internet.